zkAssess Security Report

Collatz Labs / zkAssess Protocol · Generated 2026-08-06 13:41 UTC

Application	TLS1.3
Overall security score	0 / 100
Critical backdoor detected	YES
Total findings	2343
Files scanned	89
Pattern set version	2026.08.1

■ **CRITICAL: one or more backdoor indicators were detected in this repository.**

Findings

[CRITICAL] HARDCODED_CLOUD_CREDENTIAL

Hardcoded cloud provider access key, private key material, or platform token committed directly to source.

```
Location: makecertchain/makeleafcert.cpp:1175
Snippet: fputs("-----BEGIN PRIVATE KEY-----\n",fp);
```

Remediation: Revoke and rotate the credential immediately; move secrets to a managed secret store or environment variables, never source control.

[CRITICAL] HARDCODED_CLOUD_CREDENTIAL

Hardcoded cloud provider access key, private key material, or platform token committed directly to source.

```
Location: makecertchain/makeintercert.cpp:1152
Snippet: fputs("-----BEGIN PRIVATE KEY-----\n",fp);
```

Remediation: Revoke and rotate the credential immediately; move secrets to a managed secret store or environment variables, never source control.

[CRITICAL] HARDCODED_CLOUD_CREDENTIAL

Hardcoded cloud provider access key, private key material, or platform token committed directly to source.

```
Location: makecertchain/makerootcert.cpp:961
Snippet: fputs("-----BEGIN PRIVATE KEY-----\n",fp);
```

Remediation: Revoke and rotate the credential immediately; move secrets to a managed secret store or environment variables, never source control.

[CRITICAL] HARDCODED_CLOUD_CREDENTIAL

Hardcoded cloud provider access key, private key material, or platform token committed directly to source.

```
Location: makecertchain/makeselfcert.cpp:987
Snippet: fputs("-----BEGIN PRIVATE KEY-----\n",fp);
```

Remediation: Revoke and rotate the credential immediately; move secrets to a managed secret store or environment variables, never source control.

[CRITICAL] SIKE_SIDH_USAGE

SIKE/SIDH, a NIST Round 4 post-quantum key-encapsulation candidate, was completely broken in 2022 (the Castryck-Decru attack recovers the private key using only classical computation, in about an hour on a single core)

and withdrawn from standardization.
`Location: rust/server/src/config.rs:160`
`Snippet: pub const SIDH: u16 = 0x4243;`
Remediation: Migrate immediately -- this scheme is fully broken, not merely weakened. Use ML-KEM (FIPS 203) for key encapsulation instead.

[CRITICAL] SIKE_SIDH_USAGE
SIKE/SIDH, a NIST Round 4 post-quantum key-encapsulation candidate, was completely broken in 2022 (the Castryck-Decru attack recovers the private key using only classical computation, in about an hour on a single core) and withdrawn from standardization.
`Location: rust/server/src/main.rs:51`
`Snippet: SIDH => println!("SIDH")...`
Remediation: Migrate immediately -- this scheme is fully broken, not merely weakened. Use ML-KEM (FIPS 203) for key encapsulation instead.

[CRITICAL] SIKE_SIDH_USAGE
SIKE/SIDH, a NIST Round 4 post-quantum key-encapsulation candidate, was completely broken in 2022 (the Castryck-Decru attack recovers the private key using only classical computation, in about an hour on a single core) and withdrawn from standardization.
`Location: rust/server/src/tls13/logger.rs:86`
`Snippet: SIDH => log(IO_PROTOCOL,...`
Remediation: Migrate immediately -- this scheme is fully broken, not merely weakened. Use ML-KEM (FIPS 203) for key encapsulation instead.

[CRITICAL] SIKE_SIDH_USAGE
SIKE/SIDH, a NIST Round 4 post-quantum key-encapsulation candidate, was completely broken in 2022 (the Castryck-Decru attack recovers the private key using only classical computation, in about an hour on a single core) and withdrawn from standardization.
`Location: rust/client/src/config.rs:170`
`Snippet: pub const SIDH: u16 = 0x4243;`
Remediation: Migrate immediately -- this scheme is fully broken, not merely weakened. Use ML-KEM (FIPS 203) for key encapsulation instead.

[CRITICAL] SIKE_SIDH_USAGE
SIKE/SIDH, a NIST Round 4 post-quantum key-encapsulation candidate, was completely broken in 2022 (the Castryck-Decru attack recovers the private key using only classical computation, in about an hour on a single core) and withdrawn from standardization.
`Location: rust/client/src/main.rs:179`
`Snippet: SIDH => println!("SIDH")...`
Remediation: Migrate immediately -- this scheme is fully broken, not merely weakened. Use ML-KEM (FIPS 203) for key encapsulation instead.

[CRITICAL] SIKE_SIDH_USAGE
SIKE/SIDH, a NIST Round 4 post-quantum key-encapsulation candidate, was completely broken in 2022 (the Castryck-Decru attack recovers the private key using only classical computation, in about an hour on a single core) and withdrawn from standardization.
`Location: rust/client/src/tls13/logger.rs:87`
`Snippet: SIDH => log(level,"SIDH\...`
Remediation: Migrate immediately -- this scheme is fully broken, not merely weakened. Use ML-KEM (FIPS 203) for key encapsulation instead.

[CRITICAL] HARDCODED_CLOUD_CREDENTIAL
Hardcoded cloud provider access key, private key material, or platform token committed directly to source.
`Location: cpp/lib/tls_client_cert.cpp:18`
`Snippet: "-----BEGIN PRIVATE KEY-----\n"`

Remediation: Revoke and rotate the credential immediately; move secrets to a managed secret store or environment variables, never source control.

[CRITICAL] HARDCODED_CLOUD_CREDENTIAL

Hardcoded cloud provider access key, private key material, or platform token committed directly to source.

```
Location: cpp/lib/tls_client_cert.cpp:49
Snippet: "-----BEGIN PRIVATE KEY-----\n"
```

Remediation: Revoke and rotate the credential immediately; move secrets to a managed secret store or environment variables, never source control.

[CRITICAL] HARDCODED_CLOUD_CREDENTIAL

Hardcoded cloud provider access key, private key material, or platform token committed directly to source.

```
Location: cpp/lib/tls_client_cert.cpp:122
Snippet: "-----BEGIN PRIVATE KEY-----\n"
```

Remediation: Revoke and rotate the credential immediately; move secrets to a managed secret store or environment variables, never source control.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: clientcert/client/client.py:36
Snippet: ...enssl ecparam -name secp384r1 -out ecparam.pem",s...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: clientcert/client/client.py:41
Snippet: ...sl req -new -newkey rsa:1024 -keyout client...
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: clientcert/client/client.py:45
Snippet: ...sl req -new -newkey rsa:2048 -keyout client...
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: clientcert/client/client.py:49
Snippet: ...sl req -new -newkey rsa:4096 -keyout client...
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: clientcert/client/client.py:52
Snippet: if method=="ED25519" :
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: clientcert/client/client.py:53
Snippet: ...genpkey -algorithm ED25519 -out eddsa.key",she...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: clientcert/client/client.py:54`
`Snippet: ...enssl req -new -key eddsa.key -keyout client....`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: clientcert/client/client.py:57`
`Snippet: if method=="ED448" :`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: clientcert/client/client.py:58`
`Snippet: ...genpkey -algorithm ED448 -out eddsa.key",she...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: clientcert/client/client.py:59`
`Snippet: ...enssl req -new -key eddsa.key -keyout client....`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: clientcert/client/intermediate/intermediate.py:37`
`Snippet: ...enssl ecparam -name secp384r1 -out ecparam.pem",s...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: clientcert/client/intermediate/intermediate.py:42`
`Snippet: ...sl req -new -newkey rsa:1024 -keyout inter....`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: clientcert/client/intermediate/intermediate.py:46`
`Snippet: ...sl req -new -newkey rsa:2048 -keyout inter....`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: clientcert/client/intermediate/intermediate.py:50`
`Snippet: ...sl req -new -newkey rsa:4096 -keyout inter....`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: clientcert/client/intermediate/intermediate.py:53`

Snippet: if method=="ED25519" :

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: clientcert/client/intermediate/intermediate.py:54

Snippet: ...genpkey -algorithm ED25519 -out eddsa.key",she...

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: clientcert/client/intermediate/intermediate.py:55

Snippet: ...enssl req -new -key eddsa.key -keyout inter.k...

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: clientcert/client/intermediate/intermediate.py:58

Snippet: if method=="ED448" :

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: clientcert/client/intermediate/intermediate.py:59

Snippet: ...genpkey -algorithm ED448 -out eddsa.key",she...

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: clientcert/client/intermediate/intermediate.py:60

Snippet: ...enssl req -new -key eddsa.key -keyout inter.k...

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: clientcert/client/intermediate/root/root.py:26

Snippet: ...enssl ecparam -name secp384r1 -out ecparam.pem",s...

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

Location: clientcert/client/intermediate/root/root.py:30

Snippet: ...-days 3650 -newkey rsa:1024 -keyout root.k...

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

Location: clientcert/client/intermediate/root/root.py:33

Snippet: ...-days 3650 -newkey rsa:2048 -keyout root.k...

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: clientcert/client/intermediate/root/root.py:36`
`Snippet: ...-days 3650 -newkey rsa:4096 -keyout root.k...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: clientcert/client/intermediate/root/root.py:38`
`Snippet: if method=="ED25519" :`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: clientcert/client/intermediate/root/root.py:39`
`Snippet: ...genpkey -algorithm ED25519 -out eddsa.key",she...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: clientcert/client/intermediate/root/root.py:40`
`Snippet: ...509 -days 3650 -key eddsa.key -keyout root.ke...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: clientcert/client/intermediate/root/root.py:41`
`Snippet: if method=="ED448" :`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: clientcert/client/intermediate/root/root.py:42`
`Snippet: ...genpkey -algorithm ED448 -out eddsa.key",she...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: clientcert/client/intermediate/root/root.py:43`
`Snippet: ...509 -days 3650 -key eddsa.key -keyout root.ke...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:16`
`Snippet: #define ED25519_PK 7`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: makecertchain/makeleafcert.cpp:17
Snippet: #define ED448_PK 8
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: makecertchain/makeleafcert.cpp:20
Snippet: #define ED25519_MLDSA44_PK 11 // H...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: makecertchain/makeleafcert.cpp:32
Snippet: #define ED25519_SIG 7
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: makecertchain/makeleafcert.cpp:33
Snippet: #define ED448_SIG 8
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: makecertchain/makeleafcert.cpp:36
Snippet: #define ED25519_MLDSA44_SIG 11
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: makecertchain/makeleafcert.cpp:59
Snippet: ...KEYLEN (2048/8) // RSA only - issuer key l...
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: makecertchain/makeleafcert.cpp:62
Snippet: ...KEYLEN (2048/8) // RSA only - subject key...
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: makecertchain/makeleafcert.cpp:107
Snippet: #if PKTYPE==ED25519_PK
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:113`
`Snippet: #if PKTYPE==ED448_PK`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:133`
`Snippet: #if PKTYPE==ED25519_MLDSA44_PK`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:171`
`Snippet: ...fine SIG_TYPE ECDSA_SECP256R1_SHA256 // signature...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:178`
`Snippet: ...fine SIG_TYPE ECDSA_SECP384R1_SHA384`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:201`
`Snippet: #if SIGTYPE==ED25519_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:206`
`Snippet: #define SIG_TYPE ED25519`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:208`
`Snippet: #if SIGTYPE==ED448_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:213`
`Snippet: #define SIG_TYPE ED448`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:231`
`Snippet: #if SIGTYPE==ED25519_MLDSA44_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:240`
`Snippet: #define SIG_TYPE_1 ED25519`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: makecertchain/makeleafcert.cpp:502`
`Snippet: // RSA`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:515`
`Snippet: // MLDSA or EDDSA`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:516`
`Snippet: ...DSA44_PK || PKTYPE==ED25519_PK || PKTYPE==ED448...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:536`
`Snippet: #if PKTYPE==ED25519_MLDSA44_PK`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:564`
`Snippet: // ECDSA`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeleafcert.cpp:756`
`Snippet: #if SIGTYPE==ED25519_MLDSA44_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: makecertchain/makeleafcert.cpp:818
Snippet: #if PKTYPE==ED25519_PK || PKTYPE==ED448...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeleafcert.cpp:1010
Snippet: ...if (ret.curve==USE_ED25519)
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeleafcert.cpp:1012
Snippet: ...#if SIGTYPE==ED25519_SIG
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeleafcert.cpp:1016
Snippet: ...if (ret.curve==USE_ED448)
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeleafcert.cpp:1018
Snippet: ...#if SIGTYPE==ED448_SIG
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeleafcert.cpp:1043
Snippet: ...if (ret.curve==USE_ED25519)
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeleafcert.cpp:1045
Snippet: ...#if SIGTYPE==ED25519_MLDSA44_SIG
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeintercert.cpp:15
Snippet: #define ED25519_PK 7
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: makecertchain/makeintercert.cpp:16
Snippet: #define ED448_PK 8
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeintercert.cpp:19
Snippet: #define ED25519_MLDSA44_PK 11 // H...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeintercert.cpp:31
Snippet: #define ED25519_SIG 7
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeintercert.cpp:32
Snippet: #define ED448_SIG 8
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeintercert.cpp:35
Snippet: #define ED25519_MLDSA44_SIG 11 //...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: makecertchain/makeintercert.cpp:57
Snippet: ...KEYLEN (2048/8) // RSA only - issuer key l...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: makecertchain/makeintercert.cpp:60
Snippet: ...KEYLEN (2048/8) // RSA only - subject key...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeintercert.cpp:105
Snippet: #if PKTYPE==ED25519_PK
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeintercert.cpp:111
Snippet: #if PKTYPE==ED448_PK

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:131`
`Snippet: #if PKTYPE==ED25519_MLDSA44_PK`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:169`
`Snippet: ...fine SIG_TYPE ECDSA_SECP256R1_SHA256 // signature...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:176`
`Snippet: ...fine SIG_TYPE ECDSA_SECP384R1_SHA384`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:199`
`Snippet: #if SIGTYPE==ED25519_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:204`
`Snippet: #define SIG_TYPE ED25519`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:206`
`Snippet: #if SIGTYPE==ED448_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:211`
`Snippet: #define SIG_TYPE ED448`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:229`
`Snippet: #if SIGTYPE==ED25519_MLDSA44_SIG`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:238`
`Snippet: #define SIG_TYPE_1 ED25519`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: makecertchain/makeintercert.cpp:498`
`Snippet: // RSA`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:511`
`Snippet: // MLDSA or EDDSA`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:512`
`Snippet: ...DSA44_PK || PKTYPE==ED25519_PK || PKTYPE==ED448...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:532`
`Snippet: #if PKTYPE==ED25519_MLDSA44_PK`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:560`
`Snippet: // ECDSA`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:732`
`Snippet: #if SIGTYPE==ED25519_MLDSA44_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:794`
`Snippet: #if PKTYPE==ED25519_PK || PKTYPE==ED448...`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:986`
`Snippet: ...if (ret.curve==USE_ED25519)`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:988`
`Snippet: ...#if SIGTYPE==ED25519_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:992`
`Snippet: ...if (ret.curve==USE_ED448)`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:994`
`Snippet: ...#if SIGTYPE==ED448_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:1019`
`Snippet: ...if (ret.curve==USE_ED25519)`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeintercert.cpp:1021`
`Snippet: ...#if SIGTYPE==ED25519_MLDSA44_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:15`
`Snippet: #define ED25519_PK 7`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:16`
`Snippet: #define ED448_PK 8`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:19`
`Snippet: #define ED25519_MLDSA44_PK 11 // H...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:31`
`Snippet: #define ED25519_SIG 7`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:32`
`Snippet: #define ED448_SIG 8`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:35`
`Snippet: #define ED25519_MLDSA44_SIG 11 //...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: makecertchain/makerootcert.cpp:53`
`Snippet: ...KEYLEN (2048/8) // RSA only - issuer key l...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: makecertchain/makerootcert.cpp:56`
`Snippet: ...KEYLEN (2048/8) // RSA only - subject key...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:101`
`Snippet: #if PKTYPE==ED25519_PK`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:107`
`Snippet: #if PKTYPE==ED448_PK`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:127`
`Snippet: #if PKTYPE==ED25519_MLDSA44_PK`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:165`
`Snippet: ...fine SIG_TYPE ECDSA_SECP256R1_SHA256 // signature...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:171`
`Snippet: ...fine SIG_TYPE ECDSA_SECP384R1_SHA384`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:191`
`Snippet: #if SIGTYPE==ED25519_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:195`
`Snippet: #define SIG_TYPE ED25519`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:197`
`Snippet: #if SIGTYPE==ED448_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:201`
`Snippet: #define SIG_TYPE ED448`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:218`
`Snippet: #if SIGTYPE==ED25519_MLDSA44_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:225`
`Snippet: #define SIG_TYPE_1 ED25519`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: makecertchain/makerootcert.cpp:456`
`Snippet: // RSA`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:469`
`Snippet: // MLDSA or EDDSA`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:470`
`Snippet: ...DSA44_PK || PKTYPE==ED25519_PK || PKTYPE==ED448...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:491`
`Snippet: #if PKTYPE==ED25519_MLDSA44_PK`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:519`
`Snippet: // ECDSA`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:689`
`Snippet: #if SIGTYPE==ED25519_MLDSA44_SIG`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makerootcert.cpp:752`
`Snippet: #if PKTYPE==ED25519_PK || PKTYPE==ED448...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeselfcert.cpp:15`
`Snippet: #define ED25519_PK 7`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeselfcert.cpp:16`
`Snippet: #define ED448_PK 8`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeselfcert.cpp:19`
`Snippet: #define ED25519_MLDSA44_PK 11 // H...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeselfcert.cpp:31`
`Snippet: #define ED25519_SIG 7`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeselfcert.cpp:32`
`Snippet: #define ED448_SIG 8`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: makecertchain/makeselfcert.cpp:35`
`Snippet: #define ED25519_MLDSA44_SIG 11 //...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: makecertchain/makeselfcert.cpp:54`
`Snippet: ...KEYLEN (2048/8) // RSA only - issuer key l...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: makecertchain/makeselfcert.cpp:57`
`Snippet: ...KEYLEN (2048/8) // RSA only - subject key...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: makecertchain/makeselfcert.cpp:102
Snippet: #if PKTYPE==ED25519_PK
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:108
Snippet: #if PKTYPE==ED448_PK
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:128
Snippet: #if PKTYPE==ED25519_MLDSA44_PK
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:148
Snippet: #if PKTYPE==ED448_SQISIGN3_PK
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:166
Snippet: ...fine SIG_TYPE ECDSA_SECP256R1_SHA256 // signature...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:172
Snippet: ...fine SIG_TYPE ECDSA_SECP384R1_SHA384
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:192
Snippet: #if SIGTYPE==ED25519_SIG
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:196
Snippet: #define SIG_TYPE ED25519
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: makecertchain/makeselfcert.cpp:198
Snippet: #if SIGTYPE==ED448_SIG
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:202
Snippet: #define SIG_TYPE ED448
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:219
Snippet: #if SIGTYPE==ED25519_MLDSA44_SIG
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:226
Snippet: #define SIG_TYPE_1 ED25519
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: makecertchain/makeselfcert.cpp:460
Snippet: // RSA
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:473
Snippet: // MLDSA or EDDSA
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:474
Snippet: ...DSA44_PK || PKTYPE==ED25519_PK || PKTYPE==ED448...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:495
Snippet: #if PKTYPE==ED25519_MLDSA44_PK
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:523

Snippet: // ECDSA
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:714
Snippet: #if SIGTYPE==ED25519_MLDSA44_SIG
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: makecertchain/makeselfcert.cpp:777
Snippet: #if PKTYPE==ED25519_PK || PKTYPE==ED448...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/config.rs:21
Snippet: /// Supported ECDSA Signature algorithm...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/config.rs:22
Snippet: pub const ECDSA_SECP256R1_SHA256: u16 = 0x040...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/config.rs:23
Snippet: /// Supported ECDSA Signature algorithm...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/config.rs:24
Snippet: pub const ECDSA_SECP384R1_SHA384: u16 = 0x050...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/config.rs:25
Snippet: /// Supported RSA Signature algorithm...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/config.rs:27
Snippet: /// Supported RSA Signature algorithm...

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/config.rs:29`
`Snippet: /// Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/config.rs:31`
`Snippet: /// Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/config.rs:33`
`Snippet: /// Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/config.rs:35`
`Snippet: /// Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/config.rs:42`
`Snippet: pub const ECDSA_SECP256R1_SHA384: u16 = 0x041...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/config.rs:44`
`Snippet: pub const MLDSA44_ED25519: u16 = 0xFF86; //0x...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/config.rs:45`
`Snippet: // Ed25519 EdDSA Signature alg...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/config.rs:46`
`Snippet: pub const ED25519: u16 = 0x0807;...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: rust/server/src/config.rs:47
Snippet: // Ed448 EdDSA Signature alg...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/config.rs:48
Snippet: pub const ED448: u16 = 0x0808;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/config.rs:49
Snippet: // Ed376 EdDSA Signature algorithm...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/config.rs:73
Snippet: ...ly support only (No RSA!)
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/config.rs:75
Snippet: /// ECC + RSA support
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/config.rs:81
Snippet: /// EDDSA support
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/config.rs:82
Snippet: pub const EDDSA: usize = 2;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/config.rs:96
Snippet: ...uch smaller for ECC/RSA
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/config.rs:156
Snippet: pub const X25519: u16 = 0x001d;

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/config.rs:157`
`Snippet: pub const SECP256R1: u16 = 0x0017;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/config.rs:158`
`Snippet: pub const SECP384R1: u16 = 0x0018;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/config.rs:162`
`Snippet: //pub const SECP521R1: u16 = 0x0019;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/config.rs:163`
`Snippet: //pub const X448: u16 = 0x001e;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] DIFFIE_HELLMAN
(Finite-field) Diffie-Hellman key exchange.
`Location: rust/server/src/config.rs:193`
`Snippet: ...Preshared Key with Diffie-Hellman key exchange mode`
Remediation: Replace with ML-KEM (Kyber) or a hybrid classical/PQC key exchange during migration.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/main.rs:46`
`Snippet: ...> println!("MLKEM768X25519"),`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/main.rs:47`
`Snippet: X25519 => println!("X25519...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/main.rs:48`
`Snippet: SECP256R1 => println!("SECP25...`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/main.rs:49`
`Snippet: SECP384R1 => println!("SECP38...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/main.rs:59`
`Snippet: ECDSA_SECP256R1_SHA256 => println!(...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/main.rs:62`
`Snippet: ECDSA_SECP384R1_SHA384 => println!(...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/main.rs:68`
`Snippet: ED25519 => println!("ED2551...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/main.rs:69`
`Snippet: ED448 => println!("ED448"...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:57`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:60`
`Snippet: ...if group==config::SECP256R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:63`
`Snippet: ...if group==config::SECP384R1 {`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:78`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:81`
`Snippet: ...if group==config::SECP256R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:84`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:98`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:101`
`Snippet: ...if group==config::SECP256R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:104`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:118`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:121`
`Snippet: ...if group==config::SECP256R1 {`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:124`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:148`
`Snippet: ...groups[0]=config::X25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:149`
`Snippet: ...groups[1]=config::SECP256R1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:150`
`Snippet: ...groups[2]=config::SECP384R1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:159`
`Snippet: ...gs[0]=config::ECDSA_SECP256R1_SHA256;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:160`
`Snippet: ...gs[1]=config::ECDSA_SECP384R1_SHA384;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:162`
`Snippet: ...sig_algs[3]=config::ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:163`
`Snippet: ...sig_algs[4]=config::ED448;`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:167`
`Snippet: ...[8]=config::MLDSA44_ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:168`
`Snippet: ...gs[9]=config::ECDSA_SECP256R1_SHA384;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:175`
`Snippet: ...rt[0]=config::ECDSA_SECP256R1_SHA256;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:176`
`Snippet: ...rt[1]=config::ECDSA_SECP384R1_SHA384;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:180`
`Snippet: ...lgs_cert[5]=config::ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:181`
`Snippet: ...lgs_cert[6]=config::ED448;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:185`
`Snippet: ...10]=config::MLDSA44_ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:186`
`Snippet: ...t[11]=config::ECDSA_SECP256R1_SHA384;`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:375`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:376`
`Snippet: use tlsecc::x25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:378`
`Snippet: x25519::KEY_PAIR(&csk;[0..3...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:380`
`Snippet: ...if group==config::SECP256R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:385`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:402`
`Snippet: use tlsecc::x25519;// append an X25519...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:406`
`Snippet: x25519::KEY_PAIR(&csk;[star...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:413`
`Snippet: ...if group==config::X25519 {`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:414`
`Snippet: use tlsecc::x25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:417`
`Snippet: x25519::KEY_PAIR(&csk;[0..3...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:418`
`Snippet: success=x25519::SHARED_SECRET(&csk...;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:421`
`Snippet: ...if group==config::SECP256R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:429`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:452`
`Snippet: use tlsecc::x25519; // append an X2551...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:459`
`Snippet: x25519::KEY_PAIR(&csk;, &mu...;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:460`
`Snippet: success=x25519::SHARED_SECRET(&csk...;`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:478`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:479`
`Snippet: use tlsecc::x25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:480`
`Snippet: success=x25519::SHARED_SECRET(&sk;[...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:482`
`Snippet: ...if group==config::SECP256R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:486`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:498`
`Snippet: use tlsecc::x25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:502`
`Snippet: success=x25519::SHARED_SECRET(&sk;[...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:541`
`Snippet: /// RSA 2048-bit PKCS1.5 si...`

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:543`
`Snippet: ...use mcore::rsa2048::rsa;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:544`
`Snippet: let mut ms: [u8;rsa::RFS]=[0;rsa::RFS];...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:545`
`Snippet: let mut cs: [u8;rsa::RFS]=[0;rsa::RFS];...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:546`
`Snippet: let mut pk=rsa::new_public_key();`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:547`
`Snippet: rsa::set_public_key(&mu...;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:548`
`Snippet: rsa::encrypt(&pk;,sig,&m...;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:549`
`Snippet: ...5(hlen,cert,&mut; cs,rsa::RFS);`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:552`
`Snippet: ...b(hlen,cert,&mut; cs,rsa::RFS);`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:558`
`Snippet: /// RSA 4096-bit PKCS1.5 si...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:560`
`Snippet: ...use mcore::rsa4096::rsa;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:561`
`Snippet: let mut ms: [u8;rsa::RFS]=[0;rsa::RFS];...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:562`
`Snippet: let mut cs: [u8;rsa::RFS]=[0;rsa::RFS];...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:563`
`Snippet: let mut pk=rsa::new_public_key();`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:564`
`Snippet: rsa::set_public_key(&mu...;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:565`
`Snippet: rsa::encrypt(&pk;,sig,&m...;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:566`
`Snippet: ...5(hlen,cert,&mut; cs,rsa::RFS);`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:569`
`Snippet: ...b(hlen,cert,&mut; cs,rsa::RFS);`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:575`
`Snippet: /// RSA PKCS15 signature ve...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.

Location: rust/server/src/sal_m/sal.rs:586
Snippet: /// RSA 2048-bit PSS signat...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/sal_m/sal.rs:588
Snippet: ...use mcore::rsa2048::rsa;
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/sal_m/sal.rs:589
Snippet: let mut ms: [u8;rsa::RFS]=[0;rsa::RFS];...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/sal_m/sal.rs:590
Snippet: let mut pk=rsa::new_public_key();
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/sal_m/sal.rs:591
Snippet: rsa::set_public_key(&mu...;
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/sal_m/sal.rs:592
Snippet: rsa::encrypt(&pk;,sig,&m...;
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/sal_m/sal.rs:599
Snippet: /// RSA 4096-bit PSS signat...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/sal_m/sal.rs:601
Snippet: ...use mcore::rsa4096::rsa;
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/sal_m/sal.rs:602
Snippet: let mut ms: [u8;rsa::RFS]=[0;rsa::RFS];...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/sal_m/sal.rs:603
Snippet: let mut pk=rsa::new_public_key();

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:604`
`Snippet: rsa::set_public_key(&mu...;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:605`
`Snippet: rsa::encrypt(&pk;,sig,&m...;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:612`
`Snippet: /// RSA PSS signature verif...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:623`
`Snippet: /// NIST secp256r1 curve signature ver...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:624`
`Snippet: pub fn secp256r1_ecdsa_verify(hlen:...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:630`
`Snippet: /// NIST secp384r1 curve signature ver...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:631`
`Snippet: pub fn secp384r1_ecdsa_verify(hlen:...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:637`
`Snippet: pub fn ed25519_verify(cert: &[u8],...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:638`
`Snippet: use tlsecc::ed25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:639`
`Snippet: return ed25519::VERIFY(pubkey,cert...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:640`
`Snippet: //use mcore::ed25519::eddsa;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:641`
`Snippet: //return eddsa::verify(false,pubke...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:644`
`Snippet: pub fn ed448_verify(cert: &[u8],...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:645`
`Snippet: use tlsecc::ed448;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:646`
`Snippet: return ed448::VERIFY(pubkey,cert...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:654`
`Snippet: /// Use Curve SECP256R1 ECDSA to digitally...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:655`
`Snippet: pub fn secp256r1_ecdsa_sign(hlen:usi...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:664`
`Snippet: /// Use Curve SECP384R1 ECDSA to digitally...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:665`
`Snippet: pub fn secp384r1_ecdsa_sign(hlen:usi...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:674`
`Snippet: pub fn ed25519_sign(key: &[u8],mes...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:675`
`Snippet: use tlsecc::ed25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:676`
`Snippet: ed25519::SIGN(key,None,mess...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:680`
`Snippet: pub fn ed448_sign(key: &[u8],mes...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:681`
`Snippet: use tlsecc::ed448;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:682`
`Snippet: ed448::SIGN(key,None,mess...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:692`
`Snippet: /// Use RSA-2048 PSS-RSAE to di...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:694`
`Snippet: ...use mcore::rsa2048::rsa;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:701`
`Snippet: let mut sk=rsa::new_private_key();...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:702`
`Snippet: rsa::rsa_private_key_fr...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:710`
`Snippet: rsa::decrypt(&sk;,&enc;,s...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:714`
`Snippet: /// Use RSA-4096 PSS-RSAE to di...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:716`
`Snippet: ...use mcore::rsa4096::rsa;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/sal_m/sal.rs:723`
`Snippet: let mut sk=rsa::new_private_key();...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.

Location: rust/server/src/sal_m/sal.rs:724
Snippet: rsa::rsa_private_key_fr...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/sal_m/sal.rs:732
Snippet: rsa::decrypt(&sk;,&enc;,s...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/sal_m/sal.rs:736
Snippet: /// RSA PSS transcript sign...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/sal_m/sal.rs:750
Snippet: ...ficates), and ecdsa_secp256r1_sha256."
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/sal_m/sal.rs:754
Snippet: ...config::ECDSA_SECP256R1_SHA256 => {return s...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/sal_m/sal.rs:755
Snippet: ...config::ECDSA_SECP256R1_SHA384 => {return s...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/sal_m/sal.rs:756
Snippet: ...config::ECDSA_SECP384R1_SHA384 => {return s...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/sal_m/sal.rs:761
Snippet: config::ED25519 => { return ed25519...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/sal_m/sal.rs:762
Snippet: config::ED448 => { return ed448_v...

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:774`
`Snippet: ...config:: ECDSA_SECP256R1_SHA256 => {return s...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:775`
`Snippet: ...config:: ECDSA_SECP256R1_SHA384 => {return s...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:776`
`Snippet: ...config:: ECDSA_SECP384R1_SHA384 => {return s...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:778`
`Snippet: config:: ED25519 => { return ed25519...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/sal.rs:779`
`Snippet: config:: ED448 => { return ed448_s...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/sal_m/pqibe.rs:24`
`Snippet: ...03d094,0x4ee931e,0x4ed448b,0x1115636,0x1c1891...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/keys.rs:204`
`Snippet: ...if sigalg==MLDSA44_ED25519, extract both signi...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/keys.rs:205`
`Snippet: ...if sigalg==MLDSA44_ED25519 {`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/keys.rs:207`
`Snippet: ...=sal::tls_signature(ED25519,&key;[0..32],&scv;[0....`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/keys.rs:214`
`Snippet: ...if sigalg==ECDSA_SECP256R1_SHA256 || sigalg==E...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/keys.rs:230`
`Snippet: ...if sigalg==MLDSA44_ED25519, extract two signat...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/keys.rs:231`
`Snippet: ...if sigalg==MLDSA44_ED25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/keys.rs:240`
`Snippet: ...ls_signature_verify(ED25519,&ccv;[0..ptr],&ccvsi...;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/keys.rs:244`
`Snippet: ...g required here for ECDSA signatures - scvsi...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/keys.rs:245`
`Snippet: ...if sigalg==ECDSA_SECP256R1_SHA256 || sigalg==E...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:60`
`Snippet: ECDSA_SECP256R1_SHA256 => log(IO_PR...`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:63`
`Snippet: ECDSA_SECP384R1_SHA384 => log(IO_PR...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:70`
`Snippet: MLDSA44_ED25519 => log(IO_PROTOCOL,...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:71`
`Snippet: ED25519 => log(IO_PROTOCOL,...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:72`
`Snippet: ED448 => log(IO_PROTOCOL,...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:81`
`Snippet: X25519 => log(IO_PROTOCOL,...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:82`
`Snippet: SECP256R1 => log(IO_PROTOCOL,...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:83`
`Snippet: SECP384R1 => log(IO_PROTOCOL,...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:85`
`Snippet: ...OL,"HYBRID MLKEM768+X25519\n",-1,None),`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:100`
`Snippet: ...ficate signature is ECDSA - ",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:102`
`Snippet: ..._PROTOCOL,"Curve is SECP256R1 ",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:105`
`Snippet: ..._PROTOCOL,"Curve is SECP384R1 ",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:108`
`Snippet: ..._PROTOCOL,"Curve is SECP521R1 ",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:115`
`Snippet: ...ficate signature is EdDSA - ",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:116`
`Snippet: ...gt.curve==x509::USE_ED25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:117`
`Snippet: ..._PROTOCOL,"Curve is ED25519\n",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:119`
`Snippet: ...gt.curve==x509::USE_ED448 {`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/server/src/tls13/logger.rs:120
Snippet: ..._PROTOCOL,"Curve is ED448\n",-1,None);
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: rust/server/src/tls13/logger.rs:123
Snippet: ...f d.sgt.kind==x509::RSA {
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: rust/server/src/tls13/logger.rs:124
Snippet: ...ficate signature is RSA of length (bits) ",...
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/server/src/tls13/logger.rs:130
Snippet: ...is Hybrid (MLDSA44+ED25519) of length (bits) "...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/server/src/tls13/logger.rs:137
Snippet: ..._PROTOCOL,"Curve is SECP256R1\n",-1,None);
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/server/src/tls13/logger.rs:140
Snippet: ..._PROTOCOL,"Curve is SECP384R1\n",-1,None);
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/server/src/tls13/logger.rs:143
Snippet: ..._PROTOCOL,"Curve is SECP521R1\n",-1,None);
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/server/src/tls13/logger.rs:148
Snippet: ...kt.curve==x509::USE_ED25519 {
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:149`
`Snippet: ..._PROTOCOL,"Curve is ED25519\n",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:151`
`Snippet: ...kt.curve==x509::USE_ED448 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:152`
`Snippet: ..._PROTOCOL,"Curve is ED448\n",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/logger.rs:155`
`Snippet: ...f d.pkt.kind==x509::RSA {`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/logger.rs:156`
`Snippet: ...icate public key is RSA of length (bits) ",...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/logger.rs:162`
`Snippet: ...is Hybrid (MLDSA44+ED25519) of length (bits) "...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/servercert.rs:20`
`Snippet: // Uses RSA 2048 bit SHA256`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/servercert.rs:61`
`Snippet: ...chain based on the secp256r1 elliptic curve. Exp...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: rust/server/src/tls13/servercert.rs:87
Snippet: ...ain is based on the ed448+ed25519 elliptic cu...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:115
Snippet: ...Experimental hybrid ED25519_MLDSA44 Server key...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:674
Snippet: ...return ECDSA_SECP256R1_SHA256; // as long...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:677
Snippet: ...return ECDSA_SECP384R1_SHA384; // as long...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/tls13/servercert.rs:680
Snippet: ...if pk.kind==x509::RSA {
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:692
Snippet: ...return MLDSA44_ED25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:695
Snippet: ...pk.curve==x509::USE_ED25519 {
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:696
Snippet: return ED25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:698

Snippet: ...pk.curve==x509::USE_ED448 {
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:699
Snippet: return ED448;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:710
Snippet: ...irements[len]=ECDSA_SECP256R1_SHA256; // as long...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:714
Snippet: ...irements[len]=ECDSA_SECP384R1_SHA384; // as long...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/server/src/tls13/servercert.rs:719
Snippet: ...if pk.kind==x509::RSA {
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:746
Snippet: ...ements[len]=MLDSA44_ED25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:750
Snippet: ...requirements[len]=ED25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:752
Snippet: ...need to check that ed25519 is supported - kind...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/server/src/tls13/servercert.rs:755
Snippet: ...pk.curve==x509::USE_ED25519 {

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/servercert.rs:756`
`Snippet: ...requirements[len]=ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/servercert.rs:760`
`Snippet: ...pk.curve==x509::USE_ED448 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/servercert.rs:761`
`Snippet: ...requirements[len]=ED448;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/servercert.rs:813`
`Snippet: EDDSA => {privkey=ED_PRIV...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:5`
`Snippet: .../ type of key (ECC, RSA etc)`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:7`
`Snippet: ...// sub-type (cuve, rsa bit length..)`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:20`
`Snippet: pub const RSA:usize = 2;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:21`
`Snippet: ...:usize = 3; // for Ed25519`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:35`
`Snippet: pub const USE_ED25519:usize = 1; // Be...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:36`
`Snippet: pub const USE_ED448:usize = 5;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:73`
`Snippet: const SECP384R1:[u8;5]=[0x2B, 0x81,...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:74`
`Snippet: const SECP521R1:[u8;5]=[0x2B, 0x81,...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:83`
`Snippet: ...0x06]; // MLDSA44 + ED25519`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:165`
`Snippet: /// in-place ECDSA signature encoding...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:213`
`Snippet: /// in-place ECDSA signature decoding...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:277`
`Snippet: ...x509 -nodes -newkey rsa:4096 -keyout key.pe...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:280`
`Snippet: // For RSA octet = p|q|dp|dq|c...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:353`
`Snippet: ...ret.curve = USE_ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:379`
`Snippet: ...ret.curve = USE_ED448;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:483`
`Snippet: ...ret.curve=USE_ED25519; // 8*(tlen+len);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:532`
`Snippet: if SECP384R1 == soid[0..slen] {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:536`
`Snippet: if SECP521R1 == soid[0..slen] {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:645`
`Snippet: ret.kind=RSA;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:652`
`Snippet: ...ic Curve signature, RSA for RSA signature`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:724`
`Snippet: ret.kind=RSA;`

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:728`
`Snippet: ret.kind=RSA;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:732`
`Snippet: ret.kind=RSA;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:748`
`Snippet: ...ret.curve=USE_ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:779`
`Snippet: ...ret.curve=USE_ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:782`
`Snippet: ...ret.curve=USE_ED448;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:857`
`Snippet: if ret.kind==RSA {`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:909`
`Snippet: ...ret.curve=USE_ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:1118`
`Snippet: ...icKey & prime256v1, secp384r1 or secp521r1 for EC...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:1119`
`Snippet: // rsapublicKey for RSA`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:1141`
`Snippet: ...=ECD; ret.curve=USE_ED25519`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:1144`
`Snippet: ...=ECD; ret.curve=USE_ED448`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:1147`
`Snippet: ret.kind=RSA;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:1182`
`Snippet: ...ret.curve=USE_ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:1187`
`Snippet: if SECP384R1==koid[0..slen] {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/x509.rs:1190`
`Snippet: if SECP521R1==koid[0..slen] {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/server/src/tls13/x509.rs:1234`
`Snippet: if ret.kind==RSA { // Key is (modulu...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/certchain.rs:110`
`Snippet: ...nature_verify(ECDSA_SECP256R1_SHA256,cert,sig,pub...`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/server/src/tls13/certchain.rs:113
Snippet: ...nature_verify(ECDSA_SECP384R1_SHA384,cert,sig,pub...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/server/src/tls13/certchain.rs:115
Snippet: ...st.curve==x509::USE_ED25519{
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/server/src/tls13/certchain.rs:116
Snippet: ...ls_signature_verify(ED25519,cert,sig,pubkey);
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/server/src/tls13/certchain.rs:118
Snippet: ...st.curve==x509::USE_ED448{
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/server/src/tls13/certchain.rs:119
Snippet: ...ls_signature_verify(ED448,cert,sig,pubkey);
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: rust/server/src/tls13/certchain.rs:121
Snippet: ...if st.kind==x509::RSA && st.hash==x509::H...
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: rust/server/src/tls13/certchain.rs:124
Snippet: ...if st.kind==x509::RSA && st.hash==x509::H...
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: rust/server/src/tls13/certchain.rs:127
Snippet: ...if st.kind==x509::RSA && st.hash==x509::H...
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/server/src/tls13/certchain.rs:143`
`Snippet: ...ls_signature_verify(ED25519,cert,sig1,pub1) &&...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/config.rs:23`
`Snippet: /// Supported ECDSA Signature algorithm...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/config.rs:24`
`Snippet: pub const ECDSA_SECP256R1_SHA256: u16 = 0x040...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/config.rs:25`
`Snippet: /// Supported ECDSA Signature algorithm...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/config.rs:26`
`Snippet: pub const ECDSA_SECP384R1_SHA384: u16 = 0x050...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/config.rs:27`
`Snippet: /// Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/config.rs:29`
`Snippet: /// Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/config.rs:31`
`Snippet: /// Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/config.rs:33`

Snippet: /// Supported RSA Signature algorithm...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/config.rs:35
Snippet: /// Supported RSA Signature algorithm...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/config.rs:37
Snippet: /// Supported RSA Signature algorithm...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/config.rs:44
Snippet: pub const ECDSA_SECP256R1_SHA384: u16 = 0x041...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/config.rs:46
Snippet: pub const MLDSA44_ED25519: u16 = 0xFF86; //0x...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/config.rs:47
Snippet: // Ed25519 EdDSA Signature alg...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/config.rs:48
Snippet: pub const ED25519: u16 = 0x0807;...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/config.rs:49
Snippet: // Ed448 EdDSA Signature alg...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/config.rs:50
Snippet: pub const ED448: u16 = 0x0808;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/config.rs:51`
`Snippet: // Ed376 EdDSA Signature algorithm...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/config.rs:75`
`Snippet: ...CC support only (No RSA!)`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/config.rs:77`
`Snippet: /// ECC + RSA support`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/config.rs:83`
`Snippet: /// EDDSA support - now inclu...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/config.rs:84`
`Snippet: //pub const EDDSA: usize = 2;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/config.rs:87`
`Snippet: /// self signed RSA cert`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/config.rs:95`
`Snippet: /// self signed Ed25519 cert`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/config.rs:108`
`Snippet: ...uch smaller for ECC/RSA`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/config.rs:118`

Snippet: ...is for hybrid with X25519

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: rust/client/src/config.rs:166

Snippet: pub const X25519: u16 = 0x001d;

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: rust/client/src/config.rs:167

Snippet: pub const SECP256R1: u16 = 0x0017;

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: rust/client/src/config.rs:168

Snippet: pub const SECP384R1: u16 = 0x0018;

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: rust/client/src/config.rs:172

Snippet: //pub const SECP521R1: u16 = 0x0019;

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: rust/client/src/config.rs:173

Snippet: //pub const X448: u16 = 0x001e;

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] DIFFIE_HELLMAN

(Finite-field) Diffie-Hellman key exchange.

Location: rust/client/src/config.rs:203

Snippet: ...Preshared Key with Diffie-Hellman key exchange mode

Remediation: Replace with ML-KEM (Kyber) or a hybrid classical/PQC key exchange during migration.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: rust/client/src/main.rs:174

Snippet: ...> println!("MLKEM768X25519"),

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: rust/client/src/main.rs:175

Snippet: X25519 => println!("X25519...

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/main.rs:176
Snippet: SECP256R1 => println!("SECP25...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/main.rs:177
Snippet: SECP384R1 => println!("SECP38...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/main.rs:187
Snippet: ECDSA_SECP256R1_SHA256 => println!(...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/main.rs:190
Snippet: ECDSA_SECP384R1_SHA384 => println!(...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/main.rs:196
Snippet: ED25519 => println!("ED2551...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/main.rs:197
Snippet: ED448 => println!("ED448"...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/main.rs:375
Snippet: ...n.t.favourite_group=X25519;
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/main.rs:384
Snippet: ...n.t.favourite_group=X25519;
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:57`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:60`
`Snippet: ...if group==config::SECP256R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:63`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:80`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:83`
`Snippet: ...if group==config::SECP256R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:86`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:102`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:105`
`Snippet: ...if group==config::SECP256R1 {`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:108`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:125`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:128`
`Snippet: ...if group==config::SECP256R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:131`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:164`
`Snippet: ...groups[0]=config::X25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:165`
`Snippet: ...groups[1]=config::SECP256R1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:166`
`Snippet: ...groups[2]=config::SECP384R1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:171`
`Snippet: ...groups[1]=config::X25519;`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:172`
`Snippet: ...groups[2]=config::SECP256R1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:173`
`Snippet: ...groups[3]=config::SECP384R1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:179`
`Snippet: ...groups[2]=config::X25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:180`
`Snippet: ...groups[3]=config::SECP256R1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:181`
`Snippet: ...groups[4]=config::SECP384R1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:190`
`Snippet: ...gs[0]=config::ECDSA_SECP256R1_SHA256;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:191`
`Snippet: ...gs[1]=config::ECDSA_SECP384R1_SHA384;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:195`
`Snippet: ...sig_algs[n]=config::ED25519; n+=1;`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:196`
`Snippet: ...sig_algs[n]=config::ED448; n+=1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:202`
`Snippet: ...[n]=config::MLDSA44_ED25519; n+=1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:203`
`Snippet: ...gs[n]=config::ECDSA_SECP256R1_SHA384; n+=1;...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:211`
`Snippet: ...rt[0]=config::ECDSA_SECP256R1_SHA256;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:212`
`Snippet: ...rt[1]=config::ECDSA_SECP384R1_SHA384;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:218`
`Snippet: ...lgs_cert[n]=config::ED25519; n+=1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:219`
`Snippet: ...lgs_cert[n]=config::ED448; n+=1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:225`
`Snippet: ...[n]=config::MLDSA44_ED25519; n+=1;`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:226`
`Snippet: ...rt[n]=config::ECDSA_SECP256R1_SHA384; n+=1;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:416`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:417`
`Snippet: use tlsecc::x25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:419`
`Snippet: x25519::KEY_PAIR(&csk;[0..3...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:421`
`Snippet: ...if group==config::SECP256R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:426`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:438`
`Snippet: use tlsecc::x25519;// append an X25519...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:442`
`Snippet: x25519::KEY_PAIR(&csk;[star...`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:457`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:458`
`Snippet: use tlsecc::x25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:461`
`Snippet: x25519::KEY_PAIR(&csk;[0..3...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:462`
`Snippet: success=x25519::SHARED_SECRET(&csk...;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:465`
`Snippet: ...if group==config::SECP256R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:473`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:489`
`Snippet: use tlsecc::x25519; // append an X2551...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:496`
`Snippet: x25519::KEY_PAIR(&csk;, &mu...;`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:497`
`Snippet: success=x25519::SHARED_SECRET(&csk...;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:524`
`Snippet: ...if group==config::X25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:525`
`Snippet: use tlsecc::x25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:526`
`Snippet: success=x25519::SHARED_SECRET(&sk;[...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:528`
`Snippet: ...if group==config::SECP256R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:532`
`Snippet: ...if group==config::SECP384R1 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:540`
`Snippet: use tlsecc::x25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:544`
`Snippet: success=x25519::SHARED_SECRET(&sk;[...`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:589`
`Snippet: /// RSA 2048-bit PKCS1.5 si...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:591`
`Snippet: ...use mcore::rsa2048::rsa;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:592`
`Snippet: let mut ms: [u8;rsa::RFS]=[0;rsa::RFS];...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:593`
`Snippet: let mut cs: [u8;rsa::RFS]=[0;rsa::RFS];...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:594`
`Snippet: let mut pk=rsa::new_public_key();`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:595`
`Snippet: rsa::set_public_key(&mu...;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:596`
`Snippet: rsa::encrypt(&pk;,sig,&m...;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:597`
`Snippet: ...5(hlen,cert,&mut; cs,rsa::RFS);`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:600`
`Snippet: ...b(hlen,cert,&mut; cs,rsa::RFS);`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:606`
`Snippet: /// RSA 4096-bit PKCS1.5 si...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:608`
`Snippet: ...use mcore::rsa4096::rsa;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:609`
`Snippet: let mut ms: [u8;rsa::RFS]=[0;rsa::RFS];...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:610`
`Snippet: let mut cs: [u8;rsa::RFS]=[0;rsa::RFS];...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:611`
`Snippet: let mut pk=rsa::new_public_key();`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:612`
`Snippet: rsa::set_public_key(&mu...;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:613`
`Snippet: rsa::encrypt(&pk;,sig,&m...;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:614`
`Snippet: ...5(hlen,cert,&mut; cs,rsa::RFS);`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:617`
`Snippet: ...b(hlen,cert,&mut; cs,rsa::RFS);`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:623`
`Snippet: /// RSA PKCS15 signature ve...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:634`
`Snippet: /// RSA 2048-bit PSS signat...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:636`
`Snippet: ...use mcore::rsa2048::rsa;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:637`
`Snippet: let mut ms: [u8;rsa::RFS]=[0;rsa::RFS];...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:638`
`Snippet: let mut pk=rsa::new_public_key();`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:639`
`Snippet: rsa::set_public_key(&mu...;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:640`
`Snippet: rsa::encrypt(&pk;,sig,&m...;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:647`
`Snippet: /// RSA 4096-bit PSS signat...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:649`
`Snippet: ...use mcore::rsa4096::rsa;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:650`

Snippet: let mut ms: [u8;rsa::RFS]=[0;rsa::RFS];...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/sal_m/sal.rs:651
Snippet: let mut pk=rsa::new_public_key();
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/sal_m/sal.rs:652
Snippet: rsa::set_public_key(&mu...;
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/sal_m/sal.rs:653
Snippet: rsa::encrypt(&pk;,sig,&m...;
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/sal_m/sal.rs:660
Snippet: /// RSA PSS signature verif...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/sal_m/sal.rs:671
Snippet: /// NIST secp256r1 curve signature ver...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/sal_m/sal.rs:672
Snippet: pub fn secp256r1_ecdsa_verify(hlen:...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/sal_m/sal.rs:678
Snippet: /// NIST secp384r1 curve signature ver...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/sal_m/sal.rs:679
Snippet: pub fn secp384r1_ecdsa_verify(hlen:...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:685`
`Snippet: pub fn ed25519_verify(cert: &[u8],...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:686`
`Snippet: use tlsecc::ed25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:687`
`Snippet: return ed25519::VERIFY(pubkey,cert...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:688`
`Snippet: //use mcore::ed25519::eddsa;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:689`
`Snippet: //return eddsa::verify(false,pubke...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:692`
`Snippet: pub fn ed448_verify(cert: &[u8],...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:693`
`Snippet: use tlsecc::ed448;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:694`
`Snippet: return ed448::VERIFY(pubkey,cert...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:702`
`Snippet: /// Use Curve SECP256R1 ECDSA to digitally...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:703`
`Snippet: pub fn secp256r1_ecdsa_sign(hlen:usi...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:712`
`Snippet: /// Use Curve SECP384R1 ECDSA to digitally...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:713`
`Snippet: pub fn secp384r1_ecdsa_sign(hlen:usi...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:722`
`Snippet: pub fn ed25519_sign(key: &[u8],mes...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:723`
`Snippet: use tlsecc::ed25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:724`
`Snippet: ed25519::SIGN(key,None,mess...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:728`
`Snippet: pub fn ed448_sign(key: &[u8],mes...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:729`
`Snippet: use tlsecc::ed448;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:730`
`Snippet: ed448::SIGN(key,None,mess...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:740`
`Snippet: /// Use RSA-2048 PSS-RSAE to di...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:742`
`Snippet: ...use mcore::rsa2048::rsa;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:749`
`Snippet: let mut sk=rsa::new_private_key();...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:750`
`Snippet: rsa::rsa_private_key_fr...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:758`
`Snippet: rsa::decrypt(&sk;,&enc;,s...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:762`
`Snippet: /// Use RSA-4096 PSS-RSAE to di...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:764`
`Snippet: ...use mcore::rsa4096::rsa;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:771`
`Snippet: let mut sk=rsa::new_private_key();...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:772`
`Snippet: rsa::rsa_private_key_fr...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:780`
`Snippet: rsa::decrypt(&sk;,&enc;,s...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/sal_m/sal.rs:784`
`Snippet: /// RSA PSS transcript sign...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:798`
`Snippet: ...ficates), and ecdsa_secp256r1_sha256."`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:803`
`Snippet: ...config::ECDSA_SECP256R1_SHA256 => {return s...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:804`
`Snippet: ...config::ECDSA_SECP256R1_SHA384 => {return s...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:805`
`Snippet: ...config::ECDSA_SECP384R1_SHA384 => {return s...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:815`
`Snippet: config::ED25519 => { return ed25519...`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/sal_m/sal.rs:816
Snippet: config::ED448 => { return ed448_v...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/sal_m/sal.rs:836
Snippet: ...config:: ECDSA_SECP256R1_SHA256 => {return s...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/sal_m/sal.rs:837
Snippet: ...config:: ECDSA_SECP256R1_SHA384 => {return s...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/sal_m/sal.rs:838
Snippet: ...config:: ECDSA_SECP384R1_SHA384 => {return s...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/sal_m/sal.rs:844
Snippet: ...config:: ED25519 => { return ed25519...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/sal_m/sal.rs:845
Snippet: ...config:: ED448 => { return ed448_s...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/sal_m/sal.rs:860
Snippet: ...config:: ECDSA_SECP256R1_SHA256 => {return s...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/sal_m/sal.rs:861
Snippet: ...config:: ECDSA_SECP256R1_SHA384 => {return s...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:862`
`Snippet: ...config:: ECDSA_SECP384R1_SHA384 => {return s...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:864`
`Snippet: config:: ED25519 => { return ed25519...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/sal.rs:865`
`Snippet: config:: ED448 => { return ed448_s...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/sal_m/pqibe.rs:24`
`Snippet: ...03d094,0x4ee931e,0x4ed448b,0x1115636,0x1c1891...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/keys.rs:183`
`Snippet: ...if sigalg==MLDSA44_ED25519, extract both signi...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/keys.rs:184`
`Snippet: ...if sigalg==MLDSA44_ED25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/keys.rs:186`
`Snippet: ...=sal::tls_signature(ED25519,&key;[0..32],&ccv;[0....`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/keys.rs:193`
`Snippet: ...if sigalg==ECDSA_SECP256R1_SHA256 || sigalg==E...`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/keys.rs:209`
`Snippet: ...if sigalg==MLDSA44_ED25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/keys.rs:218`
`Snippet: ...ls_signature_verify(ED25519,&scv;[0..ptr],&scvsi...;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/keys.rs:222`
`Snippet: ...g required here for ECDSA signatures - scvsi...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/keys.rs:223`
`Snippet: ...if sigalg==ECDSA_SECP256R1_SHA256 || sigalg==E...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:61`
`Snippet: ECDSA_SECP256R1_SHA256 => log(level...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:64`
`Snippet: ECDSA_SECP384R1_SHA384 => log(level...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:71`
`Snippet: MLDSA44_ED25519 => log(level,"MLDSA...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:72`
`Snippet: ED25519 => log(level,"Ed255...`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:73`
`Snippet: ED448 => log(level,"Ed448...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:82`
`Snippet: X25519 => log(level,"X2551...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:83`
`Snippet: SECP256R1 => log(level,"SECP2...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:84`
`Snippet: SECP384R1 => log(level,"SECP3...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:86`
`Snippet: ...OL,"HYBRID MLKEM768+X25519\n",-1,None),`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:159`
`Snippet: ...ficate signature is ECDSA - ",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:161`
`Snippet: ..._PROTOCOL,"Curve is SECP256R1 ",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:164`
`Snippet: ..._PROTOCOL,"Curve is SECP384R1 ",-1,None);`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:167`
`Snippet: ..._PROTOCOL,"Curve is SECP521R1 ",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:174`
`Snippet: ...ficate signature is EdDSA - ",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:175`
`Snippet: ...gt.curve==x509::USE_ED25519 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:176`
`Snippet: ..._PROTOCOL,"Curve is ED25519\n",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:178`
`Snippet: ...gt.curve==x509::USE_ED448 {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:179`
`Snippet: ..._PROTOCOL,"Curve is ED448\n",-1,None);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/logger.rs:182`
`Snippet: ...f d.sgt.kind==x509::RSA {`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/logger.rs:183`
`Snippet: ...ficate signature is RSA of length (bits) ",...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/logger.rs:189
Snippet: ...is Hybrid (MLDSA44+ED25519) of length (bits) "...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/logger.rs:196
Snippet: ..._PROTOCOL,"Curve is SECP256R1\n",-1,None);
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/logger.rs:199
Snippet: ..._PROTOCOL,"Curve is SECP384R1\n",-1,None);
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/logger.rs:202
Snippet: ..._PROTOCOL,"Curve is SECP521R1\n",-1,None);
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/logger.rs:207
Snippet: ...kt.curve==x509::USE_ED25519 {
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/logger.rs:208
Snippet: ..._PROTOCOL,"Curve is ED25519\n",-1,None);
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/logger.rs:210
Snippet: ...kt.curve==x509::USE_ED448 {
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/logger.rs:211
Snippet: ..._PROTOCOL,"Curve is ED448\n",-1,None);
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/logger.rs:214`
`Snippet: ...f d.pkt.kind==x509::RSA {`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/logger.rs:215`
`Snippet: ...icate public key is RSA of length (bits) ",...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/logger.rs:221`
`Snippet: ...is Hybrid (MLDSA44+ED25519) of length (bits) "...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/clientcert.rs:36`
`Snippet: ...ersonal private key RSA 2048 bits - expires...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/clientcert.rs:90`
`Snippet: // My personal EDDSA private key - Certi...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/clientcert.rs:124`
`Snippet: ...return ECDSA_SECP256R1_SHA256; // as long...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/clientcert.rs:127`
`Snippet: ...return ECDSA_SECP384R1_SHA384; // as long...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/clientcert.rs:130`
`Snippet: ...if pk.kind==x509::RSA {`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/clientcert.rs:142`

Snippet: ...return MLDSA44_ED25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/clientcert.rs:145
Snippet: ...pk.curve==x509::USE_ED25519 {
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/clientcert.rs:146
Snippet: return ED25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/clientcert.rs:148
Snippet: ...pk.curve==x509::USE_ED448 {
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/clientcert.rs:149
Snippet: return ED448;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/clientcert.rs:160
Snippet: ...irements[len]=ECDSA_SECP256R1_SHA256; // as long...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/clientcert.rs:164
Snippet: ...irements[len]=ECDSA_SECP384R1_SHA384; // as long...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/tls13/clientcert.rs:169
Snippet: ...if pk.kind==x509::RSA {
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/clientcert.rs:196
Snippet: ...ements[len]=MLDSA44_ED25519;

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/clientcert.rs:200
Snippet: ...requirements[len]=ED25519;
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/clientcert.rs:202
Snippet: ...need to check that ed25519 is supported - kind...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/clientcert.rs:205
Snippet: ...pk.curve==x509::USE_ED25519 {
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/clientcert.rs:206
Snippet: ...requirements[len]=ED25519;
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/clientcert.rs:210
Snippet: ...pk.curve==x509::USE_ED448 {
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: rust/client/src/tls13/clientcert.rs:211
Snippet: ...requirements[len]=ED448;
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: rust/client/src/tls13/x509.rs:5
Snippet: .../ type of key (ECC, RSA etc)
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: rust/client/src/tls13/x509.rs:7
Snippet: .../// sub-type (cuve, rsa bit length..)
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/x509.rs:20`
`Snippet: pub const RSA:usize = 2;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:21`
`Snippet: ...:usize = 3; // for Ed25519`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:35`
`Snippet: pub const USE_ED25519:usize = 1; // Be...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:36`
`Snippet: pub const USE_ED448:usize = 5;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:73`
`Snippet: const SECP384R1:[u8;5]=[0x2B, 0x81,...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:74`
`Snippet: const SECP521R1:[u8;5]=[0x2B, 0x81,...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:83`
`Snippet: ...0x06]; // MLDSA44 + ED25519`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:164`
`Snippet: /// in-place ECDSA signature encoding...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:212`
`Snippet: /// in-place ECDSA signature decoding...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/x509.rs:276`
`Snippet: ...x509 -nodes -newkey rsa:4096 -keyout key.pe...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/x509.rs:279`
`Snippet: // For RSA octet = p|q|dp|dq|c...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:352`
`Snippet: ...ret.curve = USE_ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:378`
`Snippet: ...ret.curve = USE_ED448;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:480`
`Snippet: ...ret.curve=USE_ED25519; // 8*(tlen+len);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:529`
`Snippet: if SECP384R1 == soid[0..slen] {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:533`
`Snippet: if SECP521R1 == soid[0..slen] {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.

Location: rust/client/src/tls13/x509.rs:642
Snippet: ret.kind=RSA;
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/tls13/x509.rs:649
Snippet: ...ic Curve signature, RSA for RSA signature
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/tls13/x509.rs:721
Snippet: ret.kind=RSA;
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/tls13/x509.rs:725
Snippet: ret.kind=RSA;
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/tls13/x509.rs:729
Snippet: ret.kind=RSA;
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/x509.rs:745
Snippet: ...ret.curve=USE_ED25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/x509.rs:776
Snippet: ...ret.curve=USE_ED25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/x509.rs:779
Snippet: ...ret.curve=USE_ED448;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/tls13/x509.rs:855
Snippet: if ret.kind==RSA {
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:907`
`Snippet: ...ret.curve=USE_ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:1116`
`Snippet: ...icKey & prime256v1, secp384r1 or secp521r1 for EC...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/x509.rs:1117`
`Snippet: // rsapublicKey for RSA`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:1139`
`Snippet: ...=ECD; ret.curve=USE_ED25519`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:1142`
`Snippet: ...=ECD; ret.curve=USE_ED448`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/x509.rs:1145`
`Snippet: ret.kind=RSA;`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:1180`
`Snippet: ...ret.curve=USE_ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:1185`
`Snippet: if SECP384R1==koid[0..slen] {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/x509.rs:1188`

Snippet: if SECP521R1==koid[0..slen] {
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/tls13/x509.rs:1232
Snippet: if ret.kind==RSA { // Key is (modulu...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/certchain.rs:120
Snippet: ...nature_verify(ECDSA_SECP256R1_SHA256,cert,sig,pub...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/certchain.rs:123
Snippet: ...nature_verify(ECDSA_SECP384R1_SHA384,cert,sig,pub...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/certchain.rs:125
Snippet: ...st.curve==x509::USE_ED25519{
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/certchain.rs:126
Snippet: ...ls_signature_verify(ED25519,cert,sig,pubkey);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/certchain.rs:128
Snippet: ...st.curve==x509::USE_ED448{
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: rust/client/src/tls13/certchain.rs:129
Snippet: ...ls_signature_verify(ED448,cert,sig,pubkey);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: rust/client/src/tls13/certchain.rs:131
Snippet: ...if st.kind==x509::RSA && st.hash==x509::H...

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/certchain.rs:134`
`Snippet: ...if st.kind==x509::RSA && st.hash==x509::H...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: rust/client/src/tls13/certchain.rs:137`
`Snippet: ...if st.kind==x509::RSA && st.hash==x509::H...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/certchain.rs:151`
`Snippet: ...ls_signature_verify(ED25519,cert,sig1,pub1) &&...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/cacerts.rs:5`
`Snippet: ...experimental hybrid ED25519+MLDSA44 root certif...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/cacerts.rs:6`
`Snippet: // then is our own secp256r1 root certificate, a...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/cacerts.rs:38`
`Snippet: // Hybrid ED25519+MLDSA44 Cert expire...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: rust/client/src/tls13/cacerts.rs:257`
`Snippet: // EDDSA expires Jan 2027`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls1_3.h:34`
`Snippet: .../**< Mixture of RSA and ECC - for use w...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:36`
`Snippet: ...brid, MLKEM/MLDSA + X25519 */`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:37`
`Snippet: //#define EDDSA 2 /**<...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls1_3.h:45`
`Snippet: ...1 /**< self signed RSA cert */`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:51`
`Snippet: ...7 /**< self signed Ed25519 cert */`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:57`
`Snippet: ...ular group (such as X25519) at top of list in...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls1_3.h:153`
`Snippet: // all certs use RSA or ECC`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls1_3.h:154`
`Snippet: ...tes 4096-bit RSA */`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls1_3.h:155`
`Snippet: ...ytes 4096-bit RSA */`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls1_3.h:156`
`Snippet: ...4096-bit RSA */`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:158`
`Snippet: ...bytes MLKEM768+X25519 */`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:159`
`Snippet: ...ize MLKEM768+X25519 */`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:160`
`Snippet: ...n bytes MLKEM768+X25519 */`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:175`
`Snippet: ...MLDSA65>MLDSA44 + P256 */`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:176`
`Snippet: ...MLDSA65>MLDSA44 + P256 (maybe includes the...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:177`
`Snippet: ...MLDSA65>MLDSA44 + P256 (DER encoding for E...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:179`
`Snippet: ...in bytes MLKEM768+X25519 */`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:180`
`Snippet: ...t size MLKEM768+X25519 */`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/include/tls1_3.h:181
Snippet: ...e in bytes MLKEM768+X25519 */
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/include/tls1_3.h:219
Snippet: #define X25519 0x001d...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/include/tls1_3.h:220
Snippet: #define SECP256R1 0x0017...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/include/tls1_3.h:221
Snippet: #define SECP384R1 0x0018...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/include/tls1_3.h:222
Snippet: #define SECP521R1 0x0019...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/include/tls1_3.h:223
Snippet: #define X448 0x001e...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/include/tls1_3.h:227
Snippet: ...exchange, MLKEM768+X25519 */
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/include/tls1_3.h:230
Snippet: #define ECDSA_SECP256R1_SHA256 0x0403 /**...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:231`
`Snippet: #define ECDSA_SECP256R1_SHA384 0x0413 /**...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:232`
`Snippet: #define ECDSA_SECP384R1_SHA384 0x0503 /**...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls1_3.h:233`
`Snippet: .../**< Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls1_3.h:234`
`Snippet: .../**< Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls1_3.h:235`
`Snippet: .../**< Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls1_3.h:236`
`Snippet: .../**< Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls1_3.h:237`
`Snippet: .../**< Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls1_3.h:238`
`Snippet: .../**< Supported RSA Signature algorithm...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:239`
`Snippet: #define ED25519 0x0807...`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:240`
`Snippet: #define ED448 0x0808...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:241`
`Snippet: .../**< Experimental EdDSA Signature algorithm...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/include/tls1_3.h:245`
`Snippet: #define MLDSA44_ED25519 0xFF86 /**...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] DIFFIE_HELLMAN
(Finite-field) Diffie-Hellman key exchange.
`Location: cpp/include/tls1_3.h:264`
`Snippet: ...Preshared Key with Diffie-Hellman key exchange mode *...`
Remediation: Replace with ML-KEM (Kyber) or a hybrid classical/PQC key exchange during migration.

[HIGH] DIFFIE_HELLMAN
(Finite-field) Diffie-Hellman key exchange.
`Location: cpp/include/tls_sal.h:194`
`Snippet: ...* @brief generate a Diffie-Hellman shared secret`
Remediation: Replace with ML-KEM (Kyber) or a hybrid classical/PQC key exchange during migration.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls_sal.h:226`
`Snippet: ...e type of the keys (RSA,ECC..)`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls_sal.h:228`
`Snippet: ...KEY the public key (RSA key size is passed...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls_x509.h:19`
`Snippet: ...509_RSA 2 /**< RSA data type detected...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: cpp/include/tls_x509.h:20
Snippet: ...**< Elliptic Curve (Ed25519) detected */
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/include/tls_x509.h:38
Snippet: #define USE_ED25519 1 /**< Bernstei...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/include/tls_x509.h:39
Snippet: #define USE_ED448 4 /**< Goldiloc...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: cpp/include/tls_x509.h:68
Snippet: ...nature type (ECC or RSA) */
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: cpp/include/tls_x509.h:70
Snippet: ...iptic curve used or RSA key length in bits...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/include/tls_x509.h:74
Snippet: /** @brief in-place ECDSA signature encoding
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/include/tls_x509.h:76
Snippet: @param c an ecdsa signature to be con...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/include/tls_x509.h:80
Snippet: /** @brief in-place ECDSA signature decoding
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/include/tls_x509.h:82
Snippet: @param c an ecdsa signature to be con...

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls_x509.h:92`
`Snippet: ...d private key - for RSA octad = p|q|dp|dq|c...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls_x509.h:93`
`Snippet: ...te key type (ECC or RSA)`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls_x509.h:101`
`Snippet: ...nature type (ECC or RSA)`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls_x509.h:125`
`Snippet: ...ic key type (ECC or RSA)`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/include/tls_x509.h:134`
`Snippet: ...ic key type (ECC or RSA)`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_keys_calc.cpp:389`
`Snippet: ...if (sigAlg==MLDSA44_ED25519)`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_keys_calc.cpp:393`
`Snippet: ...SAL_tlsSignature(ED25519,&FKEY;,&CCV;,CCVSIG);...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_keys_calc.cpp:402`
`Snippet: // adjustment for ECDSA signatures`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: cpp/lib/tls_keys_calc.cpp:403
Snippet: ...if (sigAlg==ECDSA_SECP256R1_SHA256 || sigAlg==E...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_keys_calc.cpp:424
Snippet: ...if (sigAlg==MLDSA44_ED25519)
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_keys_calc.cpp:434
Snippet: ..._tlsSignatureVerify(ED25519,&SCV;,&FSIG;,&FPUB;) &...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_keys_calc.cpp:437
Snippet: ...g required here for ECDSA signatures - SCVSI...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_keys_calc.cpp:438
Snippet: ...if (sigAlg==ECDSA_SECP256R1_SHA256 || sigAlg==E...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:94
Snippet: case X25519:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:95
Snippet: ...myprintf((char *)"X25519\n");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:97
Snippet: case SECP256R1:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: cpp/lib/tls_logger.cpp:98
Snippet: ...myprintf((char *)"SECP256R1\n");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:100
Snippet: case SECP384R1:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:101
Snippet: ...myprintf((char *)"SECP384R1\n");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:107
Snippet: ...(char *)"MLKEM768 + X25519\n");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:127
Snippet: case ECDSA_SECP256R1_SHA256:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:128
Snippet: ...intf((char *)"ECDSA_SECP256R1_SHA256\n");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:136
Snippet: case ECDSA_SECP384R1_SHA384:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:137
Snippet: ...intf((char *)"ECDSA_SECP384R1_SHA384\n");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: cpp/lib/tls_logger.cpp:151
Snippet: case ED25519:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:152
Snippet: ...myprintf((char *)"ED25519\n");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:160
Snippet: case MLDSA44_ED25519:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:161
Snippet: ...((char *)"MLDSA44 + ED25519\n");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:315
Snippet: ...ficate signature is ECDSA - ",NULL,0,NULL);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:317
Snippet: ...L,(char *)"Curve is SECP256R1 ",NULL,0,NULL);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:319
Snippet: ...L,(char *)"Curve is SECP384R1 ",NULL,0,NULL);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:321
Snippet: ...L,(char *)"Curve is SECP521R1 ",NULL,0,NULL);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: cpp/lib/tls_logger.cpp:328
Snippet: ...ficate signature is EdDSA - ",NULL,0,NULL);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:329
Snippet: ...if (sg.curve==USE_ED25519)
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:330
Snippet: ...L,(char *)"Curve is ED25519\n",NULL,0,NULL);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:331
Snippet: ...if (sg.curve==USE_ED448)
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:332
Snippet: ...L,(char *)"Curve is ED448\n",NULL,0,NULL);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: cpp/lib/tls_logger.cpp:335
Snippet: ...ficate signature is RSA of length (bits) ",...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:341
Snippet: ...is Hybrid (MLDSA44+ED25519) of length (bits) "...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:356
Snippet: ...L,(char *)"Curve is SECP256R1\n",NULL,0,NULL);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:358

Snippet: ...L,(char *)"Curve is SECP384R1\n",NULL,0,NULL);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:360
Snippet: ...L,(char *)"Curve is SECP521R1\n",NULL,0,NULL);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:365
Snippet: ...if (pk.curve==USE_ED25519)
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:366
Snippet: ...L,(char *)"Curve is ED25519\n",NULL,0,NULL);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:367
Snippet: ...if (pk.curve==USE_ED448)
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:368
Snippet: ...L,(char *)"Curve is ED448\n",NULL,0,NULL);
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: cpp/lib/tls_logger.cpp:371
Snippet: ...icate public key is RSA of length (bits) ",...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_logger.cpp:377
Snippet: ...is Hybrid (MLDSA44+ED25519) of length (bits) "...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_cacerts.cpp:13
Snippet: ...experimental hybrid ED25519+MLDSA44 root certif...

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/tls_cacerts.cpp:14
Snippet: // then is our own secp256r1 root certificate
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/tls_cacerts.cpp:45
Snippet: ...= POST_QUANTUM // ED25519 + MLDSA44 expires J...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/tls_cacerts.cpp:274
Snippet: ...// EdDSA cert Expires Jan 20...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/tls_client_cert.cpp:16
Snippet: // My personal ECDSA private key - Certi...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/tls_client_cert.cpp:47
Snippet: // My personal EDDSA private key - Certi...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/tls_client_cert.cpp:76
Snippet: ...int hwsigalg=ECDSA_SECP256R1_SHA256;
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/tls_client_cert.cpp:92
Snippet: ...int hwsigalg=ECDSA_SECP256R1_SHA256;
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: cpp/lib/tls_client_cert.cpp:120
Snippet: ...ersonal private key RSA 2048 bits - expires...
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/lib/tls_client_cert.cpp:203
Snippet: ...return ECDSA_SECP256R1_SHA256; // as long...
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/lib/tls_client_cert.cpp:206
Snippet: ...return ECDSA_SECP384R1_SHA384; // as long...
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/lib/tls_client_cert.cpp:219
Snippet: ...return MLDSA44_ED25519;
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/lib/tls_client_cert.cpp:222
Snippet: ...if (pk->curve==USE_ED25519) {
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/lib/tls_client_cert.cpp:223
Snippet: return ED25519;
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/lib/tls_client_cert.cpp:225
Snippet: ...if (pk->curve==USE_ED448) {
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/lib/tls_client_cert.cpp:226
Snippet: return ED448;
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/lib/tls_client_cert.cpp:237
Snippet: ...irements[len]=ECDSA_SECP256R1_SHA256; // as long...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_client_cert.cpp:241`
`Snippet: ...irements[len]=ECDSA_SECP384R1_SHA384; // as long...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_client_cert.cpp:274`
`Snippet: ...ements[len]=MLDSA44_ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_client_cert.cpp:278`
`Snippet: ...requirements[len]=ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_client_cert.cpp:280`
`Snippet: ...need to check that secp256r1 is supported - kind...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_client_cert.cpp:283`
`Snippet: ...if (pk->curve==USE_ED25519) {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_client_cert.cpp:284`
`Snippet: ...requirements[len]=ED25519;`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_client_cert.cpp:288`
`Snippet: ...if (pk->curve==USE_ED448) {`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_client_cert.cpp:289`
`Snippet: ...requirements[len]=ED448;`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_cert_chain.cpp:209`
`Snippet: ...gnatureVerify(ECDSA_SECP256R1_SHA256,CERT,SIG,PUB...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_cert_chain.cpp:211`
`Snippet: ...gnatureVerify(ECDSA_SECP384R1_SHA384,CERT,SIG,PUB...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_cert_chain.cpp:212`
`Snippet: ...CD && st.curve==USE_ED25519)`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_cert_chain.cpp:213`
`Snippet: ..._tlsSignatureVerify(ED25519,CERT,SIG,PUBKEY);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_cert_chain.cpp:214`
`Snippet: ...CD && st.curve==USE_ED448)`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_cert_chain.cpp:215`
`Snippet: ..._tlsSignatureVerify(ED448,CERT,SIG,PUBKEY);`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_cert_chain.cpp:232`
`Snippet: ...type==X509_HY1) // ED25519 + MLDSA44`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_cert_chain.cpp:238`
`Snippet: ..._tlsSignatureVerify(ED25519,CERT,&FSIG;,&FPUB;) &...`

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_x509.cpp:47`
`Snippet: // ED25519 Public Key - Ellipt...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_x509.cpp:51`
`Snippet: // ED448 Public Key - Ellipt...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_x509.cpp:59`
`Snippet: ...T256 curve - (NIST) P-256`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_x509.cpp:64`
`Snippet: ...tatic unsigned char secp384r1[5] = {0x2B, 0x81, 0...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_x509.cpp:65`
`Snippet: static octad SECP384R1 = {5, sizeof(secp38...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_x509.cpp:68`
`Snippet: ...tatic unsigned char secp521r1[5] = {0x2B, 0x81, 0...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_x509.cpp:69`
`Snippet: static octad SECP521R1 = {5, sizeof(secp52...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/lib/tls_x509.cpp:71`
`Snippet: // RSA Public Key - RSAES-...`

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/lib/tls_x509.cpp:75`
`Snippet: // RSA with SHA256`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/lib/tls_x509.cpp:79`
`Snippet: // RSA with SHA384`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/lib/tls_x509.cpp:83`
`Snippet: // RSA with SHA512`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_x509.cpp:96`
`Snippet: // MLDSA44+ED25519 - unofficial`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_x509.cpp:197`
`Snippet: // in-place ECDSA signature encoding...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_x509.cpp:244`
`Snippet: // in-place ECDSA signature decoding...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/lib/tls_x509.cpp:290`
`Snippet: ...x509 -nodes -newkey rsa:4096 -keyout key.pe...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/lib/tls_x509.cpp:293`
`Snippet: // For RSA octad = p|q|dp|dq|c...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/lib/tls_x509.cpp:333`

Snippet: { // Its an Ed25519 key
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:348
Snippet: ...ret.curve = USE_ED25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:351
Snippet: { // Its an Ed448 key
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:366
Snippet: ...ret.curve = USE_ED448;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:428
Snippet: { // Its an ED25519 key followed by a M...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:466
Snippet: ...ret.curve=USE_ED25519; // 8*pk->len;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:583
Snippet: ...if (OCT_compare(&SECP384R1;, &SOID;)) {
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:587
Snippet: ...if (OCT_compare(&SECP521R1;, &SOID;)) {
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: cpp/lib/tls_x509.cpp:604

Snippet: { // Its an RSA key
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: cpp/lib/tls_x509.cpp:677
Snippet: ...ic Curve signature, RSA for RSA signature
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:793
Snippet: ...ret.curve = USE_ED25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:822
Snippet: ...12) ret.curve = USE_ED25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:823
Snippet: ...56) ret.curve = USE_ED448;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:964
Snippet: ...ret.curve = USE_ED25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:1155
Snippet: ...icKey & prime256v1, secp384r1 or secp521r1 for EC...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: cpp/lib/tls_x509.cpp:1156
Snippet: // rsapublicKey for RSA
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/lib/tls_x509.cpp:1172
Snippet: ..._ECD; ret.curve=USE_ED25519;}
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/tls_x509.cpp:1173
Snippet: ...ECD; ret.curve=USE_ED448;}
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/tls_x509.cpp:1181
Snippet: ..._HY1; ret.curve=USE_ED25519;}
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/tls_x509.cpp:1210
Snippet: ...D)) ret.curve = USE_ED25519; /*****/
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/tls_x509.cpp:1212
Snippet: ...if (OCT_compare(&SECP384R1;, &KOID;)) ret.curve...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/tls_x509.cpp:1213
Snippet: ...if (OCT_compare(&SECP521R1;, &KOID;)) ret.curve...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: cpp/lib/tls_client_recv.cpp:768
Snippet: ...r example be 0804 - RSA-PSS-RSAE-SHA256
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/lib/ibe/tls_pqibe.cpp:34
Snippet: ...03d094,0x4ee931e,0x4ed448b,0x1115636,0x1c1891...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/desktop/client.cpp:116
Snippet: case X25519:
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/desktop/client.cpp:117
Snippet: printf("X25519\n");
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/desktop/client.cpp:119
Snippet: case SECP256R1:
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/desktop/client.cpp:120
Snippet: printf("SECP256R1\n");
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/desktop/client.cpp:122
Snippet: case SECP384R1:
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/desktop/client.cpp:123
Snippet: printf("SECP384R1\n");
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/desktop/client.cpp:129
Snippet: ...printf("MLKEM768+X25519\n");
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/desktop/client.cpp:160
Snippet: case ECDSA_SECP256R1_SHA256:
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/desktop/client.cpp:161
Snippet: ...printf("ECDSA_SECP256R1_SHA256\n");
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/src/desktop/client.cpp:169
Snippet: case ECDSA_SECP384R1_SHA384:
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/src/desktop/client.cpp:170
Snippet: ...printf("ECDSA_SECP384R1_SHA384\n");
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/src/desktop/client.cpp:184
Snippet: case ED25519:
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/src/desktop/client.cpp:185
Snippet: printf("ED25519\n");
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/src/desktop/client.cpp:187
Snippet: case ED448:
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/src/desktop/client.cpp:188
Snippet: printf("ED448\n");
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/src/desktop/client.cpp:196
Snippet: case MLDSA44_ED25519:
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: cpp/src/desktop/client.cpp:197
Snippet: ...printf("MLDSA44 + ED25519\n");
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/desktop/client.cpp:434
Snippet: ...->T.favourite_group=X25519;
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/desktop/client.cpp:447
Snippet: ...->T.favourite_group=X25519;
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:183
Snippet: os.system(copytext+" config_big.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:189
Snippet: os.system(copytext+" config_ff.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:197
Snippet: os.system(copytext+" big.cpp "+fnamec)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:198
Snippet: os.system(copytext+" big.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:213
Snippet: os.system(copytext+" ff.cpp "+fnamec)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:214`
`Snippet: os.system(copytext+" ff.h "+fnameh)`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/src/arduino/config.py:224`
`Snippet: ...s.system(copytext+" rsa.cpp "+fnamec)`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:224`
`Snippet: os.system(copytext+" rsa.cpp "+fnamec)`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/src/arduino/config.py:225`
`Snippet: ...s.system(copytext+" rsa.h "+fnameh)`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:225`
`Snippet: os.system(copytext+" rsa.h "+fnameh)`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:243`
`Snippet: os.system(copytext+" config_big.h "+fnameh)`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:249`
`Snippet: os.system(copytext+" config_field.h "+fnameh)`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:299`

Snippet: os.system(copytext+" config_curve.h "+fnameh)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:319
Snippet: os.system(copytext+" big.cpp "+fnamec)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:320
Snippet: os.system(copytext+" big.h "+fnameh)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:335
Snippet: os.system(copytext+" fp.cpp "+fnamec)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:336
Snippet: os.system(copytext+" fp.h "+fnameh)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:346
Snippet: os.system(copytext+" ecp.cpp "+fnamec)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:347
Snippet: os.system(copytext+" ecp.h "+fnameh)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/config.py:359

Snippet: ...s.system(copytext+" ecdh.cpp "+fnamec)
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:359
Snippet: os.system(copytext+" ecdh.cpp "+fnamec)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/config.py:360
Snippet: ...s.system(copytext+" ecdh.h "+fnameh)
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:360
Snippet: os.system(copytext+" ecdh.h "+fnameh)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/config.py:372
Snippet: ...s.system(copytext+" eddsa.cpp "+fnamec)
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:372
Snippet: os.system(copytext+" eddsa.cpp "+fnamec)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/config.py:373
Snippet: ...s.system(copytext+" eddsa.h "+fnameh)
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:373
Snippet: os.system(copytext+" eddsa.h "+fnameh)

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:386
Snippet: os.system(copytext+" hpke.cpp "+fnamec)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:387
Snippet: os.system(copytext+" hpke.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:396
Snippet: os.system(copytext+" fp2.cpp "+fnamec)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:397
Snippet: os.system(copytext+" fp2.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:406
Snippet: os.system(copytext+" fp4.cpp "+fnamec)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:407
Snippet: os.system(copytext+" fp4.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:419
```

Snippet: os.system(copytext+" fp12.cpp "+fnamec)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:420
Snippet: os.system(copytext+" fp12.h "+fnameh)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:431
Snippet: os.system(copytext+" ecp2.cpp "+fnamec)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:432
Snippet: os.system(copytext+" ecp2.h "+fnameh)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:443
Snippet: os.system(copytext+" pair.cpp "+fnamec)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:444
Snippet: os.system(copytext+" pair.h "+fnameh)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:455
Snippet: os.system(copytext+" mpin.cpp "+fnamec)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

Location: cpp/src/arduino/config.py:456
Snippet: os.system(copytext+" mpin.h "+fnameh)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:467
Snippet: os.system(copytext+" bls.cpp "+fnamec)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:468
Snippet: os.system(copytext+" bls.h "+fnameh)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:480
Snippet: os.system(copytext+" fp8.cpp "+fnamec)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:481
Snippet: os.system(copytext+" fp8.h "+fnameh)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:492
Snippet: os.system(copytext+" fp24.cpp "+fnamec)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:493
Snippet: os.system(copytext+" fp24.h "+fnameh)
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:504`
`Snippet: os.system(copytext+" ecp4.cpp "+fnamec)`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:505`
`Snippet: os.system(copytext+" ecp4.h "+fnameh)`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:516`
`Snippet: os.system(copytext+" pair4.cpp "+fnamec)`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:517`
`Snippet: os.system(copytext+" pair4.h "+fnameh)`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:528`
`Snippet: os.system(copytext+" mpin192.cpp "+fnamec)`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:529`
`Snippet: os.system(copytext+" mpin192.h "+fnameh)`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:540`
`Snippet: os.system(copytext+" bls192.cpp "+fnamec)`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:541
Snippet: os.system(copytext+" bls192.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:554
Snippet: os.system(copytext+" fp8.cpp "+fnamec)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:555
Snippet: os.system(copytext+" fp8.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:566
Snippet: os.system(copytext+" ecp8.cpp "+fnamec)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:567
Snippet: os.system(copytext+" ecp8.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:578
Snippet: os.system(copytext+" fp16.cpp "+fnamec)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:579
Snippet: os.system(copytext+" fp16.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:590
Snippet: os.system(copytext+" fp48.cpp "+fnamec)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:591
Snippet: os.system(copytext+" fp48.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:602
Snippet: os.system(copytext+" pair8.cpp "+fnamec)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:603
Snippet: os.system(copytext+" pair8.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:614
Snippet: os.system(copytext+" mpin256.cpp "+fnamec)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:615
Snippet: os.system(copytext+" mpin256.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:626
Snippet: os.system(copytext+" bls256.cpp "+fnamec)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:627
Snippet: os.system(copytext+" bls256.h "+fnameh)
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:644
Snippet: print("1. Ed25519")
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:650
Snippet: print("7. Ed448")
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:660
Snippet: print("17. SECP256K1")
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:664
Snippet: print("21. X448")
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:678
Snippet: print("32. FP256BN")
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: cpp/src/arduino/config.py:689
Snippet: print("RSA")
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:749
Snippet: ...set("255","F25519","Ed25519","29","2","1","PSEU...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:768
Snippet: ...veset("448","F448","Ed448","29","1","0","GENE...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:802
Snippet: ...curveset("256","SECP256K1","SECP256K1","28","...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:818
Snippet: ...veset("448","F448","X448","29","1","0","GENE...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:865
Snippet: ...curveset("256","FP256BN","FP256BN","28","...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE

RSA key generation, encryption, or signing.

```
Location: cpp/src/arduino/config.py:907
Snippet: # "ring" is the RSA modulus size = "big...
```

Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:938
Snippet: os.system(deltext+" *.rs")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:939
Snippet: #os.system(deltext+" fast*.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:940`
`Snippet: os.system(deltext+" big.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:941`
`Snippet: os.system(deltext+" fp.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:942`
`Snippet: os.system(deltext+" ecp.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/src/arduino/config.py:943`
`Snippet: os.system(deltext+" ecdh.*")`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:943`
`Snippet: os.system(deltext+" ecdh.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: cpp/src/arduino/config.py:944`
`Snippet: os.system(deltext+" eddsa.*")`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:944`
`Snippet: os.system(deltext+" eddsa.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:945`
`Snippet: os.system(deltext+" hpke.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:946`
`Snippet: os.system(deltext+" ff.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: cpp/src/arduino/config.py:947`
`Snippet: os.system(deltext+" rsa.*")`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:947`
`Snippet: os.system(deltext+" rsa.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:948`
`Snippet: os.system(deltext+" config_big.h")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:949`
`Snippet: os.system(deltext+" config_field.h")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:950`
`Snippet: os.system(deltext+" config_curve.h")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:951`
`Snippet: os.system(deltext+" config_ff.h")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:952`
`Snippet: os.system(deltext+" fp2.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:953`
`Snippet: os.system(deltext+" fp4.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:954`
`Snippet: os.system(deltext+" fp8.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:955`
`Snippet: os.system(deltext+" fp16.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:957`
`Snippet: os.system(deltext+" fp12.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:958`
`Snippet: os.system(deltext+" fp24.*")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:959
Snippet: os.system(deltext+" fp48.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:961
Snippet: os.system(deltext+" ecp2.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:962
Snippet: os.system(deltext+" ecp4.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:963
Snippet: os.system(deltext+" ecp8.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:965
Snippet: os.system(deltext+" pair.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:966
Snippet: os.system(deltext+" mpin.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:967
Snippet: os.system(deltext+" bls.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:969
Snippet: os.system(deltext+" pair4.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:970
Snippet: os.system(deltext+" mpin192.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:971
Snippet: os.system(deltext+" bls192.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:973
Snippet: os.system(deltext+" pair8.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:974
Snippet: os.system(deltext+" mpin256.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:975
Snippet: os.system(deltext+" bls256.*")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:977
Snippet: os.system(deltext+" blsrev*.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:978
Snippet: #os.system(deltext+ " hpke*.* ")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:983
Snippet: os.system(deltext+" newhope.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:984
Snippet: os.system(deltext+" newhope.h")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:987
Snippet: # os.system("mkdir examples"+slashtext+"testnhs")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:988
Snippet: # os.system(copytext+" testnhs.ino "+"examples"+slashtext+"testnhs"+slashtext+"testnhs.ino")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:989
Snippet: os.system(deltext+" testnhs.ino")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:992
```

Snippet: os.system(deltext+" dilithium.cpp")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:993
Snippet: os.system(deltext+" dilithium.h")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:994
Snippet: os.system(deltext+" testdlthm.ino")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:997
Snippet: os.system(deltext+" kyber.cpp")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:998
Snippet: os.system(deltext+" kyber.h")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:999
Snippet: os.system(deltext+" testkyber.ino")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1004
Snippet: # os.system("mkdir examples"+slashtext+"testecc")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

Location: cpp/src/arduino/config.py:1005
Snippet: # os.system(copytext+" testecc.ino "+"examples"+slashtext+"testecc"+slashtext+"testecc.ino")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1006
Snippet: # os.system("mkdir examples"+slashtext+"timeecc")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1007
Snippet: # os.system(copytext+" timeecc.ino "+"examples"+slashtext+"timeecc"+slashtext+"timeecc.ino")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1008
Snippet: os.system(deltext+" testecc.ino")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1009
Snippet: os.system(deltext+" timeecc.ino")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1012
Snippet: # os.system(deltext+" X509.cpp")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1013
Snippet: # os.system(deltext+" X509.h")
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1016`
`Snippet: # os.system("mkdir examples"+slashtext+"testrsa")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1017`
`Snippet: # os.system(copytext+" testrsa.ino "+"examples"+slashtext+"testrsa"+slashtext+"testrsa.ino")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1018`
`Snippet: os.system(deltext+" testrsa.ino")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1023`
`Snippet: # os.system("mkdir examples"+slashtext+"testbls")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1024`
`Snippet: # os.system(copytext+" testbls.ino "+"examples"+slashtext+"testbls"+slashtext+"testbls.ino")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1025`
`Snippet: # os.system("mkdir examples"+slashtext+"timepbc")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1026`
`Snippet: # os.system(copytext+" timepbc.ino "+"examples"+slashtext+"timepbc"+slashtext+"timepbc.ino")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:1027
Snippet: ...s.system(deltext+ " ecdh*.* ")
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1027
Snippet: # os.system(deltext+ " ecdh*.* ")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1028
Snippet: os.system(deltext+ " hpke*.* ")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1029
Snippet: os.system(deltext+" testbls.ino")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1030
Snippet: os.system(deltext+" timepbc.ino")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1032
Snippet: os.system(deltext+ " share*.* ")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1033
Snippet: os.system(deltext+ " x509.* ")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1036
Snippet: os.system("mkdir examples"+slashtext+"client")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1037
Snippet: os.system(copytext+" client.cpp "+"examples"+slashtext+"client"+slashtext+"client.ino")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1038
Snippet: os.system(deltext+" client.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1041
Snippet: # os.system(copytext+" tls_sal_m.xpp "+"tls_sal.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1042
Snippet: # os.system(deltext+ " ECC*.* ")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1045
Snippet: # os.system(copytext+" tls_sal_mh.xpp "+"tls_sal.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1048
Snippet: # os.system(copytext+" tls_sal_mhp.xpp "+"tls_sal.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:1049
Snippet: # ..and copy x25519.S into project from...
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1052
Snippet: os.system(copytext+" tls_sal_mt.xpp "+"tls_sal.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1055
Snippet: os.system(deltext+" *.md")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1056
Snippet: os.system(deltext+" *.xpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1057
Snippet: os.system(deltext+" testx509.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1060
Snippet: os.system(deltext+" rom_field_F25519.c...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:1062
Snippet: ...deltext+" rom_curve_Ed25519.cpp")
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

`Location: cpp/src/arduino/config.py:1062`
`Snippet: os.system(deltext+" rom_curve_Ed25519....`

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

`Location: cpp/src/arduino/config.py:1064`
`Snippet: os.system(deltext+" rom_curve_C25519.c...`

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

`Location: cpp/src/arduino/config.py:1066`
`Snippet: os.system(deltext+" rom_field_NIST256....`

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

`Location: cpp/src/arduino/config.py:1067`
`Snippet: os.system(deltext+" rom_curve_NIST256....`

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

`Location: cpp/src/arduino/config.py:1069`
`Snippet: os.system(deltext+" rom_field_BRAINPOO...`

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

`Location: cpp/src/arduino/config.py:1070`
`Snippet: os.system(deltext+" rom_curve_BRAINPOO...`

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

`Location: cpp/src/arduino/config.py:1072`
`Snippet: os.system(deltext+" rom_field_ANSSI.cp...`

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1073
Snippet: os.system(deltext+" rom_curve_ANSSI.cp...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1075
Snippet: os.system(deltext+" rom_field_HIFIVE.c...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1076
Snippet: os.system(deltext+" rom_curve_HIFIVE.c...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1079
Snippet: os.system(deltext+" rom_field_F448.cpp...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:1082
Snippet: ...deltext+" rom_curve_Ed448.cpp")
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1082
Snippet: os.system(deltext+" rom_curve_Ed448.cp...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1084
Snippet: os.system(deltext+" rom_field_NIST384....
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1085
Snippet: os.system(deltext+" rom_curve_NIST384....
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1087
Snippet: os.system(deltext+" rom_field_C41417.c...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1088
Snippet: os.system(deltext+" rom_curve_C41417.c...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1090
Snippet: os.system(deltext+" rom_field_NIST521....
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1091
Snippet: os.system(deltext+" rom_curve_NIST521....
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1093
Snippet: os.system(deltext+" rom_field_F256PMW....
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1094
Snippet: os.system(deltext+" rom_curve_NUMS256W...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1096
Snippet: os.system(deltext+" rom_field_F256PME....
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1097
Snippet: os.system(deltext+" rom_curve_NUMS256E...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1100
Snippet: os.system(deltext+" rom_field_F384PM.c...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1102
Snippet: os.system(deltext+" rom_curve_NUMS384W...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1104
Snippet: os.system(deltext+" rom_curve_NUMS384E...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1107
Snippet: os.system(deltext+" rom_field_F512PM.c...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1109
```

Snippet: os.system(deltext+" rom_curve_NUMS512W...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1111
Snippet: os.system(deltext+" rom_curve_NUMS512E...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/config.py:1114
Snippet: ...deltext+" rom_field_SECP256K1.cpp")
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1114
Snippet: os.system(deltext+" rom_field_SECP256K...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/config.py:1115
Snippet: ...deltext+" rom_curve_SECP256K1.cpp")
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1115
Snippet: os.system(deltext+" rom_curve_SECP256K...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1118
Snippet: os.system(deltext+" rom_field_SM2.cpp"...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1119
Snippet: os.system(deltext+" rom_curve_SM2.cpp"...

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1123
Snippet: os.system(deltext+" rom_curve_C13318.c...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1126
Snippet: os.system(deltext+" rom_field_JUBJUB.c...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1127
Snippet: os.system(deltext+" rom_curve_JUBJUB.c...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/config.py:1130
Snippet: ...deltext+" rom_curve_X448.cpp")
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1130
Snippet: os.system(deltext+" rom_curve_X448.cpp...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1132
Snippet: os.system(deltext+" rom_field_SECP160R...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1133
Snippet: os.system(deltext+" rom_curve_SECP160R...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1135
Snippet: os.system(deltext+" rom_field_C1174.cp...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1136
Snippet: os.system(deltext+" rom_curve_C1174.cp...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1138
Snippet: os.system(deltext+" rom_field_C1665.cp...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1139
Snippet: os.system(deltext+" rom_curve_C1665.cp...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1142
Snippet: os.system(deltext+" rom_field_MDC.cpp"...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1143
Snippet: os.system(deltext+" rom_curve_MDC.cpp"...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1145
```

Snippet: os.system(deltext+" rom_field_TWEEDLED...

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

Location: cpp/src/arduino/config.py:1146
Snippet: os.system(deltext+" rom_curve_TWEEDLED...

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

Location: cpp/src/arduino/config.py:1148
Snippet: os.system(deltext+" rom_field_TWEEDLED...

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

Location: cpp/src/arduino/config.py:1149
Snippet: os.system(deltext+" rom_curve_TWEEDLED...

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

Location: cpp/src/arduino/config.py:1152
Snippet: os.system(deltext+" rom_field_BN254.cp...

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

Location: cpp/src/arduino/config.py:1153
Snippet: os.system(deltext+" rom_curve_BN254.cp...

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

Location: cpp/src/arduino/config.py:1155
Snippet: os.system(deltext+" rom_field_BN254CX....

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

Location: cpp/src/arduino/config.py:1156
Snippet: os.system(deltext+" rom_curve_BN254CX....
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1158
Snippet: os.system(deltext+" rom_field_BLS12383...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1159
Snippet: os.system(deltext+" rom_curve_BLS12383...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1161
Snippet: os.system(deltext+" rom_field_BLS12381...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1162
Snippet: os.system(deltext+" rom_curve_BLS12381...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/config.py:1164
Snippet: ...eltext+" rom_field_FP256BN.cpp")
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1164
Snippet: os.system(deltext+" rom_field_FP256BN....
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/config.py:1165

Snippet: ...eltext+" rom_curve_FP256BN.cpp")
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1165
Snippet: os.system(deltext+" rom_curve_FP256BN....
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1167
Snippet: os.system(deltext+" rom_field_FP512BN....
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1168
Snippet: os.system(deltext+" rom_curve_FP512BN....
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1170
Snippet: os.system(deltext+" rom_field_BLS12443...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1171
Snippet: os.system(deltext+" rom_curve_BLS12443...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1174
Snippet: os.system(deltext+" rom_field_BLS12461...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

Location: cpp/src/arduino/config.py:1175
Snippet: os.system(deltext+" rom_curve_BLS12461...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1178
Snippet: os.system(deltext+" rom_field_BN462.cp...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1179
Snippet: os.system(deltext+" rom_curve_BN462.cp...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1182
Snippet: os.system(deltext+" rom_field_BLS24479...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1183
Snippet: os.system(deltext+" rom_curve_BLS24479...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1185
Snippet: os.system(deltext+" rom_field_BLS48556...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
Location: cpp/src/arduino/config.py:1186
Snippet: os.system(deltext+" rom_curve_BLS48556...
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1189`
`Snippet: os.system(deltext+" rom_field_BLS48581...`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1190`
`Snippet: os.system(deltext+" rom_curve_BLS48581...`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1193`
`Snippet: os.system(deltext+" rom_field_BLS48286...`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1194`
`Snippet: os.system(deltext+" rom_curve_BLS48286...`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1197`
`Snippet: os.system(deltext+" rom_field_BN158.cp...`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1198`
`Snippet: os.system(deltext+" rom_curve_BN158.cp...`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK
System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.
`Location: cpp/src/arduino/config.py:1201`
`Snippet: os.system(deltext+" testbls.cpp")`
Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1202
Snippet: os.system(deltext+" testecc.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1203
Snippet: os.system(deltext+" testmpin.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1204
Snippet: os.system(deltext+" testhpke.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1205
Snippet: os.system(deltext+" testhtp.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1206
Snippet: os.system(deltext+" testnhs.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1207
Snippet: os.system(deltext+" testdlthm.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1208
Snippet: os.system(deltext+" testkyber.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1209
Snippet: os.system(deltext+" testeddsa.cpp")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1211
Snippet: os.system(deltext+" config*.py")
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] COMMAND_INJECTION_SINK

System command built via string concatenation and executed with shell interpretation enabled — classic command-injection sink.

```
Location: cpp/src/arduino/config.py:1212
Snippet: os.system(deltext+" benchtest_all.cpp"...
```

Remediation: Use parameterized subprocess calls with shell=False (or the language equivalent) and an argument list, never string-concatenated shell commands.

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/client.cpp:57
Snippet: case X25519:
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/client.cpp:58
Snippet: ...Serial.println("X25519");
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/client.cpp:60
Snippet: case SECP256R1:
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

```
Location: cpp/src/arduino/client.cpp:61
Snippet: ...Serial.println("SECP256R1");
```

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE

Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: cpp/src/arduino/client.cpp:63
Snippet: case SECP384R1:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/client.cpp:64
Snippet: ...Serial.println("SECP384R1");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/client.cpp:70
Snippet: ...l.println("MLKEM768+X25519");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/client.cpp:101
Snippet: case ECDSA_SECP256R1_SHA256:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/client.cpp:102
Snippet: ...rial.println("ECDSA_SECP256R1_SHA256");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/client.cpp:110
Snippet: case ECDSA_SECP384R1_SHA384:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/client.cpp:111
Snippet: ...rial.println("ECDSA_SECP384R1_SHA384");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/client.cpp:125
Snippet: case ED25519:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: cpp/src/arduino/client.cpp:126
Snippet: ...Serial.println("ED25519");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/client.cpp:128
Snippet: case ED448:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/client.cpp:129
Snippet: ...Serial.println("ED448");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/client.cpp:137
Snippet: case MLDSA44_ED25519:
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/client.cpp:138
Snippet: ....println("MLDSA44 + ED25519");
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/client.cpp:262
Snippet: ...->T.favourite_group=X25519;
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/ECCX08.cpp:217
Snippet: ...s is one needed for ECDH
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: cpp/src/arduino/ECCX08.cpp:245
Snippet: // Do ECDH and calculate 32 by...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).

Location: cpp/src/arduino/ECCX08.cpp:730
Snippet: .../ Verify, external, P256
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: servercert/server/server.py:36
Snippet: ...enssl ecparam -name secp384r1 -out ecparam.pem",s...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: servercert/server/server.py:41
Snippet: ...sl req -new -newkey rsa:1024 -keyout server...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: servercert/server/server.py:45
Snippet: ...sl req -new -newkey rsa:2048 -keyout server...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
Location: servercert/server/server.py:49
Snippet: ...sl req -new -newkey rsa:4096 -keyout server...
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: servercert/server/server.py:52
Snippet: if method=="ED25519" :
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: servercert/server/server.py:53
Snippet: ...genpkey -algorithm ED25519 -out eddsa.key",she...
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: servercert/server/server.py:54
Snippet: ...enssl req -new -key eddsa.key -keyout server....
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
Location: servercert/server/server.py:57
Snippet: if method=="ED448" :

Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: servercert/server/server.py:58`
`Snippet: ...genpkey -algorithm ED448 -out eddsa.key",she...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: servercert/server/server.py:59`
`Snippet: ...enssl req -new -key eddsa.key -keyout server....`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: servercert/server/intermediate/intermediate.py:37`
`Snippet: ...enssl ecparam -name secp384r1 -out ecparam.pem",s...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: servercert/server/intermediate/intermediate.py:42`
`Snippet: ...sl req -new -newkey rsa:1024 -keyout inter....`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: servercert/server/intermediate/intermediate.py:46`
`Snippet: ...sl req -new -newkey rsa:2048 -keyout inter....`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: servercert/server/intermediate/intermediate.py:50`
`Snippet: ...sl req -new -newkey rsa:4096 -keyout inter....`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: servercert/server/intermediate/intermediate.py:53`
`Snippet: if method=="ED25519" :`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: servercert/server/intermediate/intermediate.py:54`
`Snippet: ...genpkey -algorithm ED25519 -out eddsa.key",she...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: servercert/server/intermediate/intermediate.py:55`
`Snippet: ...enssl req -new -key eddsa.key -keyout inter.k...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: servercert/server/intermediate/intermediate.py:58`
`Snippet: if method=="ED448" :`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: servercert/server/intermediate/intermediate.py:59`
`Snippet: ...genpkey -algorithm ED448 -out eddsa.key",she...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: servercert/server/intermediate/intermediate.py:60`
`Snippet: ...enssl req -new -key eddsa.key -keyout inter.k...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: servercert/server/intermediate/root/root.py:26`
`Snippet: ...enssl ecparam -name secp384r1 -out ecparam.pem",s...`
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: servercert/server/intermediate/root/root.py:30`
`Snippet: ...-days 3650 -newkey rsa:1024 -keyout root.k...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: servercert/server/intermediate/root/root.py:33`
`Snippet: ...-days 3650 -newkey rsa:2048 -keyout root.k...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] RSA_USAGE
RSA key generation, encryption, or signing.
`Location: servercert/server/intermediate/root/root.py:36`
`Snippet: ...-days 3650 -newkey rsa:4096 -keyout root.k...`
Remediation: Replace with ML-KEM (Kyber) for key exchange or ML-DSA (Dilithium) for signatures.

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
`Location: servercert/server/intermediate/root/root.py:38`

```
Snippet: if method=="ED25519" :
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: servercert/server/intermediate/root/root.py:39
Snippet: ...genpkey -algorithm ED25519 -out eddsa.key",she...
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: servercert/server/intermediate/root/root.py:40
Snippet: ...509 -days 3650 -key eddsa.key -keyout root.ke...
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: servercert/server/intermediate/root/root.py:41
Snippet: if method=="ED448" :
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: servercert/server/intermediate/root/root.py:42
Snippet: ...genpkey -algorithm ED448 -out eddsa.key",she...
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[HIGH] ECC_CURVE_USAGE
Elliptic-curve key exchange or signature scheme (ECDSA/ECDH/EdDSA and named curves).
```
Location: servercert/server/intermediate/root/root.py:43
Snippet: ...509 -days 3650 -key eddsa.key -keyout root.ke...
```
Remediation: Replace signatures with ML-DSA (Dilithium) or SLH-DSA (SPHINCS+); replace key exchange with ML-KEM (Kyber).

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: clientcert/client/client.py:2
Snippet: ...re elliptic curves, MLDSA options could be ad...
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: clientcert/client/client.py:10
Snippet: ...SSL by default uses sha256 hashing of certific...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: clientcert/client/client.py:11
Snippet: ...it from ecdsa_with_sha256 to ecdsa_with_sha38...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: clientcert/client/client.py:19
Snippet: ...3 client.py NIST256 sha384")
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: clientcert/client/client.py:23
Snippet: hash="sha256"
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: clientcert/client/client.py:62
Snippet: if method=="MLDSA65" :
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: clientcert/client/client.py:63
Snippet: ...sl req -new -newkey mldsa65 -keyout client.ke...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: clientcert/client/intermediate/intermediate.py:2
Snippet: ...re elliptic curves, MLDSA options could be ad...
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: clientcert/client/intermediate/intermediate.py:11
Snippet: ...SSL by default uses sha256 hashing of certific...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: clientcert/client/intermediate/intermediate.py:12
Snippet: ...it from ecdsa_with_sha256 to ecdsa_with_sha38...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: clientcert/client/intermediate/intermediate.py:20
Snippet: ...rmediate.py NIST384 sha384")
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: clientcert/client/intermediate/intermediate.py:24
Snippet: hash="sha256"
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: clientcert/client/intermediate/intermediate.py:63`
`Snippet: if method=="MLDSA65" :`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: clientcert/client/intermediate/intermediate.py:64`
`Snippet: ...sl req -new -newkey mldsa65 -keyout inter.key...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: clientcert/client/intermediate/root/root.py:2`
`Snippet: ...re elliptic curves, MLDSA options could be ad...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: clientcert/client/intermediate/root/root.py:45`
`Snippet: if method=="MLDSA65" :`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: clientcert/client/intermediate/root/root.py:46`
`Snippet: ...-days 3650 -newkey mldsa65 -keyout root.key...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:18`
`Snippet: #define MLDSA44_PK 9`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:19`
`Snippet: #define MLDSA65_PK 10`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:20`
`Snippet: #define ED25519_MLDSA44_PK 11 // Hybrid`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:27`
`Snippet: #define ECCSHA256_SIG 1`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:28`

Snippet: #define ECCSHA384_SIG 2
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeleafcert.cpp:29
Snippet: #define RSASHA256_SIG 4
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeleafcert.cpp:30
Snippet: #define RSASHA384_SIG 5
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeleafcert.cpp:31
Snippet: #define RSASHA512_SIG 6
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeleafcert.cpp:34
Snippet: #define MLDSA44_SIG 9
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeleafcert.cpp:35
Snippet: #define MLDSA65_SIG 10
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeleafcert.cpp:36
Snippet: #define ED25519_MLDSA44_SIG 11
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeleafcert.cpp:56
Snippet: #define SIGTYPE RSASHA256_SIG // Intermediate...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeleafcert.cpp:58
Snippet: #if SIGTYPE==RSASHA256_SIG || SIGTYPE==RSA...
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeleafcert.cpp:119
Snippet: #if PKTYPE==MLDSA65_PK
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:123`
`Snippet: #define PK_TYPE MLDSA_KP`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:126`
`Snippet: #if PKTYPE==MLDSA44_PK`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:130`
`Snippet: #define PK_TYPE MLDSA_KP`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:133`
`Snippet: #if PKTYPE==ED25519_MLDSA44_PK`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:141`
`Snippet: #define PK_TYPE_2 MLDSA_KP`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:166`
`Snippet: #if SIGTYPE==ECCSHA256_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:171`
`Snippet: ...YPE ECDSA_SECP256R1_SHA256 // signature algori...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:173`
`Snippet: #if SIGTYPE==ECCSHA384_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:178`
`Snippet: ...YPE ECDSA_SECP384R1_SHA384`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:180`
`Snippet: #if SIGTYPE==RSASHA256_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:185`
`Snippet: ...SIG_TYPE RSA_PKCS1_SHA256`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:187`
`Snippet: #if SIGTYPE==RSASHA384_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:192`
`Snippet: ...SIG_TYPE RSA_PKCS1_SHA384`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:194`
`Snippet: #if SIGTYPE==RSASHA512_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:199`
`Snippet: ...SIG_TYPE RSA_PKCS1_SHA512`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:215`
`Snippet: #if SIGTYPE==MLDSA65_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:220`
`Snippet: #define SIG_TYPE MLDSA65`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:223`
`Snippet: #if SIGTYPE==MLDSA44_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:228`

Snippet: #define SIG_TYPE MLDSA44
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeleafcert.cpp:231
Snippet: ...if SIGTYPE==ED25519_MLDSA44_SIG
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeleafcert.cpp:241
Snippet: #define SIG_TYPE_2 MLDSA44
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeleafcert.cpp:515
Snippet: // MLDSA or EDDSA
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeleafcert.cpp:516
Snippet: #if PKTYPE==MLDSA65_PK || PKTYPE==MLD...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeleafcert.cpp:536
Snippet: #if PKTYPE==ED25519_MLDSA44_PK
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeleafcert.cpp:756
Snippet: ...if SIGTYPE==ED25519_MLDSA44_SIG
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeleafcert.cpp:777
Snippet: #if SIGTYPE==ECCSHA256_SIG || SIGTYPE==ECC...
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeleafcert.cpp:830
Snippet: #if PKTYPE==MLDSA65_PK || PKTYPE==MLD...
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeleafcert.cpp:998
Snippet: ...#if SIGTYPE==ECCSHA256_SIG
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:1004`
`Snippet: ...#if SIGTYPE==ECCSHA384_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeleafcert.cpp:1024`
`Snippet: ...#if SIGTYPE==RSASHA256_SIG || SIGTYPE==RSA...`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:1029`
`Snippet: ...if (ret.curve==USE_MLDSA44)`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:1031`
`Snippet: ...#if SIGTYPE==MLDSA44_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:1035`
`Snippet: ...if (ret.curve==USE_MLDSA65)`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:1037`
`Snippet: ...#if SIGTYPE==MLDSA65_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeleafcert.cpp:1045`
`Snippet: ...if SIGTYPE==ED25519_MLDSA44_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:17`
`Snippet: #define MLDSA44_PK 9`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:18`
`Snippet: #define MLDSA65_PK 10`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:19`
`Snippet: #define ED25519_MLDSA44_PK 11 // Hybrid`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:26`
`Snippet: #define ECCSHA256_SIG 1`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:27`
`Snippet: #define ECCSHA384_SIG 2`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:28`
`Snippet: #define RSASHA256_SIG 4`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:29`
`Snippet: #define RSASHA384_SIG 5`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:30`
`Snippet: #define RSASHA512_SIG 6`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:33`
`Snippet: #define MLDSA44_SIG 9`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:34`
`Snippet: #define MLDSA65_SIG 10`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:35`
`Snippet: #define ED25519_MLDSA44_SIG 11 // Hybrid...`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:54`

Snippet: #define SIGTYPE RSASHA256_SIG // Root CA sign...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeintercert.cpp:56
Snippet: #if SIGTYPE==RSASHA256_SIG || SIGTYPE==RSA...
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeintercert.cpp:117
Snippet: #if PKTYPE==MLDSA65_PK
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeintercert.cpp:121
Snippet: #define PK_TYPE MLDSA_KP
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeintercert.cpp:124
Snippet: #if PKTYPE==MLDSA44_PK
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeintercert.cpp:128
Snippet: #define PK_TYPE MLDSA_KP
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeintercert.cpp:131
Snippet: #if PKTYPE==ED25519_MLDSA44_PK
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeintercert.cpp:139
Snippet: #define PK_TYPE_2 MLDSA_KP
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeintercert.cpp:164
Snippet: #if SIGTYPE==ECCSHA256_SIG
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeintercert.cpp:169
Snippet: ...YPE ECDSA_SECP256R1_SHA256 // signature algori...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:171`
`Snippet: #if SIGTYPE==ECCSHA384_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:176`
`Snippet: ...YPE ECDSA_SECP384R1_SHA384`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:178`
`Snippet: #if SIGTYPE==RSASHA256_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:183`
`Snippet: ...SIG_TYPE RSA_PKCS1_SHA256`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:185`
`Snippet: #if SIGTYPE==RSASHA384_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:190`
`Snippet: ...SIG_TYPE RSA_PKCS1_SHA384`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:192`
`Snippet: #if SIGTYPE==RSASHA512_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:197`
`Snippet: ...SIG_TYPE RSA_PKCS1_SHA512`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:213`
`Snippet: #if SIGTYPE==MLDSA65_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:218`
`Snippet: #define SIG_TYPE MLDSA65`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:221`
`Snippet: #if SIGTYPE==MLDSA44_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:226`
`Snippet: #define SIG_TYPE MLDSA44`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:229`
`Snippet: ...if SIGTYPE==ED25519_MLDSA44_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:239`
`Snippet: #define SIG_TYPE_2 MLDSA44`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:511`
`Snippet: // MLDSA or EDDSA`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:512`
`Snippet: #if PKTYPE==MLDSA65_PK || PKTYPE==MLD...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:532`
`Snippet: #if PKTYPE==ED25519_MLDSA44_PK`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeintercert.cpp:732`
`Snippet: ...if SIGTYPE==ED25519_MLDSA44_SIG`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeintercert.cpp:753`

Snippet: #if SIGTYPE==ECCSHA256_SIG || SIGTYPE==ECC...
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeintercert.cpp:805
Snippet: #if PKTYPE==MLDSA65_PK || PKTYPE==MLD...
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeintercert.cpp:974
Snippet: ...#if SIGTYPE==ECCSHA256_SIG
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeintercert.cpp:980
Snippet: ...#if SIGTYPE==ECCSHA384_SIG
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeintercert.cpp:1000
Snippet: ...#if SIGTYPE==RSASHA256_SIG || SIGTYPE==RSA...
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeintercert.cpp:1005
Snippet: ...if (ret.curve==USE_MLDSA44)
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeintercert.cpp:1007
Snippet: ...#if SIGTYPE==MLDSA44_SIG
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeintercert.cpp:1011
Snippet: ...if (ret.curve==USE_MLDSA65)
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeintercert.cpp:1013
Snippet: ...#if SIGTYPE==MLDSA65_SIG
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeintercert.cpp:1021
Snippet: ...if SIGTYPE==ED25519_MLDSA44_SIG
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:17`
`Snippet: #define MLDSA44_PK 9`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:18`
`Snippet: #define MLDSA65_PK 10`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:19`
`Snippet: #define ED25519_MLDSA44_PK 11 // Hybrid`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makerootcert.cpp:26`
`Snippet: #define ECCSHA256_SIG 1`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makerootcert.cpp:27`
`Snippet: #define ECCSHA384_SIG 2`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makerootcert.cpp:28`
`Snippet: #define RSASHA256_SIG 4`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makerootcert.cpp:29`
`Snippet: #define RSASHA384_SIG 5`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makerootcert.cpp:30`
`Snippet: #define RSASHA512_SIG 6`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:33`
`Snippet: #define MLDSA44_SIG 9`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:34`
`Snippet: #define MLDSA65_SIG 10`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:35`
`Snippet: #define ED25519_MLDSA44_SIG 11 // Hybrid...`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makerootcert.cpp:50`
`Snippet: #define SIGTYPE RSASHA256_SIG // for self-si...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makerootcert.cpp:52`
`Snippet: #if SIGTYPE==RSASHA256_SIG || SIGTYPE==RSA...`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:113`
`Snippet: #if PKTYPE==MLDSA65_PK`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:117`
`Snippet: #define PK_TYPE MLDSA_KP`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:120`
`Snippet: #if PKTYPE==MLDSA44_PK`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:124`
`Snippet: #define PK_TYPE MLDSA_KP`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:127`
`Snippet: #if PKTYPE==ED25519_MLDSA44_PK`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:135`

```
Snippet: #define PK_TYPE_2 MLDSA_KP
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makerootcert.cpp:161
Snippet: #if SIGTYPE==ECCSHA256_SIG
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makerootcert.cpp:165
Snippet: ...YPE ECDSA_SECP256R1_SHA256 // signature algori...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makerootcert.cpp:167
Snippet: #if SIGTYPE==ECCSHA384_SIG
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makerootcert.cpp:171
Snippet: ...YPE ECDSA_SECP384R1_SHA384
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makerootcert.cpp:173
Snippet: #if SIGTYPE==RSASHA256_SIG
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makerootcert.cpp:177
Snippet: ...SIG_TYPE RSA_PKCS1_SHA256
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makerootcert.cpp:179
Snippet: #if SIGTYPE==RSASHA384_SIG
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makerootcert.cpp:183
Snippet: ...SIG_TYPE RSA_PKCS1_SHA384
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makerootcert.cpp:185
Snippet: #if SIGTYPE==RSASHA512_SIG
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makerootcert.cpp:189`
`Snippet: ...SIG_TYPE RSA_PKCS1_SHA512`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:203`
`Snippet: #if SIGTYPE==MLDSA65_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:207`
`Snippet: #define SIG_TYPE MLDSA65`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:210`
`Snippet: #if SIGTYPE==MLDSA44_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:215`
`Snippet: #define SIG_TYPE MLDSA44`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:218`
`Snippet: ...if SIGTYPE==ED25519_MLDSA44_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:226`
`Snippet: #define SIG_TYPE_2 MLDSA44`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:469`
`Snippet: // MLDSA or EDDSA`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:470`
`Snippet: #if PKTYPE==MLDSA65_PK || PKTYPE==MLD...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:491`
`Snippet: #if PKTYPE==ED25519_MLDSA44_PK`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:689`
`Snippet: ...if SIGTYPE==ED25519_MLDSA44_SIG`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makerootcert.cpp:711`
`Snippet: #if SIGTYPE==ECCSHA256_SIG || SIGTYPE==ECC...`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makerootcert.cpp:763`
`Snippet: #if PKTYPE==MLDSA65_PK || PKTYPE==MLD...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeselfcert.cpp:17`
`Snippet: #define MLDSA44_PK 9`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeselfcert.cpp:18`
`Snippet: #define MLDSA65_PK 10`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeselfcert.cpp:19`
`Snippet: #define ED25519_MLDSA44_PK 11 // Hybrid`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:26`
`Snippet: #define ECCSHA256_SIG 1`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:27`
`Snippet: #define ECCSHA384_SIG 2`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:28`

```
Snippet: #define RSASHA256_SIG 4
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makeselfcert.cpp:29
Snippet: #define RSASHA384_SIG 5
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makeselfcert.cpp:30
Snippet: #define RSASHA512_SIG 6
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: makecertchain/makeselfcert.cpp:33
Snippet: #define MLDSA44_SIG 9
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: makecertchain/makeselfcert.cpp:34
Snippet: #define MLDSA65_SIG 10
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: makecertchain/makeselfcert.cpp:35
Snippet: #define ED25519_MLDSA44_SIG 11 // Hybrid...
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makeselfcert.cpp:51
Snippet: #define SIGTYPE RSASHA256_SIG // for self-si...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: makecertchain/makeselfcert.cpp:53
Snippet: #if SIGTYPE==RSASHA256_SIG || SIGTYPE==RSA...
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: makecertchain/makeselfcert.cpp:114
Snippet: #if PKTYPE==MLDSA65_PK
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: makecertchain/makeselfcert.cpp:118
Snippet: #define PK_TYPE MLDSA_KP
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeselfcert.cpp:121`
`Snippet: #if PKTYPE==MLDSA44_PK`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeselfcert.cpp:125`
`Snippet: #define PK_TYPE MLDSA_KP`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeselfcert.cpp:128`
`Snippet: #if PKTYPE==ED25519_MLDSA44_PK`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeselfcert.cpp:136`
`Snippet: #define PK_TYPE_2 MLDSA_KP`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:162`
`Snippet: #if SIGTYPE==ECCSHA256_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:166`
`Snippet: ...YPE ECDSA_SECP256R1_SHA256 // signature algori...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:168`
`Snippet: #if SIGTYPE==ECCSHA384_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:172`
`Snippet: ...YPE ECDSA_SECP384R1_SHA384`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:174`
`Snippet: #if SIGTYPE==RSASHA256_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:178`
`Snippet: ...SIG_TYPE RSA_PKCS1_SHA256`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:180`
`Snippet: #if SIGTYPE==RSASHA384_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:184`
`Snippet: ...SIG_TYPE RSA_PKCS1_SHA384`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:186`
`Snippet: #if SIGTYPE==RSASHA512_SIG`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: makecertchain/makeselfcert.cpp:190`
`Snippet: ...SIG_TYPE RSA_PKCS1_SHA512`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeselfcert.cpp:204`
`Snippet: #if SIGTYPE==MLDSA65_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeselfcert.cpp:208`
`Snippet: #define SIG_TYPE MLDSA65`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeselfcert.cpp:211`
`Snippet: #if SIGTYPE==MLDSA44_SIG`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeselfcert.cpp:216`
`Snippet: #define SIG_TYPE MLDSA44`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: makecertchain/makeselfcert.cpp:219`

Snippet: ...if SIGTYPE==ED25519_MLDSA44_SIG
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeselfcert.cpp:227
Snippet: #define SIG_TYPE_2 MLDSA44
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeselfcert.cpp:473
Snippet: // MLDSA or EDDSA
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeselfcert.cpp:474
Snippet: #if PKTYPE==MLDSA65_PK || PKTYPE==MLD...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeselfcert.cpp:495
Snippet: #if PKTYPE==ED25519_MLDSA44_PK
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeselfcert.cpp:714
Snippet: ...if SIGTYPE==ED25519_MLDSA44_SIG
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: makecertchain/makeselfcert.cpp:736
Snippet: #if SIGTYPE==ECCSHA256_SIG || SIGTYPE==ECC...
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: makecertchain/makeselfcert.cpp:788
Snippet: #if PKTYPE==MLDSA65_PK || PKTYPE==MLD...
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/config.rs:11
Snippet: /// AES128/SHA256/GCM cipher suite
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/config.rs:12
Snippet: ...b const AES_128_GCM_SHA256: u16 = 0x1301;...
Remediation: No action needed.

[INFO] AES256_USAGE

AES-256 symmetric encryption — considered quantum-resistant at this key size (Grover's algorithm only halves effective security).

```
Location: rust/server/src/config.rs:13
Snippet: /// AES256/SHA384/GCM cipher s...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/config.rs:13
Snippet: /// AES256/SHA384/GCM cipher suite
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/config.rs:14
Snippet: ...b const AES_256_GCM_SHA384: u16 = 0x1302;...
```

Remediation: No action needed.

[INFO] CHACHA20_USAGE

ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.

```
Location: rust/server/src/config.rs:15
Snippet: /// CHACHA20/SHA256/POLY1305 cip...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/config.rs:15
Snippet: /// CHACHA20/SHA256/POLY1305 cipher sui...
```

Remediation: No action needed.

[INFO] CHACHA20_USAGE

ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.

```
Location: rust/server/src/config.rs:16
Snippet: pub const CHACHA20_POLY1305_SHA256: u1...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/config.rs:16
Snippet: ...t CHACHA20_POLY1305_SHA256: u16 = 0x1303;
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/config.rs:17
Snippet: ...b const AES_128_CCM_SHA256: u16 = 0x1304;...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/config.rs:18
Snippet: ...const AES_128_CCM_8_SHA256: u16 = 0x1305;...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/config.rs:22`
`Snippet: ...nst ECDSA_SECP256R1_SHA256: u16 = 0x0403;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/config.rs:24`
`Snippet: ...nst ECDSA_SECP384R1_SHA384: u16 = 0x0503;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/config.rs:26`
`Snippet: ...const RSA_PSS_RSAE_SHA256: u16 = 0x0804;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/config.rs:28`
`Snippet: ...const RSA_PSS_RSAE_SHA384: u16 = 0x0805;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/config.rs:30`
`Snippet: ...const RSA_PSS_RSAE_SHA512: u16 = 0x0806;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/config.rs:32`
`Snippet: pub const RSA_PKCS1_SHA256: u16 = 0x0401;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/config.rs:34`
`Snippet: pub const RSA_PKCS1_SHA384: u16 = 0x0501;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/config.rs:36`
`Snippet: pub const RSA_PKCS1_SHA512: u16 = 0x0601;`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/config.rs:37`
`Snippet: /// Dilithium3 Signature algorith...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

Location: rust/server/src/config.rs:38
Snippet: pub const MLDSA65: u16 = 0x0905;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/config.rs:39
Snippet: /// Dilithium2 Signature algorith...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/config.rs:40
Snippet: pub const MLDSA44: u16 = 0x0904;
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/config.rs:42
Snippet: ...nst ECDSA_SECP256R1_SHA384: u16 = 0x0413;
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/config.rs:43
Snippet: /// Dilithium2 Hybrid Signature a...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/config.rs:44
Snippet: pub const MLDSA44_ED25519: u16 = 0x...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/config.rs:101
Snippet: ...ure Public key size Dilithium 3
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/config.rs:102
Snippet: ...ure Public key size Dilithium 3
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/config.rs:103
Snippet: ...ure size in bytes - Dilithium 3
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/server/src/config.rs:159
Snippet: pub const MLKEM768: u16 = 0x0201;

Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/config.rs:236`
`Snippet: pub const SHA256_T: usize = 1;...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/config.rs:237`
`Snippet: pub const SHA384_T: usize = 2;...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/config.rs:238`
`Snippet: pub const SHA512_T: usize = 3;...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/main.rs:36`
`Snippet: AES_128_GCM_SHA256 => println!("TLS_AE...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/main.rs:37`
`Snippet: AES_256_GCM_SHA384 => println!("TLS_AE...`
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
`Location: rust/server/src/main.rs:38`
`Snippet: CHACHA20_POLY1305_SHA256 =>...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/main.rs:38`
`Snippet: ...CHACHA20_POLY1305_SHA256 => println!("TLS_CH...`
Remediation: No action needed.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/main.rs:46`
`Snippet: ...RID_KX => println!("MLKEM768X25519"),`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/main.rs:50`
`Snippet: MLKEM768 => println!("MLK...`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/main.rs:59
Snippet: ...ECDSA_SECP256R1_SHA256 => println!("ECDSA_...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/main.rs:60
Snippet: ...RSA_PSS_RSAE_SHA256 => println!("RSA_PS...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/main.rs:61
Snippet: RSA_PKCS1_SHA256 => println!("RSA_PK...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/main.rs:62
Snippet: ...ECDSA_SECP384R1_SHA384 => println!("ECDSA_...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/main.rs:63
Snippet: ...RSA_PSS_RSAE_SHA384 => println!("RSA_PS...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/main.rs:64
Snippet: RSA_PKCS1_SHA384 => println!("RSA_PK...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/main.rs:65
Snippet: ...RSA_PSS_RSAE_SHA512 => println!("RSA_PS...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/main.rs:66
Snippet: RSA_PKCS1_SHA512 => println!("RSA_PK...
```

Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: rust/server/src/main.rs:67
Snippet: MLDSA65 => println!("MLDS...
```

Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

Location: rust/server/src/sal_m/bfibe.rs:6
Snippet: use mcore::sha3;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/bfibe.rs:7
Snippet: use mcore::sha3::SHA3;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/bfibe.rs:65
Snippet: ...dst="BLS12381G1_XMD:SHA-256_SVDW_NU_BFIBE".as_b...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/bfibe.rs:71
Snippet: ...hmac::xof_expand(sha3::SHAKE128,&mut; okm,...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/bfibe.rs:87
Snippet: let mut sh = SHA3::new(sha3::SHAKE256...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/bfibe.rs:105
Snippet: let mut sh = SHA3::new(sha3::HASH256)...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/bfibe.rs:121
Snippet: let mut sh = SHA3::new(sha3::SHAKE256...
Remediation: No action needed.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/server/src/sal_m/sal.rs:66
Snippet: ...if group==config::MLKEM768 {
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/server/src/sal_m/sal.rs:67
Snippet: ...return mcore::kyber::SECRET_CCA_SIZE_76...
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/server/src/sal_m/sal.rs:70
Snippet: ...return mcore::kyber::SECRET_CCA_SIZE_76...

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:87`
`Snippet: ...if group==config::MLKEM768 {`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:88`
`Snippet: ...return mcore::kyber::PUBLIC_SIZE_768;...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:91`
`Snippet: ...return mcore::kyber::PUBLIC_SIZE_768+32...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:107`
`Snippet: ...if group==config::MLKEM768 {`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:108`
`Snippet: ...return mcore::kyber::CIPHERTEXT_SIZE_76...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:111`
`Snippet: ...return mcore::kyber::CIPHERTEXT_SIZE_76...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:127`
`Snippet: ...if group==config::MLKEM768 {`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:128`
`Snippet: ...return mcore::kyber::SHARED_SECRET_768;...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:131`
`Snippet: ...return mcore::kyber::SHARED_SECRET_768+...`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:139`
`Snippet: ...config::AES_128_GCM_SHA256;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:140`
`Snippet: ...config::AES_256_GCM_SHA384;`
Remediation: No action needed.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:151`
`Snippet: ...groups[3]=config::MLKEM768;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:159`
`Snippet: ...ig::ECDSA_SECP256R1_SHA256;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:160`
`Snippet: ...ig::ECDSA_SECP384R1_SHA384;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:161`
`Snippet: ...onfig::RSA_PSS_RSAE_SHA256;`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:165`
`Snippet: ...sig_algs[6]=config::MLDSA65;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:166`
`Snippet: ...sig_algs[7]=config::MLDSA44;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:167`
`Snippet: ...sig_algs[8]=config::MLDSA44_ED25519;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

Location: rust/server/src/sal_m/sal.rs:168
Snippet: ...ig::ECDSA_SECP256R1_SHA384;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:175
Snippet: ...ig::ECDSA_SECP256R1_SHA256;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:176
Snippet: ...ig::ECDSA_SECP384R1_SHA384;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:177
Snippet: ...]=config::RSA_PKCS1_SHA256;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:178
Snippet: ...]=config::RSA_PKCS1_SHA384;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:179
Snippet: ...]=config::RSA_PKCS1_SHA512;
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/sal_m/sal.rs:183
Snippet: ...lgs_cert[8]=config::MLDSA65;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/sal_m/sal.rs:184
Snippet: ...lgs_cert[9]=config::MLDSA44;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/sal_m/sal.rs:185
Snippet: ...gs_cert[10]=config::MLDSA44_ED25519;
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:186
Snippet: ...ig::ECDSA_SECP256R1_SHA384;

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/sal_m/sal.rs:193
Snippet: ...config::AES_128_GCM_SHA256 {htype=config::SHA2...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/sal_m/sal.rs:194
Snippet: ...config::AES_256_GCM_SHA384 {htype=config::SHA3...
```

Remediation: No action needed.

[INFO] CHACHA20_USAGE

ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.

```
Location: rust/server/src/sal_m/sal.rs:195
Snippet: ...pher_suite==config::CHACHA20_POLY1305_SHA256 {ht...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/sal_m/sal.rs:195
Snippet: ...::CHACHA20_POLY1305_SHA256 {htype=config::SHA2...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/sal_m/sal.rs:202
Snippet: ...hash_type==config::SHA256_T {hlen=32;}
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/sal_m/sal.rs:203
Snippet: ...hash_type==config::SHA384_T {hlen=48;}
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/sal_m/sal.rs:204
Snippet: ...hash_type==config::SHA512_T {hlen=64;}
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/sal_m/sal.rs:211
Snippet: ...config::AES_128_GCM_SHA256 {
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/sal_m/sal.rs:214
Snippet: ...config::AES_256_GCM_SHA384 {
```

Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
`Location: rust/server/src/sal_m/sal.rs:217`
`Snippet: ...pher_suite==config::CHACHA20_POLY1305_SHA256 {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:217`
`Snippet: ...::CHACHA20_POLY1305_SHA256 {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:226`
`Snippet: ...config::AES_128_GCM_SHA256 {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:229`
`Snippet: ...config::AES_256_GCM_SHA384 {`
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
`Location: rust/server/src/sal_m/sal.rs:232`
`Snippet: ...pher_suite==config::CHACHA20_POLY1305_SHA256 {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:232`
`Snippet: ...::CHACHA20_POLY1305_SHA256 {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:262`
`Snippet: ...if htype==config::SHA256_T {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:267`
`Snippet: ...if htype==config::SHA384_T {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:272`
`Snippet: ...if htype==config::SHA512_T {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

Location: rust/server/src/sal_m/sal.rs:283
Snippet: ...if htype==config::SHA256_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:287
Snippet: ...if htype==config::SHA384_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:291
Snippet: ...if htype==config::SHA512_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:301
Snippet: ...if h.htype==config::SHA256_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:307
Snippet: ...if h.htype==config::SHA384_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:313
Snippet: ...if h.htype==config::SHA512_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:325
Snippet: ...if h.htype==config::SHA256_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:331
Snippet: ...if h.htype==config::SHA384_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/sal_m/sal.rs:337
Snippet: ...if h.htype==config::SHA512_T {
Remediation: No action needed.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/server/src/sal_m/sal.rs:390
Snippet: ...if group==config::MLKEM768 {

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:391`
`Snippet: use mcore::kyber;`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:394`
`Snippet: kyber::keypair_768(&r64;,&...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:397`
`Snippet: use mcore::kyber; // first...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:400`
`Snippet: kyber::keypair_768(&r64;,&...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:403`
`Snippet: ...et_key_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:404`
`Snippet: ...ic_key_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:437`
`Snippet: ...if group==config::MLKEM768 {`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:438`
`Snippet: use mcore::kyber;`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:441`
`Snippet: kyber::encrypt_768(&r32;,&...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:446`
`Snippet: use mcore::kyber;`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:449`
`Snippet: kyber::encrypt_768(&r32;,&...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:453`
`Snippet: ...ic_key_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:454`
`Snippet: ...ic_key_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:455`
`Snippet: ...secret_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:490`
`Snippet: ...if group==config::MLKEM768 {`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:491`
`Snippet: use mcore::kyber;`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:492`
`Snippet: kyber::decrypt_768(&sk;[0....`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/server/src/sal_m/sal.rs:496`
`Snippet: use mcore::kyber;`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).

Location: rust/server/src/sal_m/sal.rs:497
Snippet: kyber::decrypt_768(&sk;[0....
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/server/src/sal_m/sal.rs:499
Snippet: ...et_key_size(config::MLKEM768);
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/server/src/sal_m/sal.rs:500
Snippet: ...ic_key_size(config::MLKEM768);
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/server/src/sal_m/sal.rs:501
Snippet: ...secret_size(config::MLKEM768);
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/sal_m/sal.rs:515
Snippet: /// Dilithium signature verificat...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/sal_m/sal.rs:516
Snippet: fn dilithium3_verify(cert: &[u8]...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/sal_m/sal.rs:517
Snippet: use mcore::dilithium;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/sal_m/sal.rs:518
Snippet: return dilithium::verify_3(false,&pu...;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/sal_m/sal.rs:521
Snippet: /// Dilithium signature
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/sal_m/sal.rs:522
Snippet: pub fn dilithium3_sign(key: &[u8],me...

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:523`
`Snippet: use mcore::dilithium;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:524`
`Snippet: dilithium::signature_3(false,...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:525`
`Snippet: return dilithium::SIG_SIZE_3;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:528`
`Snippet: /// Dilithium signature verificat...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:529`
`Snippet: fn dilithium2_verify(cert: &[u8]...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:530`
`Snippet: use mcore::dilithium;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:531`
`Snippet: return dilithium::verify_2(false,&pu...;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:534`
`Snippet: /// Dilithium signature`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:535`
`Snippet: pub fn dilithium2_sign(key: &[u8],me...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:536`
`Snippet: use mcore::dilithium;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:537`
`Snippet: dilithium::signature_2(false,...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:538`
`Snippet: return dilithium::SIG_SIZE_2;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:749`
`Snippet: ...rsa_pkcs1_sha256 (for certificates),...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:750`
`Snippet: ...and ecdsa_secp256r1_sha256."`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:754`
`Snippet: ...ig::ECDSA_SECP256R1_SHA256 => {return secp256r...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:755`
`Snippet: ...ig::ECDSA_SECP256R1_SHA384 => {return secp256r...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:756`
`Snippet: ...ig::ECDSA_SECP384R1_SHA384 => {return secp384r...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/sal.rs:757`
`Snippet: ...config::RSA_PKCS1_SHA256 => {return rsa_pkcs...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/sal_m/sal.rs:758
Snippet: ...config::RSA_PKCS1_SHA384 => {return rsa_pkcs...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/server/src/sal_m/sal.rs:759
Snippet: ...config::RSA_PKCS1_SHA512 => {return rsa_pkcs...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/server/src/sal_m/sal.rs:760
Snippet: ...onfig::RSA_PSS_RSAE_SHA256 => {return rsa_pss_...
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/server/src/sal_m/sal.rs:764
Snippet: config::MLDSA65 => {return dilith...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/server/src/sal_m/sal.rs:765
Snippet: config::MLDSA44 => {return dilith...
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/server/src/sal_m/sal.rs:774
Snippet: ...g:: ECDSA_SECP256R1_SHA256 => {return secp256r...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/server/src/sal_m/sal.rs:775
Snippet: ...g:: ECDSA_SECP256R1_SHA384 => {return secp256r...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/server/src/sal_m/sal.rs:776
Snippet: ...g:: ECDSA_SECP384R1_SHA384 => {return secp384r...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/server/src/sal_m/sal.rs:777
Snippet: ...nfig:: RSA_PSS_RSAE_SHA256 => {return rsa_pss_...
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/server/src/sal_m/sal.rs:781
Snippet: config:: MLDSA65 => {return dilith...
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/sal_m/sal.rs:782`
`Snippet: config:: MLDSA44 => {return dilith...`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/pqibe.rs:6`
`Snippet: use mcore::sha3;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/pqibe.rs:7`
`Snippet: use mcore::sha3::SHA3;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/pqibe.rs:223`
`Snippet: let mut sh = SHA3::new(sha3::SHAKE128...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/pqibe.rs:259`
`Snippet: let mut sh = SHA3::new(sha3::HASH256)...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/pqibe.rs:389`
`Snippet: let mut sh = SHA3::new(sha3::SHAKE256...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/pqibe.rs:404`
`Snippet: sh = SHA3::new(sha3::SHAKE256...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/pqibe.rs:420`
`Snippet: sh = SHA3::new(sha3::SHAKE256...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/pqibe.rs:499`
`Snippet: let mut sh = SHA3::new(sha3::SHAKE256...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/pqibe.rs:509`
`Snippet: sh = SHA3::new(sha3::HASH256)...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/sal_m/pqibe.rs:530`
`Snippet: let mut sh = SHA3::new(sha3::HASH256)...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/keys.rs:110`
`Snippet: ...f.suite=AES_128_GCM_SHA256;`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/keys.rs:204`
`Snippet: // *** if sigalg==MLDSA44_ED25519, extract...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/keys.rs:205`
`Snippet: if sigalg==MLDSA44_ED25519 {`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/keys.rs:209`
`Snippet: ...=sal::tls_signature(MLDSA44,&key;[32..],&scv;[0...`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/keys.rs:214`
`Snippet: ...lg==ECDSA_SECP256R1_SHA256 || sigalg==ECDSA_SE...`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/keys.rs:230`
`Snippet: // *** if sigalg==MLDSA44_ED25519, extract...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/keys.rs:231`
`Snippet: if sigalg==MLDSA44_ED25519 {`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

Location: rust/server/src/tls13/keys.rs:240
Snippet: ...ls_signature_verify(MLDSA44,&ccv;[0..ptr],&ccv...;
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/tls13/keys.rs:245
Snippet: ...lg==ECDSA_SECP256R1_SHA256 || sigalg==ECDSA_SE...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/tls13/logger.rs:49
Snippet: AES_128_GCM_SHA256 => log(IO_DEBUG,"AE...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/tls13/logger.rs:50
Snippet: AES_256_GCM_SHA384 => log(IO_DEBUG,"AE...
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
Location: rust/server/src/tls13/logger.rs:51
Snippet: CHACHA20_POLY1305_SHA256 =>...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/tls13/logger.rs:51
Snippet: ...CHACHA20_POLY1305_SHA256 => log(IO_DEBUG,"CH...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/tls13/logger.rs:60
Snippet: ...ECDSA_SECP256R1_SHA256 => log(IO_PROTOCOL,...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/tls13/logger.rs:61
Snippet: ...RSA_PSS_RSAE_SHA256 => log(IO_PROTOCOL,...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/tls13/logger.rs:62
Snippet: RSA_PKCS1_SHA256 => log(IO_PROTOCOL,...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/tls13/logger.rs:63
Snippet: ...ECDSA_SECP384R1_SHA384 => log(IO_PROTOCOL,...

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/tls13/logger.rs:64
Snippet: ...RSA_PSS_RSAE_SHA384 => log(IO_PROTOCOL,...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/tls13/logger.rs:65
Snippet: RSA_PKCS1_SHA384 => log(IO_PROTOCOL,...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/tls13/logger.rs:66
Snippet: ...RSA_PSS_RSAE_SHA512 => log(IO_PROTOCOL,...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/tls13/logger.rs:67
Snippet: RSA_PKCS1_SHA512 => log(IO_PROTOCOL,...
```

Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: rust/server/src/tls13/logger.rs:68
Snippet: MLDSA65 => log(IO_PROTOCO...
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: rust/server/src/tls13/logger.rs:69
Snippet: MLDSA44 => log(IO_PROTOCO...
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: rust/server/src/tls13/logger.rs:70
Snippet: MLDSA44_ED25519 => log(IO...
```

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

```
Location: rust/server/src/tls13/logger.rs:84
Snippet: MLKEM768 => log(IO_PROTOC...
```

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

```
Location: rust/server/src/tls13/logger.rs:85
Snippet: ...IO_PROTOCOL,"HYBRID MLKEM768+X25519\n",-1,Non...
```

Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/logger.rs:110`
`Snippet: ...OTOCOL,"Hashed with SHA256\n",-1,None);}`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/logger.rs:111`
`Snippet: ...OTOCOL,"Hashed with SHA384\n",-1,None);}`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/logger.rs:112`
`Snippet: ...OTOCOL,"Hashed with SHA512\n",-1,None);}`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/logger.rs:127`
`Snippet: ...re is Post Quantum (MLDSA) of length (bits) "...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/logger.rs:130`
`Snippet: ...ignature is Hybrid (MLDSA44+ED25519) of lengt...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/logger.rs:159`
`Snippet: ...ey is Post Quantum (MLDSA) of length (bits) "...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/logger.rs:162`
`Snippet: ...blic key is Hybrid (MLDSA44+ED25519) of lengt...`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/servercert.rs:20`
`Snippet: .../ Uses RSA 2048 bit SHA256`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/servercert.rs:115`
`Snippet: ...ntal hybrid ED25519_MLDSA44 Server key + Inte...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

Location: rust/server/src/tls13/servercert.rs:347
Snippet: // Experimental MLDSA65 Server key + Serv...
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/tls13/servercert.rs:674
Snippet: ...urn ECDSA_SECP256R1_SHA256; // as long as this...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/tls13/servercert.rs:677
Snippet: ...urn ECDSA_SECP384R1_SHA384; // as long as thi...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/tls13/servercert.rs:681
Snippet: ...return RSA_PSS_RSAE_SHA256;
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/servercert.rs:684
Snippet: ...pk.curve==x509::USE_MLDSA65 {
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/servercert.rs:685
Snippet: return MLDSA65;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/servercert.rs:687
Snippet: ...pk.curve==x509::USE_MLDSA44 {
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/servercert.rs:688
Snippet: return MLDSA44;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/servercert.rs:692
Snippet: return MLDSA44_ED25519;
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/server/src/tls13/servercert.rs:710
Snippet: ...en]=ECDSA_SECP256R1_SHA256; // as long as this...

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/tls13/servercert.rs:714
Snippet: ...en]=ECDSA_SECP384R1_SHA384; // as long as thi...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/tls13/servercert.rs:721
Snippet: ...ents[len]=RSA_PKCS1_SHA256;
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/tls13/servercert.rs:725
Snippet: ...ents[len]=RSA_PKCS1_SHA384;
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/tls13/servercert.rs:729
Snippet: ...ents[len]=RSA_PKCS1_SHA512;
```

Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: rust/server/src/tls13/servercert.rs:736
Snippet: ...pk.curve==x509::USE_MLDSA65 {
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: rust/server/src/tls13/servercert.rs:737
Snippet: ...requirements[len]=MLDSA65;
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: rust/server/src/tls13/servercert.rs:739
Snippet: ...pk.curve==x509::USE_MLDSA44 {
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: rust/server/src/tls13/servercert.rs:740
Snippet: ...requirements[len]=MLDSA44;
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: rust/server/src/tls13/servercert.rs:746
Snippet: ...requirements[len]=MLDSA44_ED25519;
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/servercert.rs:748`
`Snippet: ...requirements[len]=MLDSA44;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/x509.rs:41`
`Snippet: pub const USE_MLDSA44:usize = 13; // F...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/x509.rs:42`
`Snippet: pub const USE_MLDSA65:usize = 14 ; // F...`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/x509.rs:63`
`Snippet: ...Elliptic Curve with SHA256`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/x509.rs:65`
`Snippet: const ECCSHA256:[u8;8]=[0x2a, 0x86,...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/x509.rs:66`
`Snippet: const ECCSHA384:[u8;8]=[0x2a, 0x86,...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/x509.rs:67`
`Snippet: const ECCSHA512:[u8;8]=[0x2a, 0x86,...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/x509.rs:76`
`Snippet: const RSASHA256:[u8;9]=[0x2a, 0x86,...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/x509.rs:77`
`Snippet: const RSASHA384:[u8;9]=[0x2a, 0x86,...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

Location: rust/server/src/tls13/x509.rs:78
Snippet: const RSASHA512:[u8;9]=[0x2a, 0x86,...
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/x509.rs:79
Snippet: //const MLDSA65:[u8;11]=[0x2b, 0x...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/x509.rs:81
Snippet: const MLDSA65:[u8;9]=[0x60,0x86...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/x509.rs:82
Snippet: const MLDSA44:[u8;9]=[0x60,0x86...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/x509.rs:83
Snippet: ...0x0F,0x0C,0x06]; // MLDSA44 + ED25519
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/x509.rs:382
Snippet: if MLDSA65 == soid[0..slen]...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/x509.rs:416
Snippet: if MLDSA65 == soid[0..slen]...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/x509.rs:417
Snippet: ...ret.curve=USE_MLDSA65;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/x509.rs:419
Snippet: if MLDSA44 == soid[0..slen]...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/server/src/tls13/x509.rs:420
Snippet: ...ret.curve=USE_MLDSA44;

Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/x509.rs:711`
`Snippet: if ECCSHA256 == soid[0..slen] {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/x509.rs:715`
`Snippet: if ECCSHA384 == soid[0..slen] {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/x509.rs:719`
`Snippet: if ECCSHA512 == soid[0..slen] {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/x509.rs:723`
`Snippet: if RSASHA256 == soid[0..slen] {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/x509.rs:727`
`Snippet: if RSASHA384 == soid[0..slen] {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/x509.rs:731`
`Snippet: if RSASHA512 == soid[0..slen] {`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/x509.rs:735`
`Snippet: if MLDSA65 == soid[0..slen]...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/x509.rs:738`
`Snippet: ...ret.curve=USE_MLDSA65;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/x509.rs:740`
`Snippet: if MLDSA44 == soid[0..slen]...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/x509.rs:743`
`Snippet: ...ret.curve=USE_MLDSA44;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/x509.rs:1149`
`Snippet: if MLDSA65 == koid[0..slen]...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/x509.rs:1150`
`Snippet: ...=DLM; ret.curve=USE_MLDSA65;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/x509.rs:1152`
`Snippet: if MLDSA44 == koid[0..slen]...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/server/src/tls13/x509.rs:1153`
`Snippet: ...=DLM; ret.curve=USE_MLDSA44;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/certchain.rs:110`
`Snippet: ...ify(ECDSA_SECP256R1_SHA256,cert,sig,pubkey);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/certchain.rs:113`
`Snippet: ...ify(ECDSA_SECP384R1_SHA384,cert,sig,pubkey);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/certchain.rs:122`
`Snippet: ...re_verify(RSA_PKCS1_SHA256,cert,sig,pubkey);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/server/src/tls13/certchain.rs:125`
`Snippet: ...re_verify(RSA_PKCS1_SHA384,cert,sig,pubkey);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

```
Location: rust/server/src/tls13/certchain.rs:128
Snippet: ...re_verify(RSA_PKCS1_SHA512,cert,sig,pubkey);
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/server/src/tls13/certchain.rs:130
Snippet: ...st.curve==x509::USE_MLDSA65 {
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/server/src/tls13/certchain.rs:131
Snippet: ...ls_signature_verify(MLDSA65,cert,sig,pubkey);...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/server/src/tls13/certchain.rs:133
Snippet: ...st.curve==x509::USE_MLDSA44 {
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/server/src/tls13/certchain.rs:134
Snippet: ...ls_signature_verify(MLDSA44,cert,sig,pubkey);...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/server/src/tls13/certchain.rs:143
Snippet: ...ls_signature_verify(MLDSA44,cert,sig2,pub2);
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/server/src/tls13/connection.rs:163
Snippet: ...: 0, //AES_128_GCM_SHA256,
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/server/src/tls13/connection.rs:1576
Snippet: ...uite != AES_128_GCM_SHA256 {
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/config.rs:11
Snippet: /// AES128/SHA256/GCM cipher suite
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/config.rs:12
Snippet: ...b const AES_128_GCM_SHA256: u16 = 0x1301;...
```

Remediation: No action needed.

[INFO] AES256_USAGE
AES-256 symmetric encryption — considered quantum-resistant at this key size (Grover's algorithm only halves effective security).
`Location: rust/client/src/config.rs:13`
`Snippet: /// AES256/SHA384/GCM cipher s...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:13`
`Snippet: /// AES256/SHA384/GCM cipher suite`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:14`
`Snippet: ...b const AES_256_GCM_SHA384: u16 = 0x1302;...`
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
`Location: rust/client/src/config.rs:15`
`Snippet: /// CHACHA20/SHA256/POLY1305 cip...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:15`
`Snippet: /// CHACHA20/SHA256/POLY1305 cipher sui...`
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
`Location: rust/client/src/config.rs:16`
`Snippet: pub const CHACHA20_POLY1305_SHA256: u1...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:16`
`Snippet: ...t CHACHA20_POLY1305_SHA256: u16 = 0x1303;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:17`
`Snippet: // AES/SHA256/CCM cipher suite -...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:18`
`Snippet: ...b const AES_128_CCM_SHA256: u16 = 0x1304;...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:19`
`Snippet: // AES/SHA256/CCM 8 cipher suite...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:20`
`Snippet: ...const AES_128_CCM_8_SHA256: u16 = 0x1305;...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:24`
`Snippet: ...nst ECDSA_SECP256R1_SHA256: u16 = 0x0403;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:26`
`Snippet: ...nst ECDSA_SECP384R1_SHA384: u16 = 0x0503;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:28`
`Snippet: ...const RSA_PSS_RSAE_SHA256: u16 = 0x0804;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:30`
`Snippet: ...const RSA_PSS_RSAE_SHA384: u16 = 0x0805;...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:32`
`Snippet: ...const RSA_PSS_RSAE_SHA512: u16 = 0x0806;...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:34`
`Snippet: pub const RSA_PKCS1_SHA256: u16 = 0x0401;...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:36`
`Snippet: pub const RSA_PKCS1_SHA384: u16 = 0x0501;...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:38`
`Snippet: pub const RSA_PKCS1_SHA512: u16 = 0x0601;`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/config.rs:39`
`Snippet: /// Dilithium3 Signature algorith...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/config.rs:40`
`Snippet: pub const MLDSA65: u16 = 0x0905;...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/config.rs:41`
`Snippet: /// Dilithium2 Signature algorith...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/config.rs:42`
`Snippet: pub const MLDSA44: u16 = 0x0904;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/config.rs:44`
`Snippet: ...nst ECDSA_SECP256R1_SHA384: u16 = 0x0413;`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/config.rs:45`
`Snippet: /// Dilithium2 Hybrid Signature a...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/config.rs:46`
`Snippet: pub const MLDSA44_ED25519: u16 = 0x...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/config.rs:91`
`Snippet: /// self signed MLDSA cert`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/config.rs:93`

Snippet: ...signed Hybrid cert (MLDSA+ECC)
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/config.rs:113
Snippet: ...ure Public key size Dilithium 3
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/config.rs:114
Snippet: ...ure Public key size Dilithium 3
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/config.rs:115
Snippet: ...ure size in bytes - Dilithium 3
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/client/src/config.rs:169
Snippet: pub const MLKEM768: u16 = 0x0201;
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/config.rs:243
Snippet: pub const SHA256_T: usize = 1;...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/config.rs:244
Snippet: pub const SHA384_T: usize = 2;...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/config.rs:245
Snippet: pub const SHA512_T: usize = 3;...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/main.rs:164
Snippet: AES_128_GCM_SHA256 => println!("TLS_AE...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/main.rs:165
Snippet: AES_256_GCM_SHA384 => println!("TLS_AE...
Remediation: No action needed.

[INFO] CHACHA20_USAGE

ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.

`Location: rust/client/src/main.rs:166`

`Snippet: CHACHA20_POLY1305_SHA256 =>...`

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

`Location: rust/client/src/main.rs:166`

`Snippet: ...CHACHA20_POLY1305_SHA256 => println!("TLS_CH...`

Remediation: No action needed.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

`Location: rust/client/src/main.rs:174`

`Snippet: ...RID_KX => println!("MLKEM768X25519"),`

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

`Location: rust/client/src/main.rs:178`

`Snippet: MLKEM768 => println!("MLK...`

Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

`Location: rust/client/src/main.rs:187`

`Snippet: ...ECDSA_SECP256R1_SHA256 => println!("ECDSA_...`

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

`Location: rust/client/src/main.rs:188`

`Snippet: ...RSA_PSS_RSAE_SHA256 => println!("RSA_PS...`

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

`Location: rust/client/src/main.rs:189`

`Snippet: RSA_PKCS1_SHA256 => println!("RSA_PK...`

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

`Location: rust/client/src/main.rs:190`

`Snippet: ...ECDSA_SECP384R1_SHA384 => println!("ECDSA_...`

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

`Location: rust/client/src/main.rs:191`

`Snippet: ...RSA_PSS_RSAE_SHA384 => println!("RSA_PS...`

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.
`Location: rust/client/src/main.rs:192`
`Snippet: RSA_PKCS1_SHA384 => println!("RSA_PK...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/main.rs:193`
`Snippet: ...RSA_PSS_RSAE_SHA512 => println!("RSA_PS...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/main.rs:194`
`Snippet: RSA_PKCS1_SHA512 => println!("RSA_PK...`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/main.rs:195`
`Snippet: MLDSA65 => println!("MLDS...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/main.rs:394`
`Snippet: ...n.t.favourite_group=MLKEM768;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/main.rs:409`
`Snippet: ...r_suite=AES_128_GCM_SHA256;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/bfibe.rs:6`
`Snippet: use mcore::sha3;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/bfibe.rs:7`
`Snippet: use mcore::sha3::SHA3;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/bfibe.rs:64`
`Snippet: ...dst="BLS12381G1_XMD:SHA-256_SVDW_NU_BFIBE".as_b...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/bfibe.rs:70`

Snippet: ...hmac::xof_expand(sha3::SHAKE128,&mut; okm,...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/bfibe.rs:86
Snippet: let mut sh = SHA3::new(sha3::SHAKE256...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/bfibe.rs:104
Snippet: let mut sh = SHA3::new(sha3::HASH256)...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/bfibe.rs:119
Snippet: let mut sh = SHA3::new(sha3::SHAKE256...
Remediation: No action needed.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/client/src/sal_m/sal.rs:68
Snippet: ...return mcore::kyber::SECRET_CCA_SIZE_76...
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/client/src/sal_m/sal.rs:71
Snippet: ...if group==config::MLKEM768 {
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/client/src/sal_m/sal.rs:72
Snippet: ...return mcore::kyber::SECRET_CCA_SIZE_76...
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/client/src/sal_m/sal.rs:91
Snippet: ...return mcore::kyber::PUBLIC_SIZE_768+32...
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/client/src/sal_m/sal.rs:94
Snippet: ...if group==config::MLKEM768 {
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/client/src/sal_m/sal.rs:95
Snippet: ...return mcore::kyber::PUBLIC_SIZE_768;...
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

`Location: rust/client/src/sal_m/sal.rs:113`

`Snippet: ...return mcore::kyber::CIPHERTEXT_SIZE_76...`

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

`Location: rust/client/src/sal_m/sal.rs:116`

`Snippet: ...if group==config::MLKEM768 {`

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

`Location: rust/client/src/sal_m/sal.rs:117`

`Snippet: ...return mcore::kyber::CIPHERTEXT_SIZE_76...`

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

`Location: rust/client/src/sal_m/sal.rs:136`

`Snippet: ...return mcore::kyber::SHARED_SECRET_768+...`

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

`Location: rust/client/src/sal_m/sal.rs:137`

`Snippet: ...new(kem::Algorithm::Kyber768).unwrap();`

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

`Location: rust/client/src/sal_m/sal.rs:141`

`Snippet: ...if group==config::MLKEM768 {`

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

`Location: rust/client/src/sal_m/sal.rs:142`

`Snippet: ...return mcore::kyber::SHARED_SECRET_768;...`

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

`Location: rust/client/src/sal_m/sal.rs:143`

`Snippet: ...new(kem::Algorithm::Kyber768).unwrap();`

Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

`Location: rust/client/src/sal_m/sal.rs:152`

`Snippet: ...config::AES_128_GCM_SHA256;`

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:153`
`Snippet: ...config::AES_256_GCM_SHA384;`
Remediation: No action needed.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:178`
`Snippet: ...groups[1]=config::MLKEM768;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:190`
`Snippet: ...ig::ECDSA_SECP256R1_SHA256;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:191`
`Snippet: ...ig::ECDSA_SECP384R1_SHA384;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:194`
`Snippet: ...onfig::RSA_PSS_RSAE_SHA256; n+=1;`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:200`
`Snippet: ...sig_algs[n]=config::MLDSA65; n+=1;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:201`
`Snippet: ...sig_algs[n]=config::MLDSA44; n+=1;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:202`
`Snippet: ...sig_algs[n]=config::MLDSA44_ED25519; n+=1;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:203`
`Snippet: ...ig::ECDSA_SECP256R1_SHA384; n+=1;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:211`

Snippet: ...ig::ECDSA_SECP256R1_SHA256;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:212
Snippet: ...ig::ECDSA_SECP384R1_SHA384;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:215
Snippet: ...]=config::RSA_PKCS1_SHA256; n+=1;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:216
Snippet: ...]=config::RSA_PKCS1_SHA384; n+=1;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:217
Snippet: ...]=config::RSA_PKCS1_SHA512; n+=1;
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/sal_m/sal.rs:223
Snippet: ...lgs_cert[n]=config::MLDSA65; n+=1;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/sal_m/sal.rs:224
Snippet: ...lgs_cert[n]=config::MLDSA44; n+=1;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/sal_m/sal.rs:225
Snippet: ...lgs_cert[n]=config::MLDSA44_ED25519; n+=1;
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:226
Snippet: ...ig::ECDSA_SECP256R1_SHA384; n+=1;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:234
Snippet: ...config::AES_128_GCM_SHA256 {htype=config::SHA2...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:235`
`Snippet: ...config::AES_256_GCM_SHA384 {htype=config::SHA3...`
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
`Location: rust/client/src/sal_m/sal.rs:236`
`Snippet: ...pher_suite==config::CHACHA20_POLY1305_SHA256 {ht...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:236`
`Snippet: ...::CHACHA20_POLY1305_SHA256 {htype=config::SHA2...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:243`
`Snippet: ...hash_type==config::SHA256_T {hlen=32;}`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:244`
`Snippet: ...hash_type==config::SHA384_T {hlen=48;}`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:245`
`Snippet: ...hash_type==config::SHA512_T {hlen=64;}`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:252`
`Snippet: ...config::AES_128_GCM_SHA256 {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:255`
`Snippet: ...config::AES_256_GCM_SHA384 {`
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
`Location: rust/client/src/sal_m/sal.rs:258`
`Snippet: ...pher_suite==config::CHACHA20_POLY1305_SHA256 {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:258`
`Snippet: ...::CHACHA20_POLY1305_SHA256 {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:267`
`Snippet: ...config::AES_128_GCM_SHA256 {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:270`
`Snippet: ...config::AES_256_GCM_SHA384 {`
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
`Location: rust/client/src/sal_m/sal.rs:273`
`Snippet: ...pher_suite==config::CHACHA20_POLY1305_SHA256 {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:273`
`Snippet: ...::CHACHA20_POLY1305_SHA256 {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:303`
`Snippet: ...if htype==config::SHA256_T {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:308`
`Snippet: ...if htype==config::SHA384_T {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:313`
`Snippet: ...if htype==config::SHA512_T {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:324`
`Snippet: ...if htype==config::SHA256_T {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:328`

Snippet: ...if htype==config::SHA384_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:332
Snippet: ...if htype==config::SHA512_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:342
Snippet: ...if h.htype==config::SHA256_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:348
Snippet: ...if h.htype==config::SHA384_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:354
Snippet: ...if h.htype==config::SHA512_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:366
Snippet: ...if h.htype==config::SHA256_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:372
Snippet: ...if h.htype==config::SHA384_T {
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/sal_m/sal.rs:378
Snippet: ...if h.htype==config::SHA512_T {
Remediation: No action needed.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/client/src/sal_m/sal.rs:433
Snippet: use mcore::kyber; // first...
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/client/src/sal_m/sal.rs:436
Snippet: kyber::keypair_768(&r64;,&...
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:439`
`Snippet: ...et_key_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:440`
`Snippet: ...ic_key_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:445`
`Snippet: ...if group==config::MLKEM768 {`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:446`
`Snippet: use mcore::kyber;`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:449`
`Snippet: kyber::keypair_768(&r64;,&...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:483`
`Snippet: use mcore::kyber;`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:486`
`Snippet: kyber::encrypt_768(&r32;,&...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:490`
`Snippet: ...ic_key_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:491`
`Snippet: ...ic_key_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:492`
`Snippet: ...secret_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:501`
`Snippet: ...if group==config::MLKEM768 {`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:502`
`Snippet: use mcore::kyber;`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:505`
`Snippet: kyber::encrypt_768(&r32;,&...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:538`
`Snippet: use mcore::kyber;`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:539`
`Snippet: kyber::decrypt_768(&sk;[0....`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:541`
`Snippet: ...et_key_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:542`
`Snippet: ...ic_key_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:543`
`Snippet: ...secret_size(config::MLKEM768);`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/sal_m/sal.rs:547`

Snippet: ...if group==config::MLKEM768 {
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/client/src/sal_m/sal.rs:548
Snippet: use mcore::kyber;
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: rust/client/src/sal_m/sal.rs:549
Snippet: kyber::decrypt_768(&sk;[0....
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/sal_m/sal.rs:563
Snippet: /// Dilithium signature verificat...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/sal_m/sal.rs:564
Snippet: fn dilithium3_verify(cert: &[u8]...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/sal_m/sal.rs:565
Snippet: use mcore::dilithium;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/sal_m/sal.rs:566
Snippet: return dilithium::verify_3(false,&pu...;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/sal_m/sal.rs:569
Snippet: /// Dilithium signature
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/sal_m/sal.rs:570
Snippet: pub fn dilithium3_sign(key: &[u8],me...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/sal_m/sal.rs:571
Snippet: use mcore::dilithium;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:572`
`Snippet: dilithium::signature_3(false,...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:573`
`Snippet: return dilithium::SIG_SIZE_3;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:576`
`Snippet: /// Dilithium signature verificat...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:577`
`Snippet: fn dilithium2_verify(cert: &[u8]...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:578`
`Snippet: use mcore::dilithium;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:579`
`Snippet: return dilithium::verify_2(false,&pu...;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:582`
`Snippet: /// Dilithium signature`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:583`
`Snippet: pub fn dilithium2_sign(key: &[u8],me...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:584`
`Snippet: use mcore::dilithium;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:585`
`Snippet: dilithium::signature_2(false,...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/sal_m/sal.rs:586`
`Snippet: return dilithium::SIG_SIZE_2;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:797`
`Snippet: ...rsa_pkcs1_sha256 (for certificates),...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:798`
`Snippet: ...and ecdsa_secp256r1_sha256."`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:803`
`Snippet: ...ig::ECDSA_SECP256R1_SHA256 => {return secp256r...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:804`
`Snippet: ...ig::ECDSA_SECP256R1_SHA384 => {return secp256r...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:805`
`Snippet: ...ig::ECDSA_SECP384R1_SHA384 => {return secp384r...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:811`
`Snippet: ...config::RSA_PKCS1_SHA256 => {return rsa_pkcs...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:812`
`Snippet: ...config::RSA_PKCS1_SHA384 => {return rsa_pkcs...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/sal.rs:813`

```
Snippet: ...config::RSA_PKCS1_SHA512 => {return rsa_pkcs...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/sal_m/sal.rs:814
Snippet: ...onfig::RSA_PSS_RSAE_SHA256 => {return rsa_pss_...
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/sal_m/sal.rs:824
Snippet: config::MLDSA65 => {return dilith...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/sal_m/sal.rs:825
Snippet: config::MLDSA44 => {return dilith...
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/sal_m/sal.rs:836
Snippet: ...g:: ECDSA_SECP256R1_SHA256 => {return secp256r...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/sal_m/sal.rs:837
Snippet: ...g:: ECDSA_SECP256R1_SHA384 => {return secp256r...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/sal_m/sal.rs:838
Snippet: ...g:: ECDSA_SECP384R1_SHA384 => {return secp384r...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/sal_m/sal.rs:843
Snippet: ...nfig:: RSA_PSS_RSAE_SHA256 => {return rsa_pss_...
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/sal_m/sal.rs:852
Snippet: ...config:: MLDSA65 => {return dilith...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/sal_m/sal.rs:853
Snippet: ...config:: MLDSA44 => {return dilith...
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/client/src/sal_m/sal.rs:860
Snippet: ...g:: ECDSA_SECP256R1_SHA256 => {return secp256r...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/client/src/sal_m/sal.rs:861
Snippet: ...g:: ECDSA_SECP256R1_SHA384 => {return secp256r...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/client/src/sal_m/sal.rs:862
Snippet: ...g:: ECDSA_SECP384R1_SHA384 => {return secp384r...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/client/src/sal_m/sal.rs:863
Snippet: ...nfig:: RSA_PSS_RSAE_SHA256 => {return rsa_pss_...
```

Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: rust/client/src/sal_m/sal.rs:866
Snippet: config:: MLDSA65 => {return dilith...
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: rust/client/src/sal_m/sal.rs:867
Snippet: config:: MLDSA44 => {return dilith...
```

Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/client/src/sal_m/pqibe.rs:6
Snippet: use mcore::sha3;
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/client/src/sal_m/pqibe.rs:7
Snippet: use mcore::sha3::SHA3;
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: rust/client/src/sal_m/pqibe.rs:223
Snippet: let mut sh = SHA3::new(sha3::SHAKE128...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/pqibe.rs:259`
`Snippet: let mut sh = SHA3::new(sha3::HASH256)...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/pqibe.rs:388`
`Snippet: let mut sh = SHA3::new(sha3::SHAKE256...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/pqibe.rs:403`
`Snippet: sh = SHA3::new(sha3::SHAKE256...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/pqibe.rs:419`
`Snippet: sh = SHA3::new(sha3::SHAKE256...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/pqibe.rs:497`
`Snippet: let mut sh = SHA3::new(sha3::SHAKE256...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/pqibe.rs:507`
`Snippet: sh = SHA3::new(sha3::HASH256)...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/sal_m/pqibe.rs:529`
`Snippet: let mut sh = SHA3::new(sha3::HASH256)...`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/keys.rs:183`
`Snippet: // *** if sigalg==MLDSA44_ED25519, extract...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/keys.rs:184`
`Snippet: if sigalg==MLDSA44_ED25519 {`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/keys.rs:188`

```
Snippet: ...=sal::tls_signature(MLDSA44,&key;[32..],&ccv;[0...
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/tls13/keys.rs:193
Snippet: ...lg==ECDSA_SECP256R1_SHA256 || sigalg==ECDSA_SE...
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/tls13/keys.rs:209
Snippet: if sigalg==MLDSA44_ED25519 {
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/tls13/keys.rs:218
Snippet: ...ls_signature_verify(MLDSA44,&scv;[0..ptr],&scv...;
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/tls13/keys.rs:223
Snippet: ...lg==ECDSA_SECP256R1_SHA256 || sigalg==ECDSA_SE...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/tls13/logger.rs:50
Snippet: AES_128_GCM_SHA256 => log(IO_DEBUG,"AE...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/tls13/logger.rs:51
Snippet: AES_256_GCM_SHA384 => log(IO_DEBUG,"AE...
```
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
```
Location: rust/client/src/tls13/logger.rs:52
Snippet: CHACHA20_POLY1305_SHA256 =>...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/tls13/logger.rs:52
Snippet: ...CHACHA20_POLY1305_SHA256 => log(IO_DEBUG,"CH...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: rust/client/src/tls13/logger.rs:61
Snippet: ...ECDSA_SECP256R1_SHA256 => log(level,"ECDSA...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/logger.rs:62`
`Snippet: ...RSA_PSS_RSAE_SHA256 => log(level,"RSA_P...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/logger.rs:63`
`Snippet: RSA_PKCS1_SHA256 => log(level,"RSA_P...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/logger.rs:64`
`Snippet: ...ECDSA_SECP384R1_SHA384 => log(level,"ECDSA...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/logger.rs:65`
`Snippet: ...RSA_PSS_RSAE_SHA384 => log(level,"RSA_P...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/logger.rs:66`
`Snippet: RSA_PKCS1_SHA384 => log(level,"RSA_P...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/logger.rs:67`
`Snippet: ...RSA_PSS_RSAE_SHA512 => log(level,"RSA_P...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/logger.rs:68`
`Snippet: RSA_PKCS1_SHA512 => log(level,"RSA_P...`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/logger.rs:69`
`Snippet: MLDSA65 => log(level,"MLD...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/logger.rs:70`
`Snippet: MLDSA44 => log(level,"MLD...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/logger.rs:71`
`Snippet: MLDSA44_ED25519 => log(le...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/tls13/logger.rs:85`
`Snippet: MLKEM768 => log(level,"ML...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: rust/client/src/tls13/logger.rs:86`
`Snippet: ...IO_PROTOCOL,"HYBRID MLKEM768+X25519\n",-1,Non...`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/logger.rs:169`
`Snippet: ...OTOCOL,"Hashed with SHA256\n",-1,None);}`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/logger.rs:170`
`Snippet: ...OTOCOL,"Hashed with SHA384\n",-1,None);}`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/logger.rs:171`
`Snippet: ...OTOCOL,"Hashed with SHA512\n",-1,None);}`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/logger.rs:186`
`Snippet: ...re is Post Quantum (MLDSA) of length (bits) "...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/logger.rs:189`
`Snippet: ...ignature is Hybrid (MLDSA44+ED25519) of lengt...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/logger.rs:218`
`Snippet: ...ey is Post Quantum (MLDSA) of length (bits) "...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/logger.rs:221`

Snippet: ...blic key is Hybrid (MLDSA44+ED25519) of lengt...
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/tls13/clientcert.rs:124
Snippet: ...urn ECDSA_SECP256R1_SHA256; // as long as this...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/tls13/clientcert.rs:127
Snippet: ...urn ECDSA_SECP384R1_SHA384; // as long as thi...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/tls13/clientcert.rs:131
Snippet: ...return RSA_PSS_RSAE_SHA256;
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/clientcert.rs:134
Snippet: ...pk.curve==x509::USE_MLDSA65 {
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/clientcert.rs:135
Snippet: return MLDSA65;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/clientcert.rs:137
Snippet: ...pk.curve==x509::USE_MLDSA44 {
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/clientcert.rs:138
Snippet: return MLDSA44;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/clientcert.rs:142
Snippet: return MLDSA44_ED25519;
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/tls13/clientcert.rs:160
Snippet: ...en]=ECDSA_SECP256R1_SHA256; // as long as this...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/clientcert.rs:164`
`Snippet: ...en]=ECDSA_SECP384R1_SHA384; // as long as thi...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/clientcert.rs:171`
`Snippet: ...ents[len]=RSA_PKCS1_SHA256;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/clientcert.rs:175`
`Snippet: ...ents[len]=RSA_PKCS1_SHA384;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/clientcert.rs:179`
`Snippet: ...ents[len]=RSA_PKCS1_SHA512;`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/clientcert.rs:186`
`Snippet: ...pk.curve==x509::USE_MLDSA65 {`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/clientcert.rs:187`
`Snippet: ...requirements[len]=MLDSA65;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/clientcert.rs:189`
`Snippet: ...pk.curve==x509::USE_MLDSA44 {`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/clientcert.rs:190`
`Snippet: ...requirements[len]=MLDSA44;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/clientcert.rs:196`
`Snippet: ...requirements[len]=MLDSA44_ED25519;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/clientcert.rs:198`
`Snippet: ...requirements[len]=MLDSA44;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/x509.rs:41`
`Snippet: pub const USE_MLDSA44:usize = 13; // F...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/x509.rs:42`
`Snippet: pub const USE_MLDSA65:usize = 14 ; // F...`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:63`
`Snippet: ...Elliptic Curve with SHA256`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:65`
`Snippet: const ECCSHA256:[u8;8]=[0x2a, 0x86,...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:66`
`Snippet: const ECCSHA384:[u8;8]=[0x2a, 0x86,...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:67`
`Snippet: const ECCSHA512:[u8;8]=[0x2a, 0x86,...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:76`
`Snippet: const RSASHA256:[u8;9]=[0x2a, 0x86,...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:77`
`Snippet: const RSASHA384:[u8;9]=[0x2a, 0x86,...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:78`

```
Snippet: const RSASHA512:[u8;9]=[0x2a, 0x86,...
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/tls13/x509.rs:79
Snippet: //const MLDSA65:[u8;11]=[0x2b, 0x...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/tls13/x509.rs:81
Snippet: const MLDSA65:[u8;9]=[0x60,0x86...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/tls13/x509.rs:82
Snippet: const MLDSA44:[u8;9]=[0x60,0x86...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/tls13/x509.rs:83
Snippet: ...0x0F,0x0C,0x06]; // MLDSA44 + ED25519
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/tls13/x509.rs:381
Snippet: if MLDSA65 == soid[0..slen]...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/tls13/x509.rs:415
Snippet: if MLDSA65 == soid[0..slen]...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/tls13/x509.rs:416
Snippet: ...ret.curve=USE_MLDSA65;
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/tls13/x509.rs:418
Snippet: if MLDSA44 == soid[0..slen]...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: rust/client/src/tls13/x509.rs:419
Snippet: ...ret.curve=USE_MLDSA44;
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:708`
`Snippet: if ECCSHA256 == soid[0..slen] {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:712`
`Snippet: if ECCSHA384 == soid[0..slen] {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:716`
`Snippet: if ECCSHA512 == soid[0..slen] {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:720`
`Snippet: if RSASHA256 == soid[0..slen] {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:724`
`Snippet: if RSASHA384 == soid[0..slen] {`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/x509.rs:728`
`Snippet: if RSASHA512 == soid[0..slen] {`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/x509.rs:732`
`Snippet: if MLDSA65 == soid[0..slen]...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/x509.rs:735`
`Snippet: ...ret.curve=USE_MLDSA65;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/x509.rs:737`
`Snippet: if MLDSA44 == soid[0..slen]...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/x509.rs:740`
`Snippet: ...ret.curve=USE_MLDSA44;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/x509.rs:1147`
`Snippet: if MLDSA65 == koid[0..slen]...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/x509.rs:1148`
`Snippet: ...=DLM; ret.curve=USE_MLDSA65;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/x509.rs:1150`
`Snippet: if MLDSA44 == koid[0..slen]...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: rust/client/src/tls13/x509.rs:1151`
`Snippet: ...=DLM; ret.curve=USE_MLDSA44;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/certchain.rs:120`
`Snippet: ...ify(ECDSA_SECP256R1_SHA256,cert,sig,pubkey);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/certchain.rs:123`
`Snippet: ...ify(ECDSA_SECP384R1_SHA384,cert,sig,pubkey);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/certchain.rs:132`
`Snippet: ...re_verify(RSA_PKCS1_SHA256,cert,sig,pubkey);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/certchain.rs:135`
`Snippet: ...re_verify(RSA_PKCS1_SHA384,cert,sig,pubkey);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: rust/client/src/tls13/certchain.rs:138`

Snippet: ...re_verify(RSA_PKCS1_SHA512,cert,sig,pubkey);
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/certchain.rs:140
Snippet: ...st.curve==x509::USE_MLDSA65 {
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/certchain.rs:141
Snippet: ...ls_signature_verify(MLDSA65,cert,sig,pubkey);...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/certchain.rs:143
Snippet: ...st.curve==x509::USE_MLDSA44 {
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/certchain.rs:144
Snippet: ...ls_signature_verify(MLDSA44,cert,sig,pubkey);...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/certchain.rs:151
Snippet: ...ls_signature_verify(MLDSA44,cert,sig2,pub2);
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: rust/client/src/tls13/connection.rs:108
Snippet: ...: 0, //AES_128_GCM_SHA256,
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/cacerts.rs:5
Snippet: ...ntal hybrid ED25519+MLDSA44 root certificate,...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/cacerts.rs:38
Snippet: // Hybrid ED25519+MLDSA44 Cert expires Jan...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: rust/client/src/tls13/cacerts.rs:127
Snippet: // MLDSA65 cert, expires Jan...
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/include/tls1_3.h:35`
`Snippet: ...quantum (MLDSA and MLKEM?) */`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:35`
`Snippet: .../**< Post quantum (MLDSA and MLKEM?) */`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/include/tls1_3.h:36`
`Snippet: .../**< Hybrid, MLKEM/MLDSA + X25519 */`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:36`
`Snippet: .../**< Hybrid, MLKEM/MLDSA + X25519 */`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:47`
`Snippet: ...3 /**< self signed MLDSA cert */`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:48`
`Snippet: ...signed Hybrid cert (MLDSA+ECC) */`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:110`
`Snippet: #define TLS_SHA256_T 1 /**<...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:111`
`Snippet: #define TLS_SHA384_T 2 /**<...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:112`
`Snippet: #define TLS_SHA512_T 3 /**<...`
Remediation: No action needed.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/include/tls1_3.h:158`
`Snippet: ...y size in bytes MLKEM768+X25519 */`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/include/tls1_3.h:159`
`Snippet: ...hertext size MLKEM768+X25519 */`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/include/tls1_3.h:160`
`Snippet: ...ey size in bytes MLKEM768+X25519 */`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:169`
`Snippet: ...be much bigger for MLDSA!*/`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:174`
`Snippet: // all certs use ML-DSA65 or smaller`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:175`
`Snippet: ...y size in bytes MLDSA65>MLDSA44 + P256 */...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:176`
`Snippet: ...ey size in bytes MLDSA65>MLDSA44 + P256 (m...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:177`
`Snippet: ...ytes MLDSA65>MLDSA44 + P256 (D...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/include/tls1_3.h:179`
`Snippet: ...key size in bytes MLKEM768+X25519 */`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/include/tls1_3.h:180`

```
Snippet: ...ciphertext size MLKEM768+X25519 */
```
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
```
Location: cpp/include/tls1_3.h:181
Snippet: ...e key size in bytes MLKEM768+X25519 */
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/include/tls1_3.h:212
Snippet: ...ine TLS_AES_128_GCM_SHA256 0x1301 /**< AES12...
```
Remediation: No action needed.

[INFO] AES256_USAGE
AES-256 symmetric encryption — considered quantum-resistant at this key size (Grover's algorithm only halves effective security).
```
Location: cpp/include/tls1_3.h:213
Snippet: ...HA384 0x1302 /**< AES256/SHA384/GCM cipher s...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/include/tls1_3.h:213
Snippet: ...ine TLS_AES_256_GCM_SHA384 0x1302 /**< AES25...
```
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
```
Location: cpp/include/tls1_3.h:214
Snippet: #define TLS_CHACHA20_POLY1305_SHA256 0x1...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/include/tls1_3.h:214
Snippet: ...S_CHACHA20_POLY1305_SHA256 0x1303 /**< CHACHA2...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/include/tls1_3.h:215
Snippet: ...ine TLS_AES_128_CCM_SHA256 0x1304 /**< AES/S...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/include/tls1_3.h:216
Snippet: ...e TLS_AES_128_CCM_8_SHA256 0x1305 /**< AES/SHA...
```
Remediation: No action needed.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
```
Location: cpp/include/tls1_3.h:224
Snippet: #define MLKEM512 0x0200...
```

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/include/tls1_3.h:225`
`Snippet: #define MLKEM768 0x0201...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/include/tls1_3.h:226`
`Snippet: #define MLKEM1024 0x0202...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/include/tls1_3.h:227`
`Snippet: ...ybrid key exchange, MLKEM768+X25519 */`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:230`
`Snippet: ...ine ECDSA_SECP256R1_SHA256 0x0403 /**< Suppo...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:231`
`Snippet: ...ine ECDSA_SECP256R1_SHA384 0x0413 /**< Non-s...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:232`
`Snippet: ...ine ECDSA_SECP384R1_SHA384 0x0503 /**< Suppo...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:233`
`Snippet: ...define RSA_PSS_RSAE_SHA256 0x0804 /**< Su...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:234`
`Snippet: ...define RSA_PSS_RSAE_SHA384 0x0805 /**< Su...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:235`
`Snippet: ...define RSA_PSS_RSAE_SHA512 0x0806 /**< Su...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:236`
`Snippet: #define RSA_PKCS1_SHA256 0x0401 /**<...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:237`
`Snippet: #define RSA_PKCS1_SHA384 0x0501 /**<...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:238`
`Snippet: #define RSA_PKCS1_SHA512 0x0601 /**<...`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:242`
`Snippet: #define MLDSA44 0x0904...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:243`
`Snippet: #define MLDSA65 0x0905...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:244`
`Snippet: #define MLDSA87 0x0906...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:245`
`Snippet: #define MLDSA44_ED25519 0xFF86...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/include/tls1_3.h:256`
`Snippet: #define MLDSA_KP 4`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/include/tls1_3.h:464`
`Snippet: ...ash type (typically SHA256) */`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

```
Location: cpp/include/tls_x509.h:31
Snippet: ...509_H256 2 /**< SHA256 hash algorithm used...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/include/tls_x509.h:32
Snippet: ...509_H384 3 /**< SHA384 hash algorithm used...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/include/tls_x509.h:33
Snippet: ...509_H512 4 /**< SHA512 hash algorithm used...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/include/tls_x509.h:34
Snippet: ...509_SHAKE256 5 /**< SHA3 SHAKE256 algorithm...
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/include/tls_x509.h:45
Snippet: #define USE_MLDSA44 13 /**< For ML-...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/include/tls_x509.h:46
Snippet: #define USE_MLDSA65 14 /**< For ML-...
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/lib/tls_keys_calc.cpp:83
Snippet: ...ite=TLS_AES_128_GCM_SHA256; // default
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_keys_calc.cpp:389
Snippet: if (sigAlg==MLDSA44_ED25519)
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_keys_calc.cpp:396
Snippet: ...SAL_tlsSignature(MLDSA44,&SKEY;,&CCV;,&SSIG;)...
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/lib/tls_keys_calc.cpp:403
Snippet: ...lg==ECDSA_SECP256R1_SHA256 || sigAlg==ECDSA_SE...
```

Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/lib/tls_keys_calc.cpp:424`
`Snippet: if (sigAlg==MLDSA44_ED25519)`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/lib/tls_keys_calc.cpp:434`
`Snippet: ..._tlsSignatureVerify(MLDSA44,&SCV;,&SSIG;,&SPUB;)...`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_keys_calc.cpp:438`
`Snippet: ...lg==ECDSA_SECP256R1_SHA256 || sigAlg==ECDSA_SE...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_logger.cpp:66`
`Snippet: ...ase TLS_AES_128_GCM_SHA256:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_logger.cpp:67`
`Snippet: ...*)"TLS_AES_128_GCM_SHA256\n");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_logger.cpp:69`
`Snippet: ...ase TLS_AES_256_GCM_SHA384:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_logger.cpp:70`
`Snippet: ...*)"TLS_AES_256_GCM_SHA384\n");`
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
`Location: cpp/lib/tls_logger.cpp:72`
`Snippet: case TLS_CHACHA20_POLY1305_SHA256:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_logger.cpp:72`
`Snippet: ...S_CHACHA20_POLY1305_SHA256:`
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
`Location: cpp/lib/tls_logger.cpp:73`
`Snippet: ...printf((char *)"TLS_CHACHA20_POLY1305_SHA256\n")...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_logger.cpp:73`
`Snippet: ...S_CHACHA20_POLY1305_SHA256\n");`
Remediation: No action needed.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/lib/tls_logger.cpp:103`
`Snippet: case MLKEM768:`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/lib/tls_logger.cpp:104`
`Snippet: ...myprintf((char *)"MLKEM768\n");`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/lib/tls_logger.cpp:107`
`Snippet: ...myprintf((char *)"MLKEM768 + X25519\n");`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_logger.cpp:127`
`Snippet: ...ase ECDSA_SECP256R1_SHA256:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_logger.cpp:128`
`Snippet: ...*)"ECDSA_SECP256R1_SHA256\n");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_logger.cpp:130`
`Snippet: ...case RSA_PSS_RSAE_SHA256:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_logger.cpp:131`
`Snippet: ...har *)"RSA_PSS_RSAE_SHA256\n");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

Location: cpp/lib/tls_logger.cpp:133
Snippet: case RSA_PKCS1_SHA256:
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_logger.cpp:134
Snippet: ...((char *)"RSA_PKCS1_SHA256\n");
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_logger.cpp:136
Snippet: ...ase ECDSA_SECP384R1_SHA384:
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_logger.cpp:137
Snippet: ...*)"ECDSA_SECP384R1_SHA384\n");
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_logger.cpp:139
Snippet: ...case RSA_PSS_RSAE_SHA384:
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_logger.cpp:140
Snippet: ...har *)"RSA_PSS_RSAE_SHA384\n");
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_logger.cpp:142
Snippet: case RSA_PKCS1_SHA384:
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_logger.cpp:143
Snippet: ...((char *)"RSA_PKCS1_SHA384\n");
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_logger.cpp:145
Snippet: ...case RSA_PSS_RSAE_SHA512:
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_logger.cpp:146
Snippet: ...har *)"RSA_PSS_RSAE_SHA512\n");

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_logger.cpp:148
Snippet: case RSA_PKCS1_SHA512:
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_logger.cpp:149
Snippet: ...((char *)"RSA_PKCS1_SHA512\n");
```

Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: cpp/lib/tls_logger.cpp:154
Snippet: case MLDSA44:
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: cpp/lib/tls_logger.cpp:155
Snippet: ...myprintf((char *)"MLDSA44\n");
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: cpp/lib/tls_logger.cpp:157
Snippet: case MLDSA65:
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: cpp/lib/tls_logger.cpp:158
Snippet: ...myprintf((char *)"MLDSA65\n");
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: cpp/lib/tls_logger.cpp:160
Snippet: case MLDSA44_ED25519:
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: cpp/lib/tls_logger.cpp:161
Snippet: ...myprintf((char *)"MLDSA44 + ED25519\n");...
```

Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_logger.cpp:322
Snippet: ...char *)"Hashed with SHA256\n",NULL,0,NULL);
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/lib/tls_logger.cpp:323
Snippet: ...char *)"Hashed with SHA384\n",NULL,0,NULL);
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/lib/tls_logger.cpp:324
Snippet: ...char *)"Hashed with SHA512\n",NULL,0,NULL);
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_logger.cpp:338
Snippet: ...re is Post Quantum (MLDSA) of length (bits) "...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_logger.cpp:341
Snippet: ...ignature is Hybrid (MLDSA44+ED25519) of lengt...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_logger.cpp:374
Snippet: ...ey is Post Quantum (MLDSA) of length (bits) "...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_logger.cpp:377
Snippet: ...blic key is Hybrid (MLDSA44+ED25519) of lengt...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_cacerts.cpp:13
Snippet: ...ntal hybrid ED25519+MLDSA44 root certificate,...
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_cacerts.cpp:45
Snippet: ...NTUM // ED25519 + MLDSA44 expires Jan 2027
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_cacerts.cpp:135
Snippet: // MLDSA65 root certificate...
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

Location: cpp/lib/tls_client_cert.cpp:76
Snippet: ...alg=ECDSA_SECP256R1_SHA256;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_client_cert.cpp:92
Snippet: ...alg=ECDSA_SECP256R1_SHA256;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_client_cert.cpp:203
Snippet: ...urn ECDSA_SECP256R1_SHA256; // as long as this...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_client_cert.cpp:206
Snippet: ...urn ECDSA_SECP384R1_SHA384; // as long as thi...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_client_cert.cpp:210
Snippet: ...return RSA_PSS_RSAE_SHA256;
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_client_cert.cpp:213
Snippet: ...if (pk->curve==USE_MLDSA65)
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_client_cert.cpp:214
Snippet: return MLDSA65;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_client_cert.cpp:215
Snippet: ...if (pk->curve==USE_MLDSA44)
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_client_cert.cpp:216
Snippet: return MLDSA44;
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_client_cert.cpp:219
Snippet: return MLDSA44_ED25519;

Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/lib/tls_client_cert.cpp:237
Snippet: ...en]=ECDSA_SECP256R1_SHA256; // as long as this...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/lib/tls_client_cert.cpp:241
Snippet: ...en]=ECDSA_SECP384R1_SHA384; // as long as thi...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/lib/tls_client_cert.cpp:248
Snippet: ...ents[len]=RSA_PKCS1_SHA256;
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/lib/tls_client_cert.cpp:252
Snippet: ...ents[len]=RSA_PKCS1_SHA384;
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/lib/tls_client_cert.cpp:256
Snippet: ...ents[len]=RSA_PKCS1_SHA512;
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_client_cert.cpp:263
Snippet: ...if (pk->curve==USE_MLDSA65) {
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_client_cert.cpp:264
Snippet: ...requirements[len]=MLDSA65;
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_client_cert.cpp:267
Snippet: ...if (pk->curve==USE_MLDSA44) {
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: cpp/lib/tls_client_cert.cpp:268
Snippet: ...requirements[len]=MLDSA44;
```
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/lib/tls_client_cert.cpp:274`
`Snippet: ...requirements[len]=MLDSA44_ED25519;`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/lib/tls_client_cert.cpp:276`
`Snippet: ...requirements[len]=MLDSA44;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_protocol.cpp:23`
`Snippet: ...0;//TLS_AES_128_GCM_SHA256; // cipher...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_cert_chain.cpp:209`
`Snippet: ...ify(ECDSA_SECP256R1_SHA256,CERT,SIG,PUBKEY);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_cert_chain.cpp:211`
`Snippet: ...ify(ECDSA_SECP384R1_SHA384,CERT,SIG,PUBKEY);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_cert_chain.cpp:217`
`Snippet: ...ureVerify(RSA_PKCS1_SHA256,CERT,SIG,PUBKEY);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_cert_chain.cpp:219`
`Snippet: ...ureVerify(RSA_PKCS1_SHA384,CERT,SIG,PUBKEY);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_cert_chain.cpp:221`
`Snippet: ...ureVerify(RSA_PKCS1_SHA512,CERT,SIG,PUBKEY);`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/lib/tls_cert_chain.cpp:222`
`Snippet: ...LM && st.curve==USE_MLDSA65)`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

Location: cpp/lib/tls_cert_chain.cpp:223
Snippet: ..._tlsSignatureVerify(MLDSA65,CERT,SIG,PUBKEY);...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_cert_chain.cpp:224
Snippet: ...LM && st.curve==USE_MLDSA44)
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_cert_chain.cpp:225
Snippet: ..._tlsSignatureVerify(MLDSA44,CERT,SIG,PUBKEY);...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_cert_chain.cpp:230
Snippet: ..... (especially for MLDSA)
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_cert_chain.cpp:232
Snippet: ..._HY1) // ED25519 + MLDSA44
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_cert_chain.cpp:238
Snippet: ..._tlsSignatureVerify(MLDSA44,CERT,&SSIG;,&SPUB;)...
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_x509.cpp:31
Snippet: ...Elliptic Curve with SHA256
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_x509.cpp:32
Snippet: ...ic unsigned char eccsha256[8] = {0x2a, 0x86, 0...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_x509.cpp:33
Snippet: static octad ECCSHA256 = {8, sizeof(eccsha...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_x509.cpp:35
Snippet: ...Elliptic Curve with SHA384

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:36
Snippet: ...ic unsigned char eccsha384[8] = {0x2a, 0x86, 0...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:37
Snippet: static octad ECCSHA384 = {8, sizeof(eccsha...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:39
Snippet: ...Elliptic Curve with SHA512
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:40
Snippet: ...ic unsigned char eccsha512[8] = {0x2a, 0x86, 0...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:41
Snippet: static octad ECCSHA512 = {8, sizeof(eccsha...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:75
Snippet: // RSA with SHA256
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:76
Snippet: ...ic unsigned char rsasha256[9] = {0x2a, 0x86, 0...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:77
Snippet: static octad RSASHA256 = {9, sizeof(rsasha...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:79
Snippet: // RSA with SHA384
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_x509.cpp:80`
`Snippet: ...ic unsigned char rsasha384[9] = {0x2a, 0x86, 0...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_x509.cpp:81`
`Snippet: static octad RSASHA384 = {9, sizeof(rsasha...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_x509.cpp:83`
`Snippet: // RSA with SHA512`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_x509.cpp:84`
`Snippet: ...ic unsigned char rsasha512[9] = {0x2a, 0x86, 0...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_x509.cpp:85`
`Snippet: static octad RSASHA512 = {9, sizeof(rsasha...`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/lib/tls_x509.cpp:87`
`Snippet: // MLDSA65 - official`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/lib/tls_x509.cpp:88`
`Snippet: ...tatic unsigned char mldsa65[9] = {0x60,0x86,0...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/lib/tls_x509.cpp:89`
`Snippet: static octad MLDSA65 = {9, sizeof(mlds...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/lib/tls_x509.cpp:91`
`Snippet: // MLDSA44 - official`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

Location: cpp/lib/tls_x509.cpp:92
Snippet: ...tatic unsigned char mldsa44[9] = {0x60,0x86,0...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_x509.cpp:93
Snippet: static octad MLDSA44 = {9, sizeof(mlds...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_x509.cpp:96
Snippet: // MLDSA44+ED25519 - unoffic...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_x509.cpp:369
Snippet: ...if (OCT_compare(&MLDSA65;, &SOID;) || OCT_co...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_x509.cpp:370
Snippet: { // Its an MLDSA key
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_x509.cpp:393
Snippet: ...if (OCT_compare(&MLDSA65;, &SOID;)) ret.curv...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_x509.cpp:394
Snippet: ...if (OCT_compare(&MLDSA44;, &SOID;)) ret.curv...
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/lib/tls_x509.cpp:428
Snippet: ...9 key followed by a MLDSA44 key
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_x509.cpp:730
Snippet: ...if (OCT_compare(&ECCSHA256;, &SOID;))
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/tls_x509.cpp:735
Snippet: ...if (OCT_compare(&ECCSHA384;, &SOID;))

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:740
Snippet: ...if (OCT_compare(&ECCSHA512;, &SOID;))
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:745
Snippet: ...if (OCT_compare(&RSASHA256;, &SOID;))
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:750
Snippet: ...if (OCT_compare(&RSASHA384;, &SOID;))
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/tls_x509.cpp:755
Snippet: ...if (OCT_compare(&RSASHA512;, &SOID;))
```

Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: cpp/lib/tls_x509.cpp:761
Snippet: ...if (OCT_compare(&MLDSA65;, &SOID;))
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: cpp/lib/tls_x509.cpp:765
Snippet: ...ret.curve= USE_MLDSA65;
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: cpp/lib/tls_x509.cpp:767
Snippet: ...if (OCT_compare(&MLDSA44;, &SOID;))
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: cpp/lib/tls_x509.cpp:771
Snippet: ...ret.curve= USE_MLDSA44;
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: cpp/lib/tls_x509.cpp:1176
Snippet: ...if (OCT_compare(&MLDSA65;, &KOID;)) {ret.typ...
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/lib/tls_x509.cpp:1177`
`Snippet: ...if (OCT_compare(&MLDSA44;, &KOID;)) {ret.typ...`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/lib/tls_x509.cpp:1179`
`Snippet: ...//if (OCT_compare(&MLDSA65;, &KOID;) || OCT_co...`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/tls_client_recv.cpp:768`
`Snippet: ...0804 - RSA-PSS-RSAE-SHA256`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_bfibe.cpp:38`
`Snippet: ...r *)"BLS12381G1_XMD:SHA-256_SVDW_NU_BFIBE"); //...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_bfibe.cpp:81`
`Snippet: sha3 sh;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_bfibe.cpp:82`
`Snippet: SHA3_init(&sh;,SHAKE256);...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_bfibe.cpp:84`
`Snippet: SHA3_process(&sh;,U->val[...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_bfibe.cpp:89`
`Snippet: SHA3_process(&sh;,V->val[...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_bfibe.cpp:92`
`Snippet: SHA3_shake(&sh;,raw,128);...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

Location: cpp/lib/ibe/tls_bfibe.cpp:102
Snippet: sha3 sh;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_bfibe.cpp:103
Snippet: SHA3_init(&sh;,SHA3_HASH2...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_bfibe.cpp:105
Snippet: SHA3_process(&sh;,I->val[...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_bfibe.cpp:107
Snippet: SHA3_hash(&sh;,O->val);
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_bfibe.cpp:152
Snippet: sha3 sh;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_bfibe.cpp:153
Snippet: SHA3_init(&sh;,SHAKE256);...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_bfibe.cpp:155
Snippet: SHA3_process(&sh;,R32->va...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_bfibe.cpp:157
Snippet: SHA3_shake(&sh;,SIGMA.val...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_bfibe.cpp:158
Snippet: SHA3_shake(&sh;,M->val,32...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_pqibe.cpp:260
Snippet: sha3 SHA3;

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:262
Snippet: SHA3_init(&SHA3;,SHAKE128...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:265
Snippet: SHA3_process(&SHA3;,ID[i+...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:266
Snippet: SHA3_shake(&SHA3;,hash,4*...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:301
Snippet: sha3 SHA3;
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:302
Snippet: SHA3_init(&SHA3;,SHA3_HAS...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:303
Snippet: ...i=0;i<DEGREE/8;i++) SHA3_process(&SHA3;,ikey[...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:304
Snippet: SHA3_hash(&SHA3;,(char *)...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:436
Snippet: sha3 sh;
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:439
Snippet: SHA3_init(&sh;,SHAKE256);...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_pqibe.cpp:441`
`Snippet: SHA3_process(&sh;,sigma[j...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_pqibe.cpp:442`
`Snippet: SHA3_process(&sh;,0);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_pqibe.cpp:443`
`Snippet: SHA3_shake(&sh;,(char *)b...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_pqibe.cpp:453`
`Snippet: SHA3_init(&sh;,SHAKE256);...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_pqibe.cpp:455`
`Snippet: SHA3_process(&sh;,sigma[j...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_pqibe.cpp:456`
`Snippet: SHA3_process(&sh;,1);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_pqibe.cpp:457`
`Snippet: SHA3_shake(&sh;,(char *)b...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_pqibe.cpp:469`
`Snippet: SHA3_init(&sh;,SHAKE256);...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/lib/ibe/tls_pqibe.cpp:471`
`Snippet: SHA3_process(&sh;,sigma[j...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

Location: cpp/lib/ibe/tls_pqibe.cpp:472
Snippet: SHA3_process(&sh;,2);
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_pqibe.cpp:473
Snippet: SHA3_shake(&sh;,(char *)b...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_pqibe.cpp:542
Snippet: sha3 sh;
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_pqibe.cpp:543
Snippet: SHA3_init(&sh;,SHAKE256);...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_pqibe.cpp:545
Snippet: SHA3_process(&sh;,R32->va...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_pqibe.cpp:546
Snippet: SHA3_shake(&sh;,(char *)i...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_pqibe.cpp:555
Snippet: SHA3_init(&sh;,SHA3_HASH2...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_pqibe.cpp:557
Snippet: SHA3_process(&sh;,(unsign...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_pqibe.cpp:559
Snippet: SHA3_process(&sh;,ikey[i]...
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/lib/ibe/tls_pqibe.cpp:560
Snippet: SHA3_hash(&sh;,KEY->val);...

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:568
Snippet: sha3 sh;
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:575
Snippet: SHA3_init(&sh;,SHA3_HASH2...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:577
Snippet: SHA3_process(&sh;,(unsign...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:579
Snippet: SHA3_process(&sh;,ikey[i]...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:580
Snippet: SHA3_hash(&sh;,KEY->val);...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/lib/ibe/tls_pqibe.cpp:587
Snippet: sha3 SHA3;
```

Remediation: No action needed.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

```
Location: cpp/src/desktop/client.cpp:125
Snippet: case MLKEM768:
```

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

```
Location: cpp/src/desktop/client.cpp:126
Snippet: printf("MLKEM768\n");
```

Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE

Post-quantum key encapsulation mechanism (NIST FIPS 203).

```
Location: cpp/src/desktop/client.cpp:129
Snippet: printf("MLKEM768+X25519\n");
```

Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:141`
`Snippet: ...ase TLS_AES_128_GCM_SHA256:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:142`
`Snippet: ...tf("TLS_AES_128_GCM_SHA256\n");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:144`
`Snippet: ...ase TLS_AES_256_GCM_SHA384:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:145`
`Snippet: ...tf("TLS_AES_256_GCM_SHA384\n");`
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
`Location: cpp/src/desktop/client.cpp:147`
`Snippet: case TLS_CHACHA20_POLY1305_SHA256:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:147`
`Snippet: ...S_CHACHA20_POLY1305_SHA256:`
Remediation: No action needed.

[INFO] CHACHA20_USAGE
ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.
`Location: cpp/src/desktop/client.cpp:148`
`Snippet: printf("TLS_CHACHA20_POLY1305_SHA256\n")...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:148`
`Snippet: ...S_CHACHA20_POLY1305_SHA256\n");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:160`
`Snippet: ...ase ECDSA_SECP256R1_SHA256:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

Location: cpp/src/desktop/client.cpp:161
Snippet: ...tf("ECDSA_SECP256R1_SHA256\n");
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/desktop/client.cpp:163
Snippet: ...case RSA_PSS_RSAE_SHA256:
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/desktop/client.cpp:164
Snippet: ...rintf("RSA_PSS_RSAE_SHA256\n");
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/desktop/client.cpp:166
Snippet: case RSA_PKCS1_SHA256:
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/desktop/client.cpp:167
Snippet: ...printf("RSA_PKCS1_SHA256\n");
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/desktop/client.cpp:169
Snippet: ...ase ECDSA_SECP384R1_SHA384:
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/desktop/client.cpp:170
Snippet: ...tf("ECDSA_SECP384R1_SHA384\n");
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/desktop/client.cpp:172
Snippet: ...case RSA_PSS_RSAE_SHA384:
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/desktop/client.cpp:173
Snippet: ...rintf("RSA_PSS_RSAE_SHA384\n");
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/desktop/client.cpp:175
Snippet: case RSA_PKCS1_SHA384:

Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:176`
`Snippet: ...printf("RSA_PKCS1_SHA384\n");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:178`
`Snippet: ...case RSA_PSS_RSAE_SHA512:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:179`
`Snippet: ...rintf("RSA_PSS_RSAE_SHA512\n");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:181`
`Snippet: case RSA_PKCS1_SHA512:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:182`
`Snippet: ...printf("RSA_PKCS1_SHA512\n");`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/src/desktop/client.cpp:190`
`Snippet: case MLDSA44:`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/src/desktop/client.cpp:191`
`Snippet: printf("MLDSA44\n");`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/src/desktop/client.cpp:193`
`Snippet: case MLDSA65:`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/src/desktop/client.cpp:194`
`Snippet: printf("MLDSA65\n");`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/src/desktop/client.cpp:196`
`Snippet: case MLDSA44_ED25519:`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/src/desktop/client.cpp:197`
`Snippet: printf("MLDSA44 + ED25519\n");...`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/src/desktop/client.cpp:460`
`Snippet: ...->T.favourite_group=MLKEM768;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/desktop/client.cpp:493`
`Snippet: ...ite=TLS_AES_128_GCM_SHA256;`
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/src/arduino/config.py:694`
`Snippet: print("46. Dilithium\n")`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/src/arduino/config.py:695`
`Snippet: print("47. Kyber\n")`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/src/arduino/config.py:713`
`Snippet: kyber_selected=False`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/src/arduino/config.py:933`
`Snippet: kyber_selected=True`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/src/arduino/config.py:992`
`Snippet: ...os.system(deltext+" dilithium.cpp")`
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

Location: cpp/src/arduino/config.py:993
Snippet: ...os.system(deltext+" dilithium.h")
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: cpp/src/arduino/config.py:996
Snippet: if not kyber_selected :
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: cpp/src/arduino/config.py:997
Snippet: ...os.system(deltext+" kyber.cpp")
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: cpp/src/arduino/config.py:998
Snippet: ...os.system(deltext+" kyber.h")
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: cpp/src/arduino/config.py:999
Snippet: ...ystem(deltext+" testkyber.ino")
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: cpp/src/arduino/config.py:1208
Snippet: ...ystem(deltext+" testkyber.cpp")
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: cpp/src/arduino/client.cpp:66
Snippet: case MLKEM768:
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: cpp/src/arduino/client.cpp:67
Snippet: ...Serial.println("MLKEM768");
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
Location: cpp/src/arduino/client.cpp:70
Snippet: ...Serial.println("MLKEM768+X25519");
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/arduino/client.cpp:82
Snippet: ...ase TLS_AES_128_GCM_SHA256:

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/src/arduino/client.cpp:83
Snippet: ...ln("TLS_AES_128_GCM_SHA256");
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/src/arduino/client.cpp:85
Snippet: ...ase TLS_AES_256_GCM_SHA384:
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/src/arduino/client.cpp:86
Snippet: ...ln("TLS_AES_256_GCM_SHA384");
```

Remediation: No action needed.

[INFO] CHACHA20_USAGE

ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.

```
Location: cpp/src/arduino/client.cpp:88
Snippet: case TLS_CHACHA20_POLY1305_SHA256:
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/src/arduino/client.cpp:88
Snippet: ...S_CHACHA20_POLY1305_SHA256:
```

Remediation: No action needed.

[INFO] CHACHA20_USAGE

ChaCha20-Poly1305 AEAD cipher — modern, fast, quantum-resistant symmetric primitive.

```
Location: cpp/src/arduino/client.cpp:89
Snippet: ...Serial.println("TLS_CHACHA20_POLY1305_SHA256");...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/src/arduino/client.cpp:89
Snippet: ...S_CHACHA20_POLY1305_SHA256");
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/src/arduino/client.cpp:101
Snippet: ...ase ECDSA_SECP256R1_SHA256:
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: cpp/src/arduino/client.cpp:102
Snippet: ...ln("ECDSA_SECP256R1_SHA256");
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/client.cpp:104`
`Snippet: ...case RSA_PSS_RSAE_SHA256:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/client.cpp:105`
`Snippet: ...intln("RSA_PSS_RSAE_SHA256");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/client.cpp:107`
`Snippet: case RSA_PKCS1_SHA256:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/client.cpp:108`
`Snippet: ....println("RSA_PKCS1_SHA256");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/client.cpp:110`
`Snippet: ...ase ECDSA_SECP384R1_SHA384:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/client.cpp:111`
`Snippet: ...ln("ECDSA_SECP384R1_SHA384");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/client.cpp:113`
`Snippet: ...case RSA_PSS_RSAE_SHA384:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/client.cpp:114`
`Snippet: ...intln("RSA_PSS_RSAE_SHA384");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/client.cpp:116`
`Snippet: case RSA_PKCS1_SHA384:`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

Location: cpp/src/arduino/client.cpp:117
Snippet: ....println("RSA_PKCS1_SHA384");
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/arduino/client.cpp:119
Snippet: ...case RSA_PSS_RSAE_SHA512:
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/arduino/client.cpp:120
Snippet: ...intln("RSA_PSS_RSAE_SHA512");
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/arduino/client.cpp:122
Snippet: case RSA_PKCS1_SHA512:
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
Location: cpp/src/arduino/client.cpp:123
Snippet: ....println("RSA_PKCS1_SHA512");
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/src/arduino/client.cpp:131
Snippet: case MLDSA44:
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/src/arduino/client.cpp:132
Snippet: ...Serial.println("MLDSA44");
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/src/arduino/client.cpp:134
Snippet: case MLDSA65:
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/src/arduino/client.cpp:135
Snippet: ...Serial.println("MLDSA65");
Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
Location: cpp/src/arduino/client.cpp:137
Snippet: case MLDSA44_ED25519:

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
`Location: cpp/src/arduino/client.cpp:138`
`Snippet: ...Serial.println("MLDSA44 + ED25519");`
Remediation: No action needed — already migrated.

[INFO] PQC_KEM_USAGE
Post-quantum key encapsulation mechanism (NIST FIPS 203).
`Location: cpp/src/arduino/client.cpp:275`
`Snippet: ...->T.favourite_group=MLKEM768;`
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/client.cpp:307`
`Snippet: ...ite=TLS_AES_128_GCM_SHA256;`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.h:53`
`Snippet: int beginSHA256();`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.h:55`
`Snippet: int updateSHA256(const byte data[],i...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.h:56`
`Snippet: int endSHA256(byte result[]);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.h:57`
`Snippet: int endSHA256(const byte data[],...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.h:58`
`Snippet: int readSHA256(byte context[]);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.h:59`
`Snippet: int writeSHA256(byte context[],int...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.cpp:327`
`Snippet: ...t ECCX08Class::beginSHA256()`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.cpp:330`
`Snippet: ....println("Into beginSHA256");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.cpp:384`
`Snippet: ...ECCX08Class::updateSHA256(const byte data[],i...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.cpp:387`
`Snippet: ...l.print("Into updateSHA256 "); Serial.println(...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.cpp:405`
`Snippet: ...intln("Out of updateSHA256 (FAIL)");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.cpp:416`
`Snippet: ...intln("Out of updateSHA256 ");`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.cpp:420`
`Snippet: int ECCX08Class::endSHA256(byte result[])`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.cpp:422`
`Snippet: return endSHA256(NULL, 0, result);`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
`Location: cpp/src/arduino/ECCX08.cpp:425`
`Snippet: int ECCX08Class::endSHA256(const byte data[],...`
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.

```
Location: cpp/src/arduino/ECCX08.cpp:427
Snippet: ...al.println("Into endSHA256");
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/src/arduino/ECCX08.cpp:444
Snippet: ....println("Out of endSHA256");
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/src/arduino/ECCX08.cpp:448
Snippet: // Read SHA256 context
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/src/arduino/ECCX08.cpp:449
Snippet: ...nt ECCX08Class::readSHA256(byte context[])
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/src/arduino/ECCX08.cpp:452
Snippet: ...l.println("Into readSHA256");
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/src/arduino/ECCX08.cpp:477
Snippet: // write SHA256 context
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/src/arduino/ECCX08.cpp:478
Snippet: ...t ECCX08Class::writeSHA256(byte context[],int...
```
Remediation: No action needed.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: cpp/src/arduino/ECCX08.cpp:482
Snippet: ....println("Into writeSHA256");
```
Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE
Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).
```
Location: servercert/server/server.py:2
Snippet: ...re elliptic curves, MLDSA options could be ad...
```
Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER
Modern hash function with no known practical collision attacks.
```
Location: servercert/server/server.py:10
Snippet: ...SSL by default uses sha256 hashing of certific...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: servercert/server/server.py:11
Snippet: ...it from ecdsa_with_sha256 to ecdsa_with_sha38...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: servercert/server/server.py:19
Snippet: ...3 server.py NIST256 sha384")
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: servercert/server/server.py:23
Snippet: hash="sha256"
```

Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: servercert/server/server.py:62
Snippet: if method=="MLDSA65" :
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: servercert/server/server.py:63
Snippet: ...sl req -new -newkey mldsa65 -keyout server.ke...
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: servercert/server/intermediate/intermediate.py:2
Snippet: ...re elliptic curves, MLDSA options could be ad...
```

Remediation: No action needed — already migrated.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: servercert/server/intermediate/intermediate.py:11
Snippet: ...SSL by default uses sha256 hashing of certific...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: servercert/server/intermediate/intermediate.py:12
Snippet: ...it from ecdsa_with_sha256 to ecdsa_with_sha38...
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: servercert/server/intermediate/intermediate.py:20
Snippet: ...rmediate.py NIST384 sha384")
```

Remediation: No action needed.

[INFO] SHA256_OR_BETTER

Modern hash function with no known practical collision attacks.

```
Location: servercert/server/intermediate/intermediate.py:24
Snippet: hash="sha256"
```

Remediation: No action needed.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: servercert/server/intermediate/intermediate.py:63
Snippet: if method=="MLDSA65" :
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: servercert/server/intermediate/intermediate.py:64
Snippet: ...sl req -new -newkey mldsa65 -keyout inter.key...
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: servercert/server/intermediate/root/root.py:2
Snippet: ...re elliptic curves, MLDSA options could be ad...
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: servercert/server/intermediate/root/root.py:45
Snippet: if method=="MLDSA65" :
```

Remediation: No action needed — already migrated.

[INFO] PQC_SIGNATURE_USAGE

Post-quantum digital signature scheme (NIST FIPS 204/205, or Falcon).

```
Location: servercert/server/intermediate/root/root.py:46
Snippet: ...-days 3650 -newkey mldsa65 -keyout root.key...
```

Remediation: No action needed — already migrated.

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This report is accompanied by a separate, signed manifest file (`TLS1.3.sig.json`) and Collatz Labs' public key (`collatzlabs_public_key.pem`). The manifest contains a SHA-256 hash of this exact PDF and an Ed25519 signature over that hash plus the underlying finding data, made with Collatz Labs' private key. This PDF intentionally does not embed its own hash or signature -- a signed file cannot contain a correct signature of itself, since adding the signature would change the file. Verify independently with:

```
python3 verify_report.py --pdf <this file> --manifest TLS1.3.sig.json --pubkey
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